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High-Dimensional Financial Volatility: A Bellman Filter Approach to Dynamic-Factor Stochastic Volatility

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Abstract

Accurately modeling high-dimensional financial volatility requires both tractable estimation methods and correctly specified models. This thesis evaluates the Bellman Information Filter (BIF) for Dynamic Factor Stochastic Volatility (DFSV) models, adapting it via block-coordinate descent for stable, efficient ($\mathcal{O}(K^3 + N)$) estimation. While simulations confirm BIF's superior volatility tracking (+30% vs a Particle Filter with 10,000 particles) and stability (up to $N = 150$ assets and $K = 15$ factors), empirical application (95 Fama-French portfolios) reveals a critical insight: despite BIF producing plausible aggregate dynamics, residual diagnostics fail due to the standard DFSV assumption of constant idiosyncratic variance. The BIF is a promising estimator, but tackling high-dimensional volatility effectively demands addressing model specification limitations.

Contents

List of Abbreviations and Symbols

BCD	Block Coordinate Descent
BF	Bellman Filter
BIF	Bellman Information Filter
DCC	Dynamic Conditional Correlation
DFSV	Dynamic Factor Stochastic Volatility
EM	Expectation-Maximization
FIM	Fisher Information Matrix
PD	Positive Definite
PF	Particle Filter
PSD	Positive Semi-Definite
RMSE	Root Mean Squared Error
SISR	Sequential Importance Sampling with Resampling
VAR	Vector Autoregressive
ϵ	Small buffer for eigenvalue constraint
λ	Penalty coefficient for eigenvalue constraint
$\boldsymbol{\Theta}$	Parameter vector containing all model parameters
$\boldsymbol{\mathcal{I}}$	Fisher Information Matrix
$\boldsymbol{\Lambda}$	Factor loading matrix
$\boldsymbol{\Phi}_f$	Factor transition matrix
$\boldsymbol{\Phi}_h$	Log-volatility transition matrix
$\boldsymbol{Q}_h$	Log-volatility innovation covariance matrix
$\boldsymbol{\Sigma}_\epsilon$	Idiosyncratic error covariance matrix (diagonal)

$\boldsymbol{\Sigma}_t$	Marginal covariance matrix of returns
$\boldsymbol{T}$	State transition matrix
$\mathbf{O}$	Matrix of zeros
$\mathrm{cov}(\cdot, \cdot)$	Covariance operator
$\det(\cdot)$	Determinant of a matrix
$\mathrm{diag}(\cdot)$	Diagonal operator (extracts diagonal or creates diagonal matrix)
$\mathrm{E}[\cdot]$	Expectation operator
$\mathrm{softplus}(x)$	Function $\log(1 + e^x)$ that maps any real number to a positive value
$\tanh(x)$	Hyperbolic tangent function
$\mathrm{tr}(\cdot)$	Trace of a matrix
$\mathrm{var}(\cdot)$	Variance operator
$\ell(\cdot)$	Log-likelihood function
$\mathcal{L}(\boldsymbol{\Theta})$	Pseudo-likelihood function for parameter vector $\boldsymbol{\Theta}$
K	Number of latent factors
N	Number of observed series (assets)
P	Number of particles in Particle Filter
T	Number of time periods
$\boldsymbol{\alpha}_t$	State vector $[\boldsymbol{f}_t', \boldsymbol{h}_t']'$ at time t
$\boldsymbol{\epsilon}_t$	Idiosyncratic error vector in observation equation
$\boldsymbol{\eta}_{t+1}$	Innovation vector in log-volatility evolution equation
$\boldsymbol{f}_t$	Vector of latent factors at time t
$\boldsymbol{h}_t$	Vector of log-volatilities at time t
$\boldsymbol{\mu}$	Long-run mean of log-volatilities
$\boldsymbol{\nu}_{t+1}$	Innovation vector in factor evolution equation
$\boldsymbol{\imath}$	Vector of ones
$\boldsymbol{r}_t$	Vector of asset returns at time t
$\mathbf{0}$	Vector of zeros

Chapter 1

Introduction

This thesis addresses the challenge of modeling and estimating dynamic risk in high-dimensional financial systems. Specifically, I investigate the application of a Bellman Information Filter (BIF) to efficiently estimate Dynamic Factor Stochastic Volatility (DFSV) models. My primary contribution is developing and rigorously evaluating a practical BIF framework tailored for DFSV models, integrating Vector Autoregression (VAR) dynamics and proposing a Block Coordinate Descent (BCD) optimization strategy within the filter's update step. My aim is to make high-dimensional DFSV estimation feasible in practice.

1.1 Motivation: The Challenge of Modeling Dynamic Financial Risk

Static factor models ignore empirical features such as volatility clustering and time-varying correlations. The 1998 collapse of Long-Term Capital Management, where a Nobel Prize-winning team lost $4.6 billion in four months due to inadequate modeling of volatility dynamics during the Russian financial crisis, illustrates the cost of mis-modelling volatility. Asset returns are characterized by several well-documented stylized facts, including heavy tails (leptokurtosis), volatility clustering (periods of high/low volatility tend to persist), leverage effects (negative returns often correlate with increased future volatility), and time-varying correlations between assets (Cont, 2001). While models like the Arbitrage Pricing Theory or Fama-French factor models provide valuable insights into average return structures, they typically assume constant volatilities and correlations, contradicting empirical evidence (Han, 2006). Modeling the dynamic nature of risk, particularly the co-movement and persistence in volatility observed across assets (Borghi et al., 2018), enables effective portfolio allocation, risk management, and derivative pricing. This requires models explicitly accounting for evolving factor exposures and stochastic volatility.

1.2 DFSV Models: A Framework for Dynamic Co-movement

Dynamic Factor Stochastic Volatility (DFSV) models provide a tractable way to model joint, time-varying risk. They introduce latent factors driving asset returns and, importantly, allow the volatility of these factors (and potentially idiosyncratic components) to change stochastically over time. This enriches factor models with a time-varying covariance structure, aligning better with observed market phenomena like volatility clustering and dynamic correlations. Building on early work incorporating stochastic volatility into factor structures (Aguilar & West, 2000), models like the Dynamic Factor Multivariate Stochastic Volatility (DFMSV) framework proposed by Han (2006) allow both expected returns and volatilities of many assets to vary over time, driven by a few persistent latent factors with their own stochastic volatilities. Factor-based Stochastic Volatility (SV) models, such as those explored by Philipov and Glickman (2006), provide a structured way to handle covariance evolution in higher dimensions. This offers a parsimonious representation capable of describing complex patterns in asset return distributions while retaining interpretability through factor decomposition, though estimating these complex dynamics efficiently remains a significant hurdle.

1.3 The Estimation Bottleneck: Scaling DFSV Models

Despite their theoretical appeal and ability to capture key stylized facts, estimating DFSV models poses serious computational challenges, particularly in the high-dimensional settings typical of financial applications (i.e., many assets N and factors K). The core difficulties stem from the model's characteristics: a large latent state vector (including factors and their log-volatilities) and non-linear and non-Gaussian dynamics due to the stochastic evolution of volatility. Traditional multivariate GARCH models, while simpler, face parameter proliferation issues (Bollerslev et al., 1994) or impose restrictive correlation structures like in the Dynamic Conditional Correlation (DCC) model (Engle, 2002).

Classical state-space estimation methods falter when applied to DFSV models. The standard Kalman Filter (KF) is optimal only for linear and Gaussian systems, assumptions clearly violated by the SV components (Shumway & Stoffer, 1982). Approximate methods like the Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF) rely on local linearizations or specific distributional assumptions (e.g., Gaussian approximations via moment matching). These approximations can introduce significant bias and may struggle with the strong non-linearities and potentially multi-modal distributions induced by SV, leading to inaccurate state estimates or filter divergence, especially over long time series (Lange, 2024).

Simulation-based approaches offer more flexibility but face their own scalability issues. Bayesian Markov Chain Monte Carlo (MCMC) methods provide a gold standard for inference in complex latent variable models (Kim et al., 1998). However, for high-dimensional state spaces like those in DFSV models, MCMC algorithms often mix slowly and converge slowly, requiring extremely long simulation runs to explore the posterior distribution adequately, making them computationally intensive (Kastner et al., 2017). Particle Filters (PF), also known as Sequential Monte Carlo (SMC) methods, directly approximate the filtering distribution using a set of weighted samples (particles). While theoretically capable of handling arbitrary non-linearities and non-Gaussianities, PFs face an exponential particle-count growth in state dimension: the number of particles required to maintain a given level of accuracy grows exponentially with the dimension of the state vector (Lange, 2024; Malik & Pitt, 2011). In practice, for DFSV models with even moderate N and K, the required particle count becomes computationally prohibitive, and the filter often succumbs to weight degeneracy (where only a few particles have non-negligible weights), leading to unstable and unreliable estimates (Rebeschini & Handel, 2015). This computational bottleneck severely limits the practical application of sophisticated DFSV models.

1.4 The Bellman Filter: A Novel Estimation Approach

The Bellman filter, a recently developed method inspired by Bellman's dynamic programming principle, offers an alternative estimator (Lange, 2024). Instead of attempting to approximate the full filtering distribution via integration (like KF) or simulation (like PF), it recasts the filtering update as a sequential optimization problem, seeking the mode of the posterior state distribution. This mode-based approach, typically implemented using an iterative quadratic approximation to the log-posterior density (akin to a Laplace approximation), efficiently handles the non-linear and non-Gaussian dynamics inherent in DFSV models. Lange (2024) demonstrates that the Bellman filter generalizes Kalman-based filters while maintaining polynomial computational complexity, scaling cubically with the state dimension ($O(m^3)$), making it feasible for the large state vectors encountered in DFSV applications.

Because it is deterministic, BIF side-steps particle degeneracy. It provides a deterministic approximation to the posterior mode and covariance, offering a compelling speed-accuracy trade-off (Lange, 2024). Furthermore, the filter has desirable stability properties, preventing error accumulation over long time series. A key practical advantage is that the Bellman filter naturally produces an approximate pseudo-likelihood function as a by-product of its recursive updates. This function can be maximized using standard numerical optimization techniques to estimate the model's static parameters (hyperparameters). This provides a computationally feasible route to parameter estimation within a quasi-maximum likelihood framework, avoiding the complexities of likelihood evaluation via particle smoothing or the computational burden of full MCMC procedures.

This potential makes the Bellman filter a promising candidate for addressing the estimation challenges in high-dimensional DFSV models, which is the focus of the research questions presented in the following section.

1.5 Research Questions and Contributions

I adapt the Information-form Bellman filter (BIF) to high-dimensional DFSV models. My aim is to connect the expressive power of DFSV models with practical estimation methods, addressing the limitations of existing techniques highlighted in Section 1.3. Specifically, I address four inter-related research questions and the corresponding contributions:

My primary contribution is developing and rigorously evaluating a practical BIF framework tailored for DFSV models. This involves integrating the BIF with VAR dynamics and proposing a specialized BCD optimization strategy within the filter's update step to handle the specific structure of the DFSV estimation problem efficiently (addressing RQ 1). I contrast this approach with the literature, particularly the challenges identified for KF, PF, and MCMC methods. While Lange (2024) introduced the general Bellman filter, its application and adaptation to the specific structure and scale of financial DFSV models, including the BCD

RQ 1	**Method engineering:** How can the Bellman Information Filter be customized—through block-coordinate descent, VAR dynamics and Fisher-information updates—to deliver *deterministic* state *and* hyperparameter estimation for high-dimensional DFSV models, and what is the resulting computational complexity? *(Answered in Sections 2.2.2–2.7 and Appendix A)*
RQ 2	**Simulation benchmarking:** Across increasing asset (N) and factor (K) dimensions, how does DFSV-BIF compare with particle filtering in runtime scaling, numerical stability, factor-state accuracy and volatility-state accuracy? *(Answered in Chapter 4)*
RQ 3	**Empirical performance versus standard industry models:** On a panel of 95 Fama–French size/BM portfolios (1963-2023), does DFSV-BIF improve the modelling and forecasting of time-varying covariances and market co-movement relative to (i) DFSV-PF, (ii) DCC-GARCH(1,1) and (iii) a constant-volatility DFM, when judged under a six-pillar evaluation framework (statistical fit, covariance dynamics, factor loadings, residual diagnostics, parameter plausibility, computational cost)? *(Answered in Chapters 3 and 5)*
RQ 4	**Diagnostic insight and model limitations:** To what extent does the assumption of constant diagonal idiosyncratic variance in DFSV models hinder residual normality and ARCH behaviour, and what extensions (e.g., grouped or asset-level idiosyncratic SV) are indicated by the diagnostic evidence? *(Answered in Section 5.4 and Chapter 6)*

Table 1.1: Research Questions and Corresponding Contributions.

enhancement, is the core advancement here. I demonstrate through simulation studies that the BIF delivers substantial computational speedups and comparable or superior state estimation accuracy, particularly for latent log-volatilities, compared to benchmark particle filters, especially in high-dimensional scenarios. An empirical application will further demonstrate the framework's practical utility, thereby addressing the simulation benchmarking (RQ 2), empirical comparison (RQ 3), and diagnostic analysis (RQ 4) outlined above.

Chapter 2

Methodology: Estimating DFSV Models via Bellman Information Filtering

In this chapter, I detail the methodology for estimating Dynamic Factor Stochastic Volatility (DFSV) models using the Bellman Information Filter (BIF). First, I specify the DFSV model, which captures common movements in asset returns via latent factors while accommodating time-varying volatility. However, the model's non-linear, non-Gaussian structure presents estimation challenges. Subsequently, I introduce the BIF, an efficient state estimation approach offering potential computational and numerical stability advantages over traditional filters (Lange, 2024).

2.1 The Dynamic Factor Stochastic Volatility (DFSV) Model Specification

The DFSV model employs a state-space formulation comprising an observation equation and state evolution equations for latent factors and their volatilities.

Observation Equation: Asset returns, represented by the N-dimensional vector $\boldsymbol{r}_t$, are generated via a linear factor model incorporating idiosyncratic noise:

$$\boldsymbol{r}_t = \boldsymbol{\Lambda}\boldsymbol{f}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}\big(\mathbf{0},\ \boldsymbol{\Sigma}_\epsilon\big), \tag{2.1}$$

$$\boldsymbol{\Sigma}_\epsilon = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_N^2). \tag{2.2}$$

Where:

- $\boldsymbol{\Lambda}$ is an $N \times K$ matrix of factor loadings.
- $\boldsymbol{f}_t = (f_{1,t}, \ldots, f_{K,t})' \in \mathbb{R}^K$ is the vector of latent factors, with $K \ll N$, that captures the common variation among asset returns.
- $\boldsymbol{\epsilon}_t$ represents the idiosyncratic errors with a diagonal covariance matrix $\boldsymbol{\Sigma}_\epsilon$.

Factor Evolution: To capture common dynamics, the latent factor vector $\boldsymbol{f}_t$ follows a VAR(1) process featuring stochastic volatility:

$$\boldsymbol{f}_{t+1} = \boldsymbol{\Phi}_f\, \boldsymbol{f}_t + \boldsymbol{\nu}_{t+1}, \quad \boldsymbol{\nu}_{t+1} \sim \mathcal{N}\Big(\boldsymbol{0},\ \mathrm{diag}\Big(e^{h_{1,t+1}}, \ldots, e^{h_{K,t+1}}\Big)\Big). \tag{2.3}$$

Where $\boldsymbol{\Phi}_f$ governs factor persistence and spill-over. The innovation covariance is time-varying and depends on the latent log-volatility $h_{k,t+1}$ for $k = 1, \ldots, K$.

Log-Volatility Dynamics: Volatility dynamics are modeled jointly for the factors' latent log-volatilities, $\boldsymbol{h}_t = (h_{1,t}, h_{2,t}, \ldots, h_{K,t})^\top$, which also follow a VAR(1) process:

$$\boldsymbol{h}_{t+1} = \boldsymbol{\mu} + \boldsymbol{\Phi}_h\, (\boldsymbol{h}_t - \boldsymbol{\mu}) + \boldsymbol{\eta}_{t+1}, \quad \boldsymbol{\eta}_{t+1} \sim \mathcal{N}(\boldsymbol{0},\, \boldsymbol{Q}_h)\,. \tag{2.4}$$

Here,

- $\boldsymbol{\mu} \in \mathbb{R}^K$ is the long-run mean vector of the log-volatilities,
- $\boldsymbol{\Phi}_h \in \mathbb{R}^{K\times K}$ is an autoregressive coefficient matrix that captures both the persistence in each factor's volatility and cross-factor spillovers,
- $\boldsymbol{Q}_h \in \mathbb{R}^{K\times K}$ is a positive-definite covariance matrix governing the volatility shocks.

Complete State Vector: Combining these components, the full state vector $\boldsymbol{\alpha}_t$ encompasses both factors and their log-volatilities:

$$\boldsymbol{\alpha}_t = \begin{pmatrix} \boldsymbol{f}_t \\ \boldsymbol{h}_t \end{pmatrix} \in \mathbb{R}^{2K}. \tag{2.5}$$

This combines the latent factors and their corresponding log-volatilities into a single vector for state-space modeling.

2.1.1 Observation Likelihood Formulations

Given the observation equation (2.1), two distinct log-likelihood formulations are relevant, used differently by the BIF and PF.

First, the conditional log-likelihood of the observation $\boldsymbol{r}_t$ given a specific factor realization $\boldsymbol{f}_t$ depends solely on the idiosyncratic noise distribution:

$$\log p(\boldsymbol{r}_t \mid \boldsymbol{f}_t) = -\frac{1}{2}\left[N \log(2\pi) + \log|\boldsymbol{\Sigma}_\epsilon| + (\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_\epsilon^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)\right]. \tag{2.6}$$

The Particle Filter benchmark (Section 2.3) directly employs this standard conditional log-likelihood (2.6). This choice is inherent to the PF's mechanism, where importance weights for each particle $\boldsymbol{\alpha}_t^{(i)}$ are calculated based on how well that specific particle's state explains the observation $\boldsymbol{r}_t$, i.e., based on $p(\boldsymbol{r}_t|\boldsymbol{\alpha}_t^{(i)})$, which simplifies to $p(\boldsymbol{r}_t|\boldsymbol{f}_t^{(i)})$ given the model structure.

Second, the Bellman Information Filter (BIF) leverages a different calculation, denoted $\ell(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t)$, within its state update objective (Section 2.2.2) and its pseudo-likelihood for parameter estimation (Section 2.4.1). This term incorporates the covariance arising from both the idiosyncratic noise and the uncertainty about the factors $\boldsymbol{f}_t$, conditional on the volatility state $\boldsymbol{h}_t$:

$$\ell(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) = -\frac{1}{2}\left[N\log(2\pi) + \log|\boldsymbol{\Sigma}_t| + (\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_t^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)\right], \tag{2.7}$$

where the marginal observation covariance conditional on $\boldsymbol{h}_t$ is

$$\boldsymbol{\Sigma}_t = \text{Var}(\boldsymbol{r}_t \mid \boldsymbol{h}_t) = \boldsymbol{\Lambda}\text{Var}(\boldsymbol{f}_t \mid \boldsymbol{h}_t)\boldsymbol{\Lambda}' + \text{Var}(\boldsymbol{\epsilon}_t) = \boldsymbol{\Lambda}\,\text{diag}(e^{\boldsymbol{h}_t})\boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_\epsilon. \tag{2.8}$$

Using the marginal covariance $\boldsymbol{\Sigma}_t$ (2.8) within the likelihood term (2.7) is crucial for the BIF framework. It ensures that the optimization performed during the filter update correctly accounts for the information $\boldsymbol{r}_t$ provides about the full state $\boldsymbol{\alpha}_t = [\boldsymbol{f}_t', \boldsymbol{h}_t']'$, including the influence of the volatility state $\boldsymbol{h}_t$. Furthermore, it maintains consistency with the structure of the information matrix updates derived from BIF theory (Lange, 2024). Appendix A.2 provides detailed derivations for both likelihood formulations.

2.1.2 Model Assumptions

The specified DFSV model rests on several key assumptions:

- **Idiosyncratic Error Structure ($\boldsymbol{\Sigma}_\epsilon$):** Following standard practice (Aguilar & West, 2000) and to maintain tractability, I assume idiosyncratic errors are uncorrelated across assets ($\boldsymbol{\Sigma}_\epsilon$ is diagonal, Equation 2.1), relying on the latent factors to capture co-movement. The model structure intends for the latent factors and their dynamic, potentially correlated, stochastic volatilities (via $\boldsymbol{\Phi}_f$, $\boldsymbol{\Phi}_h$, and $\boldsymbol{h}_t$ influencing $\boldsymbol{\Sigma}_t$ in Equation 2.8) to capture the primary drivers of cross-asset correlation, leaving only asset-specific variation in $\boldsymbol{\epsilon}_t$. This assumption of constant, diagonal idiosyncratic variance simplifies estimation considerably. However, empirical asset returns often exhibit asset-specific time-varying volatility, a potential limitation of this specification that I revisit during the empirical evaluation (Chapter 5).

- **Factor Innovation Structure ($\boldsymbol{\nu}_{t+1}$):** The model assumes factor innovations $\boldsymbol{\nu}_{t+1}$ have a diagonal covariance matrix conditional on $\boldsymbol{h}_{t+1}$ (Equation 2.3). This interprets the latent factors $\boldsymbol{f}_t$ as distinct risk sources with uncorrelated contemporaneous shocks. Dynamic interactions and lagged cross-dependencies (between factors, and between factors and volatilities) are captured by the potentially non-diagonal autoregressive matrices $\boldsymbol{\Phi}_f$ and $\boldsymbol{\Phi}_h$.

- **Error Distributions:** I assume all error terms ($\boldsymbol{\epsilon}_t$, $\boldsymbol{\nu}_{t+1}$, $\boldsymbol{\eta}_{t+1}$) follow Gaussian distributions. While mathematically convenient, this simplifies reality, as financial returns often exhibit heavy tails and skewness. Extending the model to non-Gaussian errors (e.g., Student's t) is a potential area for future research.

- **Identification Constraint and Factor Ordering:** The lower-triangular constraint on $\boldsymbol{\Lambda}$ (Section 2.1.3) ensures model identification. This structure, however, imposes an inherent ordering on the factors. Consequently, I identify factors statistically rather than comparing them directly to pre-defined economic factors (e.g., Fama-French). The interpretation of these statistical factors and potential sensitivity to ordering are discussed further in the empirical analysis (Chapter 5).

These assumptions facilitate tractable DFSV estimation within the BIF framework while allowing the model considerable flexibility to capture complex financial time series dynamics.

2.1.3 Parameterization, Constraints, and Identification

To ensure model identifiability and stable estimation, I impose constraints on the factor loading matrix $\mathbf{\Lambda}$ and the autoregressive matrices $\mathbf{\Phi}_f, \mathbf{\Phi}_h$. Without constraints, the model suffers from rotational/scale indeterminacy ($\mathbf{\Lambda}$) and potential non-stationarity ($\mathbf{\Phi}_f, \mathbf{\Phi}_h$).

Lower-Triangular Constraint: I adopt a lower-triangular constraint with unit diagonal elements for the first K rows of $\mathbf{\Lambda}$:

$$\mathbf{\Lambda} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \lambda_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{K1} & \lambda_{K2} & \cdots & 1 \\ \lambda_{K+1,1} & \lambda_{K+1,2} & \cdots & \lambda_{K+1,K} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{N1} & \lambda_{N2} & \cdots & \lambda_{NK} \end{bmatrix} \tag{2.9}$$

This constraint ensures identification and aids interpretation by resolving factor indeterminacy and providing a clear role for each factor (e.g., the first factor affects all assets, the second affects all but the first, etc.). Such uniqueness is essential for consistent estimation.

Stability Constraints: For stationarity, the eigenvalues of both $\mathbf{\Phi}_f$ and $\mathbf{\Phi}_h$ must lie within the unit circle. Section 2.4 details how I enforce this during parameter estimation using transformations and penalties.

2.2 State Estimation via Bellman Information Filter (BIF)

I estimate the latent state $\boldsymbol{\alpha}_t$ using the Bellman Information Filter (BIF), adapting the methodology from Lange (2024). Although Lange (2024) derives both covariance (BF) and information (BIF) forms, I employ the BIF due to its enhanced numerical stability for the DFSV model, particularly with state-dependent volatility. The BIF propagates the information state (mode and precision matrix) rather than the traditional mean and covariance.

2.2.1 BIF Prediction Step

The BIF prediction step adapts the standard Kalman filter prediction equations to propagate the information state (mode and precision matrix) forward in time according to the model dynamics.

State Transition Matrix: The state transition matrix $\boldsymbol{T}$ for the DFSV model is block-diagonal, incorporating the factor dynamics and log-volatility dynamics:

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{\Phi}_f & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Phi}_h \end{bmatrix}. \tag{2.10}$$

Where $\boldsymbol{\Phi}_f$ governs the factor dynamics and $\boldsymbol{\Phi}_h$ governs the log-volatility dynamics as specified in Equations (2.3) and (2.4).

State Prediction: The predicted state mean $\hat{\boldsymbol{\alpha}}_{t|t-1}$ is calculated by applying the state transition matrix to the posterior state estimate from the previous time step. For the DFSV model, this involves separate predictions for the factors and log-volatilities:

$$\hat{\boldsymbol{f}}_{t|t-1} = \boldsymbol{\Phi}_f \, \hat{\boldsymbol{f}}_{t-1|t-1}, \tag{2.11}$$

$$\hat{\boldsymbol{h}}_{t|t-1} = \boldsymbol{\mu} + \boldsymbol{\Phi}_h \, (\hat{\boldsymbol{h}}_{t-1|t-1} - \boldsymbol{\mu}). \tag{2.12}$$

These are combined to form the complete predicted state vector:

$$\hat{\boldsymbol{\alpha}}_{t|t-1} = \begin{bmatrix} \hat{\boldsymbol{f}}_{t|t-1} \\ \hat{\boldsymbol{h}}_{t|t-1} \end{bmatrix}. \tag{2.13}$$

Innovation Covariance: The covariance matrix of state innovations, $\hat{\boldsymbol{Q}}_{t|t-1}$, is block-diagonal, incorporating the time-varying volatility of factor innovations and the constant volatility of log-volatility shocks:

$$\hat{\boldsymbol{Q}}_{t|t-1} = \begin{bmatrix} \operatorname{diag}\left(e^{\hat{h}_{1,t|t-1}}, \ldots, e^{\hat{h}_{K,t|t-1}}\right) & \mathbf{0} \\ \mathbf{0} & \boldsymbol{Q}_h \end{bmatrix}. \tag{2.14}$$

This structure reflects the model's assumption that factor innovations have stochastic volatilities determined by the current log-volatility states, while log-volatility innovations have constant covariance $\boldsymbol{Q}_h$.

Covariance Prediction: In the standard Kalman filter, the predicted covariance would be calculated as:

$$\hat{\boldsymbol{P}}_{t|t-1} = \boldsymbol{T} \, \hat{\boldsymbol{P}}_{t-1|t-1} \, \boldsymbol{T}' + \hat{\boldsymbol{Q}}_{t|t-1}. \tag{2.15}$$

However, the BIF operates in the information space, using precision matrices instead of covariance matrices for improved numerical stability.

Information Prediction (Joseph Form): The BIF propagates the precision matrix $\boldsymbol{\Omega}_{t|t-1} = \hat{\boldsymbol{P}}^{-1}_{t|t-1}$ instead of the covariance matrix. The prediction step for the information matrix employs the Joseph form for enhanced numerical stability (see Appendix A.1 for details).

2.2.2 BIF Update Step: Posterior Mode Optimization via Block Coordinate Descent

When a new observation $\boldsymbol{r}_t$ arrives, the BIF update step finds the posterior mode $\hat{\boldsymbol{\alpha}}_{t|t}$ by maximizing the log-posterior density $p(\boldsymbol{\alpha}_t|\boldsymbol{r}_t, \mathcal{F}_{t-1})$, where $\mathcal{F}_{t-1}$ denotes the information set up to time $t-1$. This log-posterior is proportional to the sum of the observation log-likelihood and the predicted state log-density:

$$\log p(\boldsymbol{\alpha}_t|\boldsymbol{r}_t, \mathcal{F}_{t-1}) \propto \ell(\boldsymbol{r}_t|\boldsymbol{\alpha}_t) + \log p(\boldsymbol{\alpha}_t|\mathcal{F}_{t-1}). \tag{2.16}$$

The BIF approximates the predicted state density $p(\boldsymbol{\alpha}_t|\mathcal{F}_{t-1})$ as Gaussian with mean $\hat{\boldsymbol{\alpha}}_{t|t-1}$ and precision $\boldsymbol{\Omega}_{t|t-1}$. Therefore, the update step maximizes the objective function:

$$J(\boldsymbol{\alpha}_t) = \ell(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) - \frac{1}{2}\left(\boldsymbol{\alpha}_t - \hat{\boldsymbol{\alpha}}_{t|t-1}\right)' \boldsymbol{\Omega}_{t|t-1}\left(\boldsymbol{\alpha}_t - \hat{\boldsymbol{\alpha}}_{t|t-1}\right). \tag{2.17}$$

The posterior mode is then:

$$\hat{\boldsymbol{\alpha}}_{t|t} = \arg\max_{\boldsymbol{\alpha}_t \in \mathbb{R}^{2K}} J(\boldsymbol{\alpha}_t). \tag{2.18}$$

This optimization combines the marginal observation log-likelihood (2.7) with a quadratic penalty incorporating the prior information from the prediction step. However, the non-linear dependence of the likelihood on $\boldsymbol{h}_t$ (through $\boldsymbol{\Sigma}_t$ in (2.8)) makes direct optimization over the full state $\boldsymbol{\alpha}_t$ challenging. To handle this efficiently, I employ a Block Coordinate Descent (BCD) algorithm, which iteratively optimizes blocks of the state vector ($\boldsymbol{f}_t$ and $\boldsymbol{h}_t$).

Block Coordinate Descent Algorithm: The BCD algorithm alternates between updating factors $\boldsymbol{f}_t$ (holding $\boldsymbol{h}_t$ fixed) and log-volatilities $\boldsymbol{h}_t$ (holding $\boldsymbol{f}_t$ fixed), as outlined in Algorithm 1. This leverages the problem's structure: the objective function is quadratic in $\boldsymbol{f}_t$ (allowing a closed-form update) but non-linear in $\boldsymbol{h}_t$ (requiring numerical optimization). Full derivations are in Appendices A.3.1 and A.3.2.

Algorithm 1 Block Coordinate Descent for BIF Update Step

Require: Factor loadings $\boldsymbol{\Lambda}$, idiosyncratic variances $\boldsymbol{\Sigma}_\epsilon$, predicted state $\hat{\boldsymbol{\alpha}}_{t|t-1}$, predicted information matrix $\boldsymbol{\Omega}_{t|t-1}$, observation $\boldsymbol{r}_t$, maximum iterations max_iters.

1: **Initialize** $\boldsymbol{f}^{(1)} \leftarrow \hat{\boldsymbol{f}}_{t|t-1}$, $\boldsymbol{h}^{(1)} \leftarrow \hat{\boldsymbol{h}}_{t|t-1}$
2: **for** $k = 1$ to max_iters **do**
3: *Update Factors:* Fix $\boldsymbol{h}^{(k)}$, compute $\boldsymbol{\Sigma}_t^{(k)} = \boldsymbol{\Lambda}\,\mathrm{diag}(e^{\boldsymbol{h}^{(k)}})\,\boldsymbol{\Lambda}^\top + \boldsymbol{\Sigma}_\epsilon$.
4: Solve linear system for $\boldsymbol{f}^{(k+1)}$:
5: $\boldsymbol{f}^{(k+1)} \leftarrow \left(\boldsymbol{\Lambda}^\top(\boldsymbol{\Sigma}_t^{(k)})^{-1}\boldsymbol{\Lambda} + \boldsymbol{\Omega}_{ff}\right)^{-1}\left(\boldsymbol{\Lambda}^\top(\boldsymbol{\Sigma}_t^{(k)})^{-1}\boldsymbol{r}_t + \boldsymbol{\Omega}_{ff}\hat{\boldsymbol{f}}_{t|t-1} + \boldsymbol{\Omega}_{fh}(\boldsymbol{h}^{(k)} - \hat{\boldsymbol{h}}_{t|t-1})\right)$ ▷ See Appendix A.3.1
6: *Update Log-Volatilities:* Fix $\boldsymbol{f}^{(k+1)}$.
7: Solve non-linear optimization for $\boldsymbol{h}^{(k+1)}$ using BFGS:
8: $\boldsymbol{h}^{(k+1)} \leftarrow \arg\max_{\boldsymbol{h}} J([\boldsymbol{f}^{(k+1)\prime}, \boldsymbol{h}']')$ ▷ See Appendix A.3.2
9: **end for**
10: **Return** $\hat{\boldsymbol{\alpha}}_{t|t} = [\boldsymbol{f}^{(\mathrm{final})\prime}, \boldsymbol{h}^{(\mathrm{final})\prime}]'$

Information Matrix Update: After finding the posterior mode $\hat{\boldsymbol{\alpha}}_{t|t}$, I update the information matrix using the Expected Fisher Information Matrix (E-FIM):

$$\boldsymbol{\Omega}_{t|t} = \boldsymbol{\Omega}_{t|t-1} + \boldsymbol{\mathcal{I}}_t, \tag{2.19}$$

where $\boldsymbol{\mathcal{I}}_t$ is the E-FIM evaluated at $\hat{\boldsymbol{\alpha}}_{t|t}$. The E-FIM exhibits a block-diagonal structure due to factors affecting only the mean and log-volatilities affecting only the covariance of the observation distribution:

$$\boldsymbol{\mathcal{I}}_t = \begin{bmatrix} \boldsymbol{\mathcal{I}}_{ff} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\mathcal{I}}_{hh} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Lambda} & \mathbf{0} \\ \mathbf{0} & \frac{1}{2}\left[e^{h_{i,t}+h_{j,t}}[\boldsymbol{\mathcal{I}}_{ff}]_{i,j}^2\right]_{i,j} \end{bmatrix}. \tag{2.20}$$

I provide a detailed derivation in Appendix A.4. Using the E-FIM instead of the Observed Fisher Information Matrix (O-FIM) guarantees positive semi-definiteness, enhancing numerical stability and avoiding the need for regularization techniques like eigenvalue clipping.

Computational Efficiency: The BCD algorithm provides significant computational advantages for high-dimensional DFSV models. The implementation leverages matrix identities (Appendix A.1), dimensionality reduction, and efficient optimization techniques to achieve stability and speed. Appendix A.3.3 discusses these computational aspects in detail.

2.3 Benchmark State Estimation via Particle Filter (PF)

To benchmark the BIF's performance, I also implement a Particle Filter (PF) for the DFSV model. The PF is a sequential Monte Carlo method that approximates the filtering distribution using a set of weighted particles.

2.3.1 Bootstrap Filter / SISR Implementation

Specifically, I use the Bootstrap Filter, a common Sequential Importance Sampling with Resampling (SISR) algorithm where the proposal distribution is simply the state transition density $p(\boldsymbol{\alpha}_t|\boldsymbol{\alpha}_{t-1})$. The standard Bootstrap filter was chosen as a widely recognized baseline for comparison due to its relative ease of implementation and established ability to handle complex dynamics in moderately dimensioned problems, despite known limitations such as potential particle impoverishment, which more advanced particle filters aim to mitigate. Algorithm 2 outlines the core steps.

Algorithm 2 Bootstrap Particle Filter for DFSV Model

Require: Model parameters $\boldsymbol{\Theta}$, observations $\{\boldsymbol{r}_t\}_{t=1}^T$, number of particles P, resampling threshold $\text{ESS}_{\text{threshold}}$

```
Initialize particles {α_0^(i)}_{i=1}^P and weights {w_0^(i) = 1/P}_{i=1}^P
for t = 1 to T do
    Predict: Propagate particles {α_{t-1}^(i)}_{i=1}^P using state transition equations (2.3) and (2.4)
  to obtain {α_t^(i)}_{i=1}^P.
    Update: Compute unnormalized importance weights using the conditional observation
  log-likelihood (Equation 2.6):
    w̃_t^(i) ← w_{t-1}^(i) · exp(log p(r_t|α_t^(i)))
    Normalize weights: w_t^(i) ← w̃_t^(i) / Σ_{j=1}^P w̃_t^(j)
    Calculate Effective Sample Size (ESS): ESS_t ← 1/ Σ_{i=1}^P (w_t^(i))^2
    if ESS_t < ESS_threshold then
        Resample particles {α_t^(i)}_{i=1}^P with weights {w_t^(i)}_{i=1}^P using systematic resampling.
        Reset weights: w_t^(i) ← 1/P for all i.
    end if
    Compute filtered state estimate: α̂_{t|t} ← Σ_{i=1}^P w_t^(i) α_t^(i)
end for
```

Systematic Resampling: To combat particle degeneracy (where a few particles dominate the weight distribution), I perform systematic resampling whenever the Effective Sample Size (ESS) falls below a threshold (e.g., $P/2$). Systematic resampling selects particles based on a single random draw mapped onto the cumulative weight distribution, generally offering better state space coverage than simpler multinomial resampling.

Likelihood Calculation for Parameter Optimization: The PF also yields an estimate of the model's likelihood, essential for parameter optimization via MLE. The log-likelihood is approximated by summing the log of the average importance weights at each time step:

$$\log p(\boldsymbol{r}_{1:T}|\boldsymbol{\Theta}) \approx \sum_{t=1}^{T} \log\left(\frac{1}{P}\sum_{i=1}^{P}\frac{\tilde{w}_t^{(i)}}{w_{t-1}^{(i)}}\right) = \sum_{t=1}^{T} \log\left(\frac{1}{P}\sum_{i=1}^{P} p(\boldsymbol{r}_t|\boldsymbol{\alpha}_t^{(i)})\right). \tag{2.21}$$

The conditional likelihood $p(\boldsymbol{r}_t|\boldsymbol{\alpha}_t^{(i)})$ is computed efficiently using Equation (2.6) and leveraging the diagonal structure of $\boldsymbol{\Sigma}_\epsilon$:

$$\log p(\boldsymbol{r}_t|\boldsymbol{\alpha}_t^{(i)}) = -\frac{1}{2}\left[N\log(2\pi) + \log|\boldsymbol{\Sigma}_\epsilon| + (\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t^{(i)})'\boldsymbol{\Sigma}_\epsilon^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t^{(i)})\right] \tag{2.22}$$

$$= -\frac{1}{2}\left[N\log(2\pi) + \sum_{j=1}^{N}\log(\sigma_j^2) + \sum_{j=1}^{N}\frac{(r_{j,t} - [\boldsymbol{\Lambda}\boldsymbol{f}_t^{(i)}]_j)^2}{\sigma_j^2}\right]. \tag{2.23}$$

This accumulated log-likelihood serves as the objective function for Maximum Likelihood Estimation (MLE) of the model parameters $\boldsymbol{\Theta}$.

2.4 Parameter Estimation

Beyond state estimation, the static parameters $\boldsymbol{\Theta} = (\boldsymbol{\Lambda}, \boldsymbol{\Phi}_f, \boldsymbol{\Phi}_h, \boldsymbol{\mu}, \boldsymbol{Q}_h, \boldsymbol{\Sigma}_\epsilon)$ must be estimated from the data. For the BIF, this involves maximizing an approximate pseudo-likelihood function derived from the filter outputs. For the PF, standard Maximum Likelihood Estimation (MLE) using the particle-based likelihood approximation (Equation 2.21) is used.

2.4.1 BIF Pseudo-Likelihood Maximization

Following Lange (2024), I estimate the parameters $\boldsymbol{\Theta}$ by maximizing the BIF's approximate pseudo-likelihood function:

$$\mathcal{L}(\boldsymbol{\Theta}) \approx \sum_{t=1}^{T} \left\{ \ell(\boldsymbol{r}_t \mid \hat{\boldsymbol{\alpha}}_{t|t}, \boldsymbol{\Theta}) - \frac{1}{2} \log \frac{\det(\hat{\boldsymbol{P}}_{t|t-1})}{\det(\hat{\boldsymbol{P}}_{t|t})} - \frac{1}{2} \left(\hat{\boldsymbol{\alpha}}_{t|t} - \hat{\boldsymbol{\alpha}}_{t|t-1}\right)' \hat{\boldsymbol{P}}_{t|t-1}^{-1} \left(\hat{\boldsymbol{\alpha}}_{t|t} - \hat{\boldsymbol{\alpha}}_{t|t-1}\right) \right\}. \tag{2.24}$$

This function combines the marginal observation likelihood $\ell(\cdot)$ (2.7) evaluated at the filtered state $\hat{\boldsymbol{\alpha}}_{t|t}$ with terms accounting for the information gain during the update step, represented by the predicted and filtered state estimates $(\hat{\boldsymbol{\alpha}}_{t|t-1}, \hat{\boldsymbol{\alpha}}_{t|t})$ and their associated covariance/information matrices. This pseudo-likelihood is exact for linear Gaussian models and provides a second-order approximation otherwise.

Rewriting Equation (2.24) using information matrices ($\boldsymbol{\Omega} = \boldsymbol{P}^{-1}$) yields the form used in the implementation:

$$\mathcal{L}(\boldsymbol{\Theta}) \approx \sum_{t=1}^{T} \left\{ \ell(\boldsymbol{r}_t \mid \hat{\boldsymbol{\alpha}}_{t|t}, \boldsymbol{\Theta}) - \frac{1}{2} \log \frac{\det(\boldsymbol{\Omega}_{t|t})}{\det(\boldsymbol{\Omega}_{t|t-1})} - \frac{1}{2} \left(\hat{\boldsymbol{\alpha}}_{t|t} - \hat{\boldsymbol{\alpha}}_{t|t-1}\right)' \boldsymbol{\Omega}_{t|t-1} \left(\hat{\boldsymbol{\alpha}}_{t|t} - \hat{\boldsymbol{\alpha}}_{t|t-1}\right) \right\}. \tag{2.25}$$

Parameter estimation maximizes this pseudo-likelihood: $\hat{\boldsymbol{\Theta}} = \arg\max_{\boldsymbol{\Theta}} \mathcal{L}(\boldsymbol{\Theta})$.

2.4.2 Parameter Transformations and Stationarity Constraints

To facilitate unconstrained optimization while respecting model constraints, I employ parameter transformations and stability enforcement techniques.[1]

Parameter Transformations:

- **Diagonal Tanh Transformation:** To constrain diagonal elements of $\boldsymbol{\Phi}_f$ and $\boldsymbol{\Phi}_h$ to $(-1, 1)$, I use the inverse hyperbolic tangent (tanh) transformation on the unconstrained parameters:

$$[\boldsymbol{\Phi}_f]_{ii} = \tanh([\boldsymbol{\Phi}_f]_{ii}^{\text{unc}}), \quad [\boldsymbol{\Phi}_h]_{ii} = \tanh([\boldsymbol{\Phi}_h]_{ii}^{\text{unc}}). \tag{2.26}$$

  This aids stability during optimization while permitting rich off-diagonal dynamics.

[1] The inverse mappings are $\operatorname{atanh}(x) = \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$ and $\operatorname{softplus}^{-1}(y) = \log(e^y - 1)$.

- **Variance Softplus Transformation:** To ensure positive variances (σ_i^2 in $\mathbf{\Sigma}_\epsilon$, diagonal elements of $\boldsymbol{Q}_h$), I apply the softplus function to the unconstrained parameters:

$$[\mathbf{\Sigma}_\epsilon]_{ii} = \text{softplus}([\mathbf{\Sigma}_\epsilon]_{ii}^{\text{unc}}), \quad [\boldsymbol{Q}_h]_{ii} = \text{softplus}([\boldsymbol{Q}_h]_{ii}^{\text{unc}}). \tag{2.27}$$

  Where $\text{softplus}(x) = \log(1 + e^x)$ maps $\mathbb{R} \to \mathbb{R}^+$.

- **Identification Constraint:** The lower-triangular structure of $\mathbf{\Lambda}$ (Section 2.1.3) is enforced directly during optimization to maintain identifiability.

Stability Enforcement via Eigenvalue Penalty: The tanh transformation alone does not guarantee stability for the full matrices $\mathbf{\Phi}_f$ and $\mathbf{\Phi}_h$. Therefore, I add a penalty term to the objective function that discourages eigenvalues from exceeding the unit circle:

$$\mathcal{L}_{\text{penalized}}(\mathbf{\Theta}) = \mathcal{L}(\mathbf{\Theta}) - \lambda \cdot \left(\sum_{j=1}^{K} \max(0, |\lambda_j(\mathbf{\Phi}_f)| - (1-\epsilon)) + \sum_{j=1}^{K} \max(0, |\lambda_j(\mathbf{\Phi}_h)| - (1-\epsilon)) \right). \tag{2.28}$$

Here, $\lambda_j(\cdot)$ are eigenvalues, λ is a penalty weight, and ϵ is a small buffer (10^{-6}). This penalty guides the optimization towards stationary solutions. I set $\lambda = 10^4$ throughout the simulations and empirical application (Chapters 4 and 5).

2.5 Computational Implementation Notes

The practical implementation leverages JAX (Bradbury et al., 2018) for automatic differentiation, JIT compilation, and vectorization, enabling efficient gradient-based optimization and execution speed. Key algorithmic choices enhance efficiency for high-dimensional settings ($N \gg K$):

- For the BIF, matrix identities (Appendix A.1) reduce the complexity of crucial update steps from $O(N^3)$ to $O(NK^2 + K^3)$.
- The BIF's Block Coordinate Descent update (Section 2.2.2) further improves efficiency by breaking the optimization into manageable subproblems.
- Both BIF and PF implementations utilize vectorized operations extensively.

Parameter estimation uses standard gradient-based optimizers, incorporating the transformations and stability penalties described in Section 2.4. All computations use 64-bit precision. This setup makes DFSV estimation feasible for the problem dimensions considered. The code is available on GitHub[2].

2.6 Benchmark Model Specifications

To assess the performance of the DFSV model estimated via BIF (henceforth DFSV-BIF), I compare it against three benchmark models. This section briefly outlines their methodologies; Chapter 5 provides specific configuration details used in the empirical study.

[2] Project repository: https://github.com/givani30/BellmanFilterDFSV

2.6.1 DFSV Model with Particle Filter (DFSV-PF)

The first benchmark uses the identical DFSV model specification (Section 2.1) but employs the Particle Filter (Section 2.3) for both state and parameter estimation (via MLE using Equation 2.21). This allows a direct comparison of the BIF and PF estimation approaches while holding the underlying dynamic model constant, isolating the impact of the chosen filtering and estimation technique.

2.6.2 Dynamic Conditional Correlation (DCC) GARCH

The Dynamic Conditional Correlation (DCC) GARCH model (Engle, 2002) represents a different paradigm for modeling time-varying covariance. Rather than relying on a factor structure, DCC-GARCH directly models the dynamics of the conditional correlations between assets after filtering out individual volatilities. The standard DCC(1,1) model involves two stages:

1. **Univariate GARCH:** Fit GARCH(1,1) models to each return series $r_{i,t}$ to estimate individual conditional variances $\sigma^2_{i,t}$:

$$\sigma^2_{i,t} = \omega_i + \alpha_i r^2_{i,t-1} + \beta_i \sigma^2_{i,t-1}. \tag{2.29}$$

2. **Dynamic Correlation:** Use standardized residuals $\boldsymbol{\varepsilon}_t$ from stage 1 to estimate a time-varying correlation matrix $\mathbf{R}_t$ via a proxy process $\mathbf{Q}_t$:

$$\mathbf{Q}_t = (1 - a - b)\bar{\mathbf{Q}} + a(\boldsymbol{\varepsilon}_{t-1}\boldsymbol{\varepsilon}'_{t-1}) + b\mathbf{Q}_{t-1} \tag{2.30}$$

$$\mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-1/2}\mathbf{Q}_t\text{diag}(\mathbf{Q}_t)^{-1/2}. \tag{2.31}$$

Where $\bar{\mathbf{Q}}$ is the unconditional correlation matrix, and a, b are scalar dynamics parameters.

The full conditional covariance is $\boldsymbol{\Sigma}_t = \text{diag}(\sigma_{1,t}, \ldots, \sigma_{N,t})\mathbf{R}_t\text{diag}(\sigma_{1,t}, \ldots, \sigma_{N,t})$. Estimation typically uses two-step MLE.

2.6.3 Dynamic Factor Model with Constant Volatility (DFM)

The third benchmark simplifies the DFSV model by removing the stochastic volatility components, resulting in a standard Dynamic Factor Model (DFM) with constant volatility. It retains the factor structure for returns but assumes constant covariance matrices for the innovations:

$$\mathbf{r}_t = \boldsymbol{\Lambda}\mathbf{f}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon) \tag{2.32}$$

$$\mathbf{f}_{t+1} = \boldsymbol{\Phi}_f\mathbf{f}_t + \boldsymbol{\nu}_{t+1}, \quad \boldsymbol{\nu}_{t+1} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\nu). \tag{2.33}$$

Here, $\boldsymbol{\Sigma}_\varepsilon$ (diagonal) and $\boldsymbol{\Sigma}_\nu$ (potentially non-diagonal) are constant. This linear Gaussian state-space model is estimable via the standard Kalman filter and MLE, often used for dimension reduction and forecasting (Bai & Ng, 2002).

2.7 Summary: Methodology and Contribution

This chapter detailed the DFSV model and the BIF estimation framework adapted for it, including BCD optimization and stability constraints. Parameter estimation via pseudo-likelihood was outlined, alongside benchmark models (PF, DCC, DFM) and computational considerations.

The complete estimation pipeline involves preprocessing data, initializing parameters and states, iteratively running the chosen filtering algorithm (BIF or PF) to estimate latent states, computing the corresponding (pseudo-)likelihood, and optimizing this objective function to estimate static parameters. Multiple optimization runs with different initializations help ensure convergence towards a global optimum. While this framework offers a computationally feasible approach for high-dimensional DFSV estimation, its empirical adequacy, particularly concerning the assumption of constant idiosyncratic variance (Aguilar & West, 2000), is rigorously tested in Chapter 5.

Addressing Research Question 1, this chapter demonstrated how the BIF can be customized for high-dimensional DFSV models. Key adaptations include the Block Coordinate Descent (BCD) algorithm for the non-linear update step, integration of VAR dynamics for both factors and log-volatilities, and the use of the Expected Fisher Information Matrix (E-FIM) for stable information matrix updates. These innovations, combined with efficient computational techniques (Section 2.5 and appendices), yield an algorithm with computational complexity scaling favorably ($O(NK^2 + K^3)$), making it practical for applications where $N \gg K$. The subsequent chapters evaluate the empirical performance and utility of this framework using simulated (Chapter 4) and real financial data (Chapter 5).

Chapter 3

Data, Evaluation, and Simulation Design

This chapter establishes the foundation for evaluating the proposed DFSV-BIF model. It introduces the empirical dataset, defines benchmark models and evaluation metrics for real-world comparison, and details simulation studies designed to test the filter's properties under controlled conditions.

3.1 Empirical Data Description

This section describes the Fama-French 100 portfolio dataset used for empirically testing the DFSV-BIF framework. The analysis centers on the monthly value-weighted returns of these portfolios, serving as the primary input.

3.1.1 Data Sources, Selection, and Relevance

This study utilizes the value-weighted monthly returns of the '100 Portfolios Formed on Size and Book-to-Market' from the Kenneth R. French Data Library[1], a standard benchmark in asset pricing Fama and French, 1993. The data span July 1963 to December 2023 ($T = 726$ observations). Its high dimensionality (initially $N = 100$) offers a suitable testbed for factor models and facilitates evaluating the BIF's scalability. This sample period aligns with standard practice Fama and French, 1993, ensuring data reliability. Appendix B.2 provides further details on portfolio characteristics.

3.1.2 Preprocessing and Exploratory Data Analysis (EDA)

This section summarizes the exploratory data analysis; Appendix B offers more detailed EDA results, including additional statistics and figures.

Preprocessing involved several steps. First, handling missing values (-99.99 converted to NaN) necessitated excluding five portfolios ('BIG HiBM', 'ME10 BM8', 'ME10 BM9', 'ME7 BM10', 'ME9 BM10') to ensure a balanced panel, resulting in $N = 95$ portfolios. Second,

[1] Available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

percentage returns were converted to decimals. Third, each series was demeaned using its full-sample arithmetic mean. Standard tests confirmed stationarity for the resulting return series, fulfilling a necessary condition for the modeling framework. The final input data for the DFSV model consist of these $N = 95$ demeaned, decimal monthly value-weighted returns, denoted $\boldsymbol{r}'_t$. Appendix B.1 contains detailed statistics and plots.

These demeaned return series ($\boldsymbol{r}'_t$) exhibit characteristics typical of financial data: means are effectively zero, standard deviations vary across portfolios, and many series display negative skewness and significant excess kurtosis (leptokurtosis). These stylized facts motivate using models that incorporate stochastic volatility (SV) and persistent dynamics, such as the DFSV framework. Appendix B.1, Table B.1, provides detailed summary statistics.

Figure 3.1: Correlation Heatmap of 95 Demeaned Portfolio Returns ($\boldsymbol{r}'_t$)

. Five portfolios ('BIG HiBM', 'ME10 BM8', 'ME10 BM9', 'ME7 BM10', 'ME9 BM10') are excluded due to missing values.

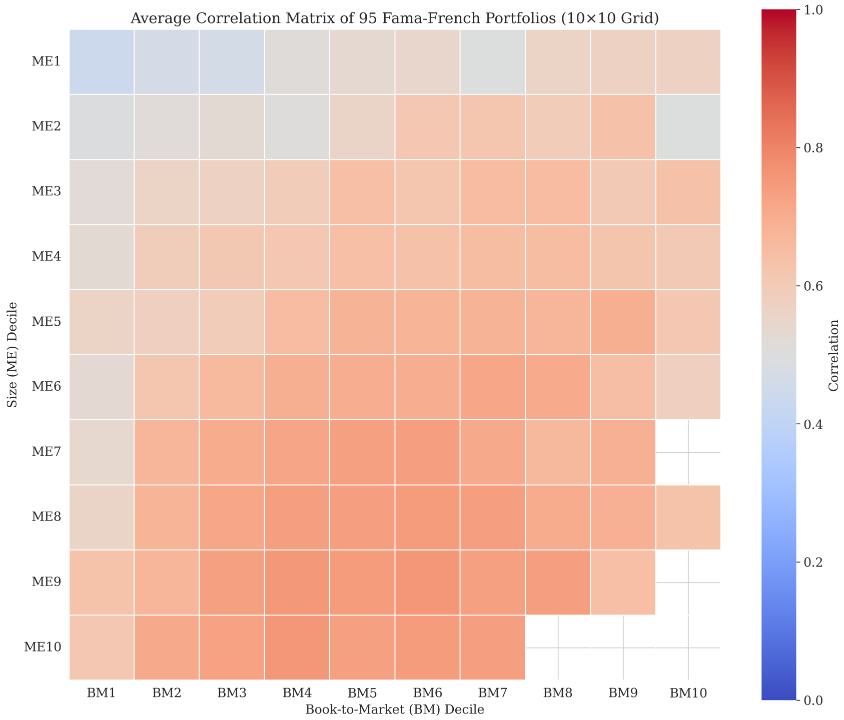


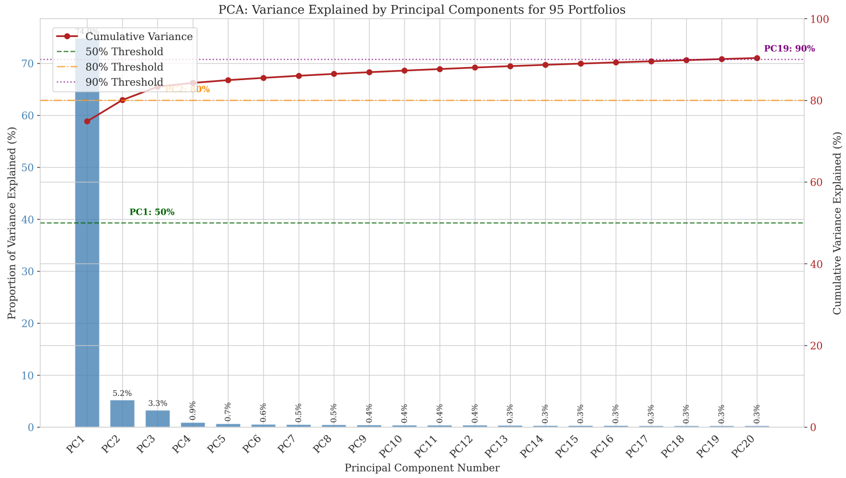
Figure 3.1 shows the sample correlation matrix, revealing strong positive correlations across most portfolio pairs. This observation supports using a factor model to capture common variation efficiently.

3.1.3 Principal Component Analysis (PCA) & Factor Selection (K)

Principal Component Analysis (PCA) on the standardized demeaned returns ($\boldsymbol{r}'_t$) helps assess the dimensionality of common variation within the $N = 95$ portfolio returns. Standardization ensures that portfolios with higher variance do not disproportionately influence the principal components.

The PCA scree plot (Figure 3.2) illustrates the variance explained by successive principal components. The first three components capture approximately 83% of the total variance, while

Figure 3.2: PCA Scree Plot: Variance Explained by Principal Components



the first five explain around 85%, indicating diminishing explanatory power beyond the initial few components.

Although Principal Component Analysis suggests that $K = 3$ factors capture the majority of common variation (approximately 83%, see Figure 3.2), we select $K = 5$ factors for the empirical application. This choice serves multiple purposes. First, it facilitates a more comprehensive evaluation of the Bellman Information Filter's (BIF) performance within a higher-dimensional latent state space ($2K = 10$). Second, while the factors in this study are latent and estimated via the DFSV model, a five-factor structure offers a conceptual parallel to established asset pricing frameworks, such as the Fama-French five-factor model Fama and French, 2015. Finally, preliminary assessments confirmed that estimating the DFSV model with $N = 95$ assets and $K = 5$ factors using the BIF remains computationally feasible given the project's resources.

3.2 Benchmark Models

To evaluate the proposed DFSV-BIF model, its performance is compared against three benchmark models. Section 2.6 provides methodological details; the specific configurations used for the empirical application ($N = 95$ assets, $K = 5$ factors where applicable) are:

- **DFSV-PF:** Employs the same DFSV specification as the primary model but is estimated using a Particle Filter (PF) with $P = 10,000$ particles (see Sections 2.3 and 3.4). This setup permits a direct comparison of BIF and PF estimation for the identical model structure.

- **DCC-GARCH:** Uses a standard DCC(1,1) specification with GARCH(1,1) for each return series ($\boldsymbol{r}_t'$). Estimation uses Maximum Likelihood via the 'mvgarch' python library (see Section 2.6.2). This benchmark contrasts the factor approach with direct covariance modeling.

- **DFM:** Represents a Dynamic Factor Model with constant volatility and $K = 5$ factors. Estimation uses the Kalman Filter and Maximum Likelihood via the 'statsmodels' python library (see Section 2.6.3). This model helps isolate the impact of incorporating stochastic volatility in the DFSV models.

3.3 Evaluation Framework

With the models, data, and benchmarks established (Chapter 2, Sections 3.1, 3.2), a comprehensive evaluation framework is necessary to assess the DFSV-BIF model's effectiveness in explaining and forecasting high-dimensional returns relative to its competitors over the full sample period. This assessment rests on six pillars: statistical fit, covariance dynamics, factor relevance, residual behavior, parameter/state plausibility, and computational efficiency. The subsequent subsections detail the specific metrics within each pillar, forming the basis for the empirical application presented in Chapter 5.

3.3.1 Scope and Estimation Window

The evaluation utilizes all $T = 726$ monthly observations from July 1963 to December 2023 (Section 3.1). Compared models include the DFSV-BIF (Bellman filter), DFSV-PF (particle filter), DFM (Dynamic Factor Model), and DCC-GARCH(1,1). All criteria are calculated in-sample to maximize information for parameter estimation; out-of-sample evaluation remains a topic for future research. Section 2.7 contains the model equations and estimation details.

3.3.2 Overview of Evaluation Pillars

Table 3.1 outlines the evaluation framework, detailing the key areas, metrics, and their purpose.

Table 3.1: Evaluation pillars and primary metrics

Pillar	Metric(s)	Purpose
Statistical fit	(Pseudo) LL, AIC, BIC	Assess overall goodness-of-fit, penalizing complexity
Covariance dynamics	Log GV ($\log \lvert\hat{\boldsymbol{\Sigma}}_t\rvert$), Avg. $\hat{\rho}_t$	Capture systemic variance and market co-movement dynamics
Residual behavior	Jarque–Bera, LB-Q^2 (lags 5,10,15,20), ARCH-LM (lags 5, 10)	Detect remaining non-normality, serial correlation, and ARCH effects
Parameter & state plausibility	Stationarity eigenvalues, economic ranges, heatmaps	Perform sanity checks on estimated parameters ($\hat{\boldsymbol{\Lambda}}$, $\hat{\boldsymbol{\Phi}}_f$, $\hat{\boldsymbol{\Phi}}_h$, $\hat{\boldsymbol{\mu}}$) and states ($\hat{\boldsymbol{f}}_t, \hat{\boldsymbol{h}}_t$)
Computational efficiency	Wall-clock minutes, optimizer iterations, convergence flag	Evaluate practical feasibility and scalability

Note: This table outlines the six key areas for model evaluation, the primary metrics associated with each, their purpose, and the section in Chapter 5 where results are presented. LL denotes Log-Likelihood, PLL denotes Pseudo-Log-Likelihood. LB-Q^2 refers to the Ljung-Box test on squared residuals.

These pillars collectively address RQ-1 (Can the model capture time-varying risk?) and RQ-2 (Is the BIF estimator computationally attractive?), forming the evaluation basis for Chapter 5.

3.3.3 Statistical-Fit Criteria

Statistical fit employs the true Log-Likelihood (LL) for DFM and DCC-GARCH, and the Pseudo-Log-Likelihood (PLL) via Laplace approximation for DFSV-BIF and DFSV-PF. Information criteria (AIC, BIC) are calculated based on these likelihoods (p parameters, $T = 726$). Direct comparison across likelihood types (LL vs. PLL) is invalid; AIC/BIC comparisons hold only within the same likelihood category.

3.3.4 Dynamic-Covariance Metrics

Dynamic covariance properties are evaluated using the model-implied conditional covariance matrix. For model $m \in \{\text{DFSV-BIF}, \text{DFSV-PF}, \text{DCC}, \text{DFM}\}$, this matrix is defined as:

$$\widehat{\mathbf{\Sigma}}_t^{(m)} := \text{Var}_m(\boldsymbol{r}_t \mid \mathcal{F}_{t-1}; \widehat{\mathbf{\Theta}}^{(m)}, \widehat{\boldsymbol{\alpha}}_{t|t}^{(m)}). \tag{3.1}$$

This matrix incorporates the final parameter estimates $\widehat{\mathbf{\Theta}}^{(m)}$ and, where applicable, the filtered states $\widehat{\boldsymbol{\alpha}}_{t|t}^{(m)}$. Key metrics derived from $\widehat{\mathbf{\Sigma}}_t^{(m)}$ include the Log Generalized Variance ($\log|\hat{\mathbf{\Sigma}}_t|$), summarizing total conditional variance, and the average conditional correlation ($\bar{\rho}_t$, the mean off-diagonal of the corresponding correlation matrix), signaling market co-movement. Time-series plots of these metrics allow visual inspection of covariance dynamics.

3.3.5 Residual Diagnostics

Standardized residuals $\hat{\boldsymbol{z}}_t = \hat{\mathbf{\Sigma}}_{t|t-1}^{-1/2}(\boldsymbol{r}'_t - \hat{\boldsymbol{r}}'_{t|t-1})$ are examined via pass rates (at the 5% significance level across the 95 series) for the Jarque-Bera test (normality), the Ljung-Box Q^2 test (lags 5, 10, 15, 20; serial correlation in squared residuals), and the ARCH-LM test (lags 5, 10; remaining ARCH effects).

3.3.6 Parameter and State Plausibility Checks

Plausibility checks include: verifying stationarity conditions ($|\lambda_{\max}(\hat{\mathbf{\Phi}}_f)| < 1$, $|\lambda_{\max}(\hat{\mathbf{\Phi}}_h)| < 1$); comparing estimated unconditional means ($\hat{\boldsymbol{\mu}}$) and idiosyncratic variances ($\hat{\mathbf{\Sigma}}_\epsilon$) against typical economic ranges; and visually inspecting heatmaps ($\hat{\mathbf{\Lambda}}$, $\hat{\mathbf{\Phi}}_f$, $\hat{\mathbf{\Phi}}_h$) and plots of smoothed states ($\hat{\boldsymbol{f}}_t, \hat{\boldsymbol{h}}_t$).

3.3.7 Computational Cost and Stability

Assessing computational cost involves recording wall-clock time, optimizer iterations, and convergence status for each estimation run conducted on identical hardware[2]. Lower computation time for comparable model fit indicates superior scalability and practical feasibility.

Chapter 5 utilizes these six pillars to evaluate whether the Bellman-filtered DFSV model provides a credible and efficient representation of dynamic market risk.

[2]Computations performed on a local machine with an Intel Ultra 7 155H processor and 16GB RAM.

3.4 Simulation Study Design

3.4.1 Motivation & Overall Goals

To complement the empirical analysis, this section details two simulation studies designed to systematically evaluate the performance of the Bellman Information Filter (BIF) and Particle Filter (PF) for DFSV models under controlled conditions.

- **Study 1 (Computational Scaling):** This study examines how the computational efficiency (time) and state estimation accuracy (RMSE, correlation) of BIF and PF scale with increasing model dimensions (N, K). The goal is assessing the practical feasibility of these methods for high-dimensional applications.

- **Study 2 (Parameter Estimation):** This study evaluates how well BIF and PF recover true model parameters $(\boldsymbol{\Theta})$ via Maximum Likelihood Estimation (MLE), addressing the critical challenge of parameter estimation in real-world scenarios where true parameters are unknown.

Together, these studies provide a comprehensive assessment of the filters' relative strengths and limitations across various dimensions and use cases, informing empirical filter selection and guiding future development. All simulations use identical hardware[3] to ensure reproducibility.

3.4.2 General Data Generating Process (DGP)

Both studies simulate synthetic data from the DFSV model detailed in Chapter 2 (visualized in Figure 3.3). For each replication, model parameters $(\boldsymbol{\Lambda}, \boldsymbol{\Phi}_f, \boldsymbol{\Phi}_h, \boldsymbol{\mu}, \sigma^2_\epsilon, \boldsymbol{Q}_h)$ are generated randomly, ensuring model stability (e.g., stationarity via eigenvalue constraints) and identifiability. Specific details on parameter generation and configurations for each study appear in their respective sections below and in Appendices C.1 and C.2.

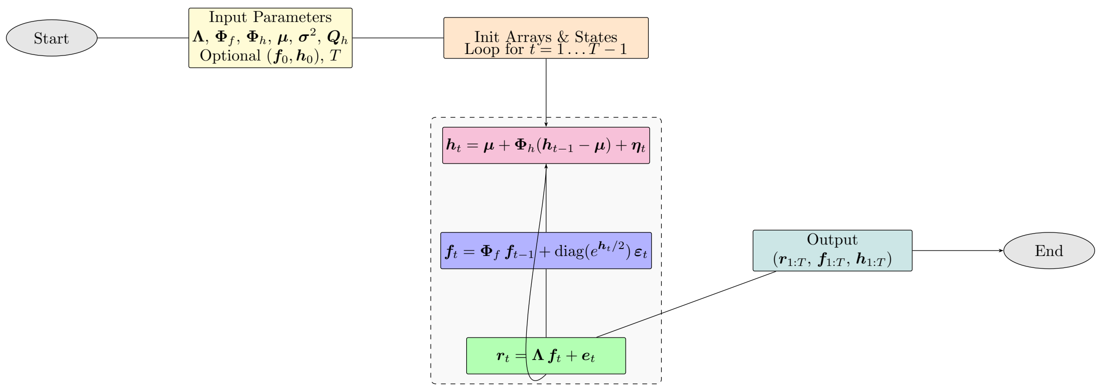


Figure 3.3: Data-generating process for the Dynamic Factor Stochastic Volatility model.

[3]Google Cloud e2-standard-4 instances (4 vCPUs, 16 GB RAM) ensure fair comparisons and allow parallel execution via Google Cloud Batch.

3.4.3 Simulation Study 1: Computational Scaling Analysis

Study 1 evaluates BIF and PF computational efficiency and scalability, addressing three key questions: (1) How does filter computation timescale with dimensions (N, K)? (2) How does BIF performance compare to PF with various particle counts (P)? (3) What are the practical dimensional limits of each filter?

Data Generating Process (DGP)

Synthetic data generation follows the general DGP (Section 3.4.2) with a time series length of $T = 1500$. The analysis varies model dimensions across assets $N \in \{5, 10, 20, 50, 100, 150\}$ and factors $K \in \{2, 3, 5, 10, 15\}$, subject to the constraint $K < N$. For each (N, K) configuration, 100 replications were generated using unique random seeds. Parameters were generated randomly, ensuring stability and identifiability (details appear in Appendix C.1, Table C.1).

Filter Configurations

The analysis compares the BIF (Section 2.2.2) against the bootstrap PF (Section 2.3) using four particle counts: $P \in \{1{,}000, 5{,}000, 10{,}000, 20{,}000\}$. For a controlled comparison focused purely on filtering (not estimation), both filters were initialized with the true initial states $\boldsymbol{\alpha}_1 = [\boldsymbol{f}_1', \boldsymbol{h}_1']'$ and the true model parameters $\boldsymbol{\Theta}$. The BIF employs a diagonal initial information matrix (precision 1.0 for factors, 0.1 for log-volatilities), while the PF draws initial particles from a Gaussian centered at $\boldsymbol{\alpha}_1$. Both implementations leverage JAX for computational efficiency.

Performance Evaluation Metrics

Performance assessment relies on two categories of metrics:

- **Computational Efficiency:** Total wall-clock time required for filtering the $T = 1500$ observations; the analysis examines how computation time scales with N, K, and P.
- **State Estimation Accuracy:** Root Mean Squared Error (RMSE) and Pearson Correlation is calculated between the true simulated states $(\boldsymbol{f}_t, \boldsymbol{h}_t)$ and their filtered estimates $(\hat{\boldsymbol{f}}_{t|t}, \hat{\boldsymbol{h}}_{t|t})$ at each time step.

Results (mean, median, standard error across 100 replications per configuration) are presented using log-scale plots for timing and comparative tables/heatmaps for accuracy metrics to illustrate scaling behavior and trade-offs.

3.4.4 Simulation Study 2: Parameter Estimation Performance

Study 2 evaluates the ability of BIF-based and PF-based Maximum Likelihood Estimation (MLE) to recover true DFSV parameters $(\boldsymbol{\Theta})$ from simulated data. It also assesses the impact of using estimated parameters $(\hat{\boldsymbol{\Theta}})$ on state filtering accuracy.

Data Generating Process (DGP)

Synthetic data generation follows the general DGP (Section 3.4.2) but uses configurations relevant to estimation challenges. Time series were simulated for lengths $T \in \{500, 1000, 2000\}$ across nine combinations of assets $N \in \{5, 10, 15\}$ and factors $K \in \{2, 3, 5\}$, plus a larger case $(N = 50, K = 5)$. These configurations cover various asset-to-factor ratios. For each (N, K, T) setting, 5 replications were generated using different random seeds. Unlike Study 1, which used more general random parameter draws, Study 2 employs specific distributions and imposes identification constraints (e.g., lower-triangular $\boldsymbol{\Lambda}$ with unit diagonal, diagonal $\boldsymbol{Q}_h$) necessary for parameter estimation. This process yields realistic simulated series (full details are in Appendix C.2, Table C.4).

Filter and Estimation Configurations

Maximum Likelihood Estimation (MLE) utilizes two approaches:

- BIF-based likelihood approximation, optimized using DampedTrustRegionBFGS. This custom algorithm combines a damped Newton descent strategy with a trust region mechanism to robustly handle the non-convex pseudo-likelihood surface (see Appendix C.3).
- PF-based likelihood approximation, optimized using ArmijoBFGS. This custom algorithm combines BFGS with an Armijo line search strategy suitable for the potentially noisy likelihood estimates from the particle filter (see Appendix C.3).

The PF approach was tested with $P \in \{1000, 5000\}$ particles (PF-1k, PF-5k). The BIF variant estimates all parameters, including the log-volatility mean $\boldsymbol{\mu}$. Optimizations were initialized with plausible parameter values, distinct from the true values, simulating a realistic estimation scenario. Appendix C.3 provides further details on optimizers, parameter transformations, and constraints.

Evaluation Themes and Metrics

Estimation performance evaluation covers five key themes:

1. **Parameter Accuracy:** Assessment of how closely estimated parameters ($\hat{\boldsymbol{\Theta}}$) match true parameters ($\boldsymbol{\Theta}$), measured by Root Mean Squared Error (RMSE) and bias for each parameter type ($\boldsymbol{\Lambda}, \boldsymbol{\Phi}_f, \boldsymbol{\Phi}_h, \boldsymbol{\mu}, \sigma_\epsilon^2, \boldsymbol{Q}_h$).
2. **State Accuracy (Post-Estimation):** Evaluation of the quality of filtered states ($\hat{\boldsymbol{f}}_{t|t}, \hat{\boldsymbol{h}}_{t|t}$) obtained using the *estimated* parameters $\hat{\boldsymbol{\Theta}}$, measured by RMSE and correlation against true states.
3. **Bias and Identification Difficulty:** Investigation of systematic deviations (bias) of $\hat{\boldsymbol{\Theta}}$ from $\boldsymbol{\Theta}$ and identification of parameters that are consistently harder to estimate accurately.
4. **Computational Cost vs. Accuracy Trade-offs:** Comparison of estimation efficiency (wall-clock time, iterations) against the resulting parameter and state accuracy.

5. **Scaling:** Examination of how estimation performance (accuracy, bias, cost) varies with model dimensions (N, K) and time series length (T).

Results are reported as means and standard errors across the 5 replications for each configuration, using comparative tables and plots for illustration.

Chapter 4

Simulation Studies: Evaluating the Bellman Information Filter

This chapter evaluates the Bellman Information Filter (BIF) through two distinct simulation studies. The first study focuses on computational performance and scalability, comparing the BIF with the Particle Filter (PF) across different model dimensions. The second study assesses the accuracy of state-estimation and parameter-recovery, providing insights into the filter's ability to recover the true model parameters and latent states. The chapter concludes with a synthesis of findings, discussing the trade-offs between computational efficiency and estimation accuracy.

4.1 Simulation Study 1: Computational Performance and Scalability

This section compares the Bellman Information Filter's (BIF) tractability against a standard Particle Filter (PF) benchmark as model dimensionality grows. I generated synthetic data from the DFSV model using stable parameters detailed in Appendix C.1 (Table C.1). The simulation covers a time-series length of $T = 1500$ and dimensions $N \in \{5, \ldots, 150\}$, $K \in \{2, \ldots, 15\}$ ($K < N$), running 100 replications per configuration. The PF uses varying particle counts ($P \in \{1{,}000, \ldots, 20{,}000\}$). I performed computations using an Intel Ultra 7 155H processor with 16GB RAM and evaluated the median computation time and median Root Mean Squared Error (RMSE) for the latent states.

4.1.1 Summary Results

Table 4.1 details the median runtime and state-estimation accuracy for representative small (N=50, K=5) and large (N=150, K=15) models, comparing the BIF to the PF-20k (20,000 particles). The BIF's speed advantage in the smaller dimension indicates that PF particle overhead can dominate the BIF's complexity in K. However, the PF-20k runs faster for the larger dimension, revealing the BIF's cubic scaling in K. Despite this scaling difference, the BIF maintains competitive factor accuracy and achieves better log-volatility accuracy than the PF-20k in the large model. The BIF's higher factor RMSE but lower log-volatility RMSE in the smaller dimension reflects different approximations: the BIF's Gaussian assumption may capture linear factor dynamics less accurately than PF particles in this case, while its structured approach proves more effective for the non-linear volatility states.

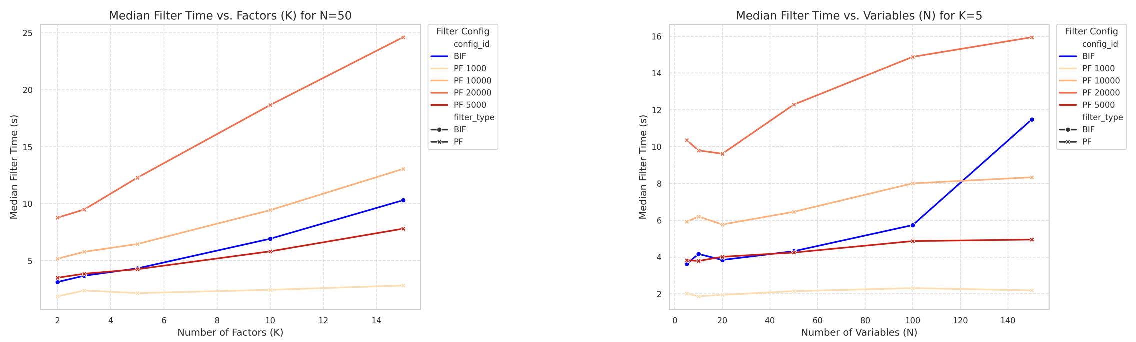
Table 4.1: Summary of Median Runtime and State-Estimation RMSE.

Filter	N	K	Median Time (s)	Median RMSE $\boldsymbol{f}_t$	Median RMSE $\boldsymbol{h}_t$
BIF	50	5	4.320	0.450	0.805
PF-20k	50	5	5.350	0.198	0.913
BIF	150	15	31.300	1.137	1.360
PF-20k	150	15	11.820	1.114	1.936

PF-20k refers to the Particle Filter with 20,000 particles. Median RMSE and time values are extracted from detailed simulation results across 100 replications.

4.1.2 Timing Results vs. Dimensionality (N, K)

Figure 4.1a plots how median computation time scales with the number of factors K (for a fixed N=50). The BIF's runtime increases more steeply than the PF variants, reflecting its higher theoretical complexity in K. PF runtimes scale more gently with K but increase substantially with the particle count. Figure 4.1b plots scaling with the number of series N (for a fixed K=5); both filters scale approximately linearly with N. However, the BIF shows a steeper slope, particularly for $N > 100$. These plots confirm the computational trade-offs between the filters.



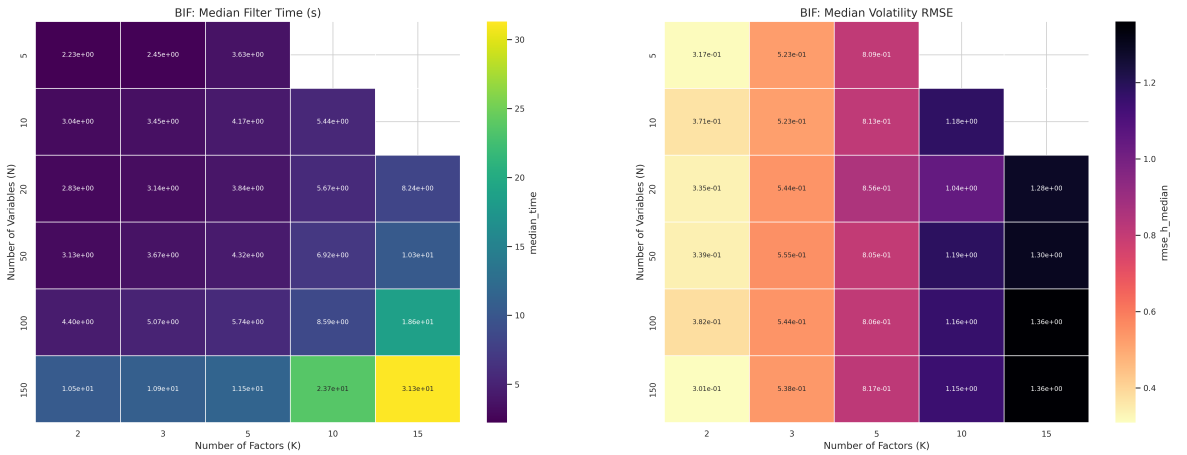
(a) Runtime scaling with factors K (N=50) **(b)** Runtime scaling with series N (K=5)

Figure 4.1: Computational scaling comparison. Left: BIF runtime scales faster with K than PF runtimes, which grow linearly with particle count. Right: Both filters scale approximately linearly with N, with BIF showing steeper growth at higher dimensions.

4.1.3 Analysis of Scalability

Scalability defines how computational requirements grow with problem size. Figure 4.1a shows the BIF's runtime growing roughly cubically with K, consistent with its $O(K^3)$ complexity. Conversely, Figure 4.1b confirms a near-linear dependence on N. In contrast, PF runtime scales approximately linearly with both the particle count P and the series count N.

Figure 4.2 plots the BIF median runtime and log-volatility RMSE across the $N \times K$ grid. The heatmaps reveal the BIF's runtime sensitivity to K (Figure 4.2a, where darker cells indicate longer runtimes) and its relatively stable log-volatility RMSE across dimensions (Figure 4.2b, where darker cells mean higher error). The BIF's deterministic nature offers advantages for reliability. Its primary computational bottleneck involves matrix operations related to the $O(K^2)$ state covariance $\hat{\boldsymbol{P}}_t$. In contrast, the PF bottleneck typically lies in resampling and likelihood evaluation, scaling with P and N.



(a) BIF median runtime increases with N and K, with steeper growth along K axis

(b) BIF median log-volatility RMSE remains stable across N and K combinations

Figure 4.2: Heatmap visualization of BIF performance metrics. Darker cells in (a) indicate longer runtimes (seconds), consistent with theoretical complexity. Darker cells in (b) indicate higher RMSE (worse accuracy) for log-volatility estimation.

4.1.4 State Estimation Accuracy

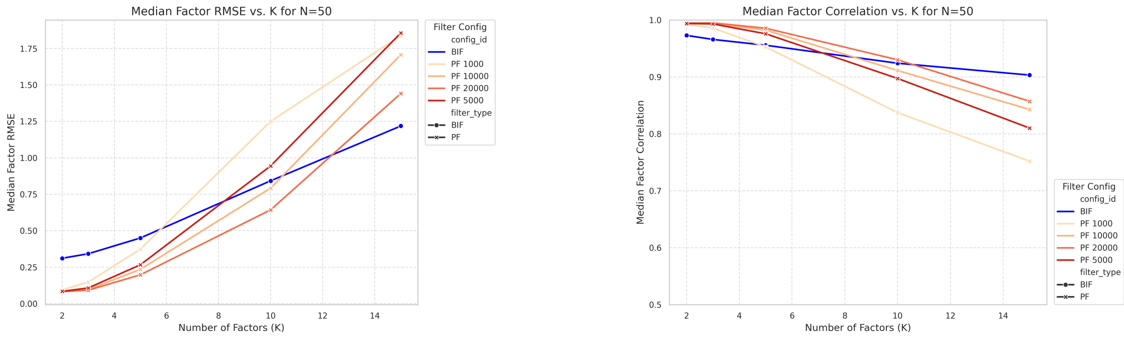
I assess the accuracy of recovering the true latent states using median RMSE and correlation as metrics.

Factor State Estimation

Figure 4.3 plots the median RMSE and correlation for factor estimation as K increases (N=50). PF variants with sufficient particles generally achieve lower factor RMSE than the BIF, particularly for smaller K. However, this advantage diminishes as K increases (Figure 4.3a). A similar pattern holds for correlation (Figure 4.3b). While the PF can yield marginally better factor estimates, the BIF's performance becomes increasingly comparable as model complexity grows.

Log-Volatility State Estimation

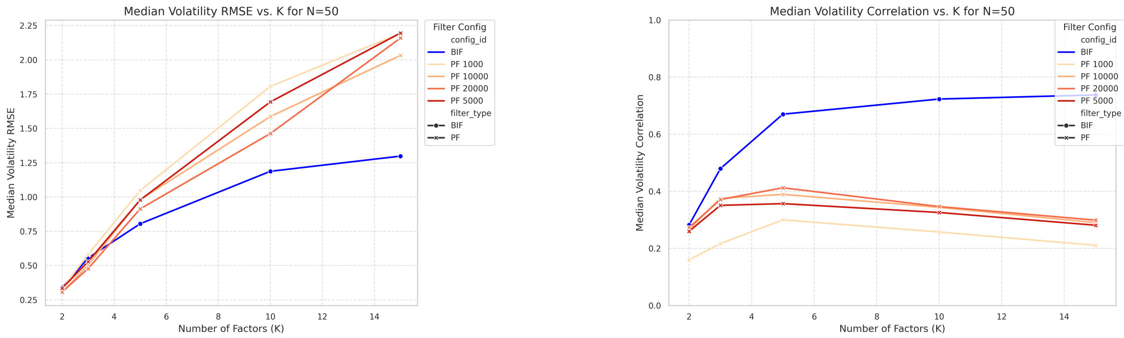
Figure 4.4 plots the corresponding metrics for log-volatility estimation. The BIF consistently achieves lower log-volatility RMSE compared to the PF variants, and this advantage widens as K increases. At $K = 15$, the BIF reduces log-volatility RMSE by more than 30% compared to the best PF variant (Figure 4.4a). The BIF also maintains higher correlations with the true log-volatilities, especially for larger K (Figure 4.4b). These findings confirm the BIF's superior ability to estimate latent log-volatility states, a key advantage in financial applications. The



(a) Factor RMSE vs. K (N=50) **(b)** Factor correlation vs. K (N=50)

Figure 4.3: Factor state-estimation accuracy. (a) PF variants show lower median factor RMSE than BIF at low K, but the gap narrows as K increases. (b) PF variants achieve slightly higher median factor correlation than BIF across K, though the difference is marginal for K=15.

combined results reveal a trade-off: the PF (with high P) offers slightly better factor estimation in lower dimensions, while the BIF provides superior log-volatility estimation, especially in higher dimensions.



(a) Log-volatility RMSE vs. K (N=50) **(b)** Log-volatility correlation vs. K (N=50)

Figure 4.4: Log-volatility state-estimation accuracy. (a) BIF consistently achieves lower median log-volatility RMSE compared to PF variants, with the advantage widening as K increases. (b) BIF maintains higher median log-volatility correlation than PF variants, particularly for larger K.

4.1.5 Discussion

Simulation Study 1 confirms trade-offs between the BIF and PF concerning computational cost, accuracy, and robustness. The BIF confirms superior log-volatility estimation and offers deterministic runtime with predictable scaling (cubic in K, linear in N), making it attractive for reliability. PF runtime scales linearly with N and particle count P. However, achieving sufficient accuracy often requires a large P, potentially negating its scaling advantage in some regimes (Table 4.1, N=50, K=5). A significant PF limitation is its susceptibility to instability, particularly with insufficient particles or in high dimensions Rebeschini and Handel, 2015. This often requires large P and assessment via median metrics (see Appendix C.1.1 for stability details). In contrast, the BIF's deterministic nature avoids the risk of filter collapse. While its cubic scaling in K limits applicability for very large K, it provides a compelling balance of efficiency and reliable state estimation (especially for log-volatilities) for the dimensions I study (up to N=150, K=15). The choice between filters depends on model dimensions, accuracy priorities, and tolerance for PF instability.

4.2 Simulation Study 2: State Estimation & Parameter Recovery

This study assesses the ability of BIF and PF estimators to recover the true DFSV parameters ($\mathbf{\Theta}$) and latent states via Maximum Likelihood Estimation (MLE) on simulated data when the parameters are unknown. The simulation design (detailed in Appendix C.2, Table C.4) covers various time-series lengths and dimensions across all N, K, T, using 5 replications per configuration. I assess parameter/state accuracy, bias, and computational scaling. My results suggest that the BIF generally achieves superior parameter-recovery and state-estimation (particularly for log-volatilities) with a favorable bias-variance trade-off.

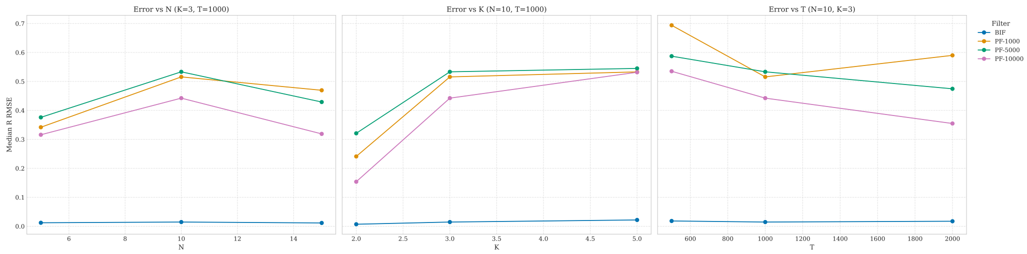
4.2.1 Parameter accuracy

Table 4.2 details the median RMSE and bias for key parameter blocks ($\mathbf{\Lambda}, \mathbf{\Phi}_f, \mathbf{\Phi}_h, \boldsymbol{\mu}, \mathbf{\Sigma}_\epsilon, \boldsymbol{Q}_h$) across the simulation grid, comparing the BIF against the PF-10k. The BIF confirms superior recovery for most blocks. Notably, the BIF achieves over 20 times lower median RMSE for factor loadings ($\mathbf{\Lambda}$) than PF-10k. The BIF also achieves substantially lower median RMSE for the volatility transition matrix ($\mathbf{\Phi}_h$) and idiosyncratic variances ($\mathbf{\Sigma}_\epsilon$), and slightly lower error for the factor transition matrix ($\mathbf{\Phi}_f$). Estimating the log-volatility mean ($\boldsymbol{\mu}$) remains challenging for both methods, showing comparable RMSE. The PF-10k confirms lower median RMSE only for the volatility innovation covariance ($\boldsymbol{Q}_h$), although the BIF exhibits lower median bias for this parameter.

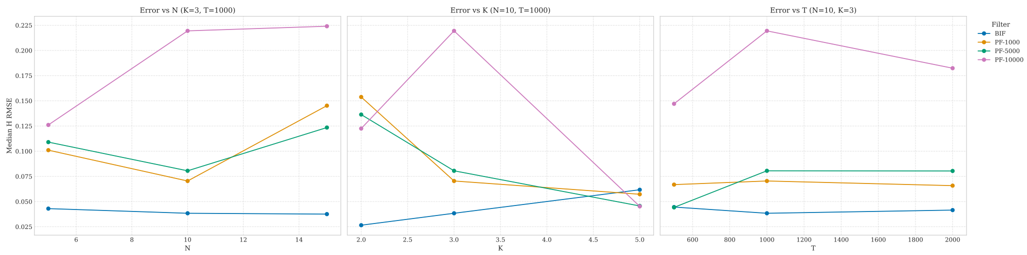
Table 4.2: Parameter-recovery Median RMSE and Median Bias. Aggregated over the full design grid of (N, K, T) configurations.

	BIF		PF-10k	
Parameter	Median RMSE	Median Bias	Median RMSE	Median Bias
Factor Loadings ($\mathbf{\Lambda}$)	0.02	0.00	0.41	-0.17
Factor Transition ($\mathbf{\Phi}_f$)	0.05	0.03	0.07	-0.03
Volatility Transition ($\mathbf{\Phi}_h$)	0.04	0.01	0.12	0.00
Log-Volatility Mean ($\boldsymbol{\mu}$)	1.08	0.89	1.01	1.01
Idiosyncratic Variance ($\mathbf{\Sigma}_\epsilon$)	0.01	0.00	0.06	0.05
Volatility Innovation Covariance ($\boldsymbol{Q}_h$)	0.21	-0.10	0.04	-0.01

Figure 4.5 plots how median RMSE scales with dimensions. The top panel reveals the BIF's accuracy for $\mathbf{\Lambda}$; it maintains near-zero RMSE irrespective of N, K, T, while the PF variants exhibit substantially higher RMSE. The bottom panel shows the BIF's superior performance for $\mathbf{\Phi}_h$, achieving lower RMSE than all PF variants across dimensions. Accuracy generally improves with T for all methods, but the BIF's advantage is clear, especially compared to the PF-10k which struggles with $\mathbf{\Phi}_h$ as N and K increase.



(a) Factor loadings ($\mathbf{\Lambda}$) RMSE



(b) Volatility transition matrix ($\mathbf{\Phi}_h$) RMSE

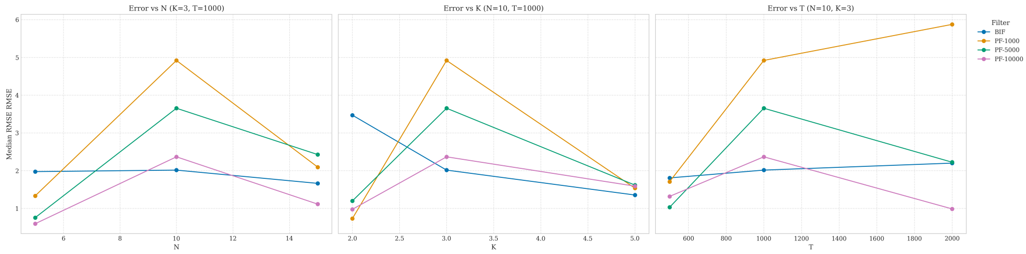
Figure 4.5: Parameter estimation error (median RMSE) scaling with model dimensions (N, K, T). BIF consistently achieves lower RMSE for both parameter blocks compared to PF variants, with the advantage being particularly pronounced for $\mathbf{\Lambda}$. Accuracy generally improves with increasing time-series length T for all methods. See Appendix C.4.9 for additional parameters.

4.2.2 State accuracy

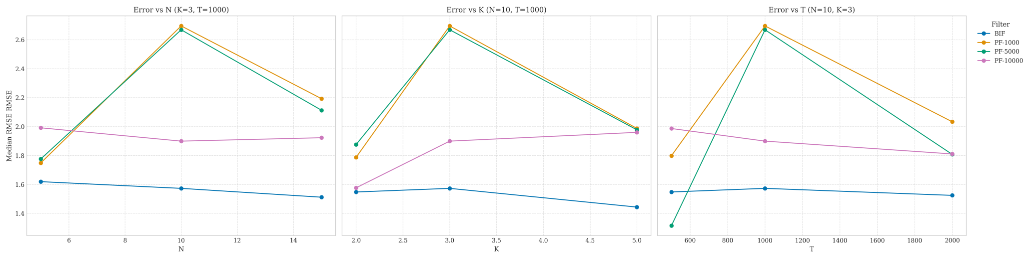
Figure 4.6 plots the median RMSE for filtered states using estimated parameters, scaling with dimensions N, K, T. For factor states, PF variants (especially PF-10k) are competitive and often outperform the BIF in smaller configurations (lower N, K). As dimensions increase, this advantage diminishes, and the BIF's performance becomes comparable or occasionally superior. The BIF's factor accuracy remains relatively stable with N, degrades slightly with K, and improves slightly with T. In contrast, Figure 4.6b confirms the BIF's significant advantage for log-volatility states. The BIF consistently achieves the lowest median log-volatility RMSE across almost all dimensions. Its performance remains remarkably stable with increasing N and T, showing only a minor error increase with K. All PF variants exhibit substantially higher log-volatility RMSE. This underscores the BIF's robustness in log-volatility estimation, even with parameter uncertainty, while acknowledging the PF's strength in factor estimation for smaller models.

4.2.3 Bias & identification difficulty

Figure 4.7 plots bias (horizontal axis) against RMSE (vertical axis) for the log-volatility transition parameters across configurations. The BIF estimates (pink circles) cluster tightly near zero bias with low RMSE, indicating low systematic bias and high precision (low variance). In contrast, the PF estimates (teal, purple, olive markers) exhibit significantly wider dispersion both horizontally and vertically. While centered around zero bias on average, their high variance means individual estimates can suffer substantial bias and large RMSE. Increasing time-series length T (circles to crosses to squares) improves performance for all filters, but the BIF maintains lower variance. This figure, focusing on the crucial $\mathbf{\Phi}_h$ parameter block, confirms the



(a) Factor states ($\hat{\boldsymbol{f}}_{t|t}$) RMSE



(b) Log-volatility states ($\hat{\boldsymbol{h}}_{t|t}$) RMSE

Figure 4.6: Median RMSE for filtered states using estimated parameters, plotted against model dimensions N, K, and T. (a) BIF factor state accuracy is competitive with PF-10000 and generally better than lower-particle PFs. (b) BIF consistently achieves the lowest log-volatility RMSE compared to all PF variants across dimensions.

BIF's superior bias-variance trade-off compared to PFs. Similar analyses for other parameters are deferred to Appendix C.4.

4.2.4 Computational cost and Accuracy Trade-offs

Computationally, Table 4.3 details cost alongside state-estimation accuracy. The PFs exhibit a clear speed-accuracy trade-off: raising the particle count from 1,000 to 10,000 increases the median runtime tenfold (from 56s to 606s) while significantly improving factor RMSE (from 4.36 to 1.54). The BIF shows a median runtime of 692s, achieving competitive factor RMSE (2.51) and the lowest volatility RMSE (1.56). Although the aggregated runtime is similar to the PF-10k here, the BIF's relative efficiency may improve in higher dimensions where PF performance degrades more rapidly. Figure 4.8 plots time scaling: the BIF confirms approximately linear growth with N and T, but cubic complexity with K, consistent with Section 4.1.3. PF variants scale more gradually with K but depend heavily on the particle count. This implies the BIF may be preferable for moderate K and large N, while PFs might be more efficient for large K relative to N, assuming stability can be maintained.

Table 4.3: Computational cost and accuracy comparison (median/mean over configurations). Aggregated across all (N, K, T) settings.

	Computational Cost			Accuracy (Avg RMSE)	
Filter	Med Time (s)	Med Time/Iter (s)	Med Std Dev Time (s)	Factor	Volatility
BIF	692.1	1.89	176.0	2.505	1.561
PF-1k	56.4	0.22	17.2	4.358	2.306
PF-5k	240.5	0.86	109.0	3.952	2.456
PF-10k	605.6	2.20	230.4	1.535	2.048

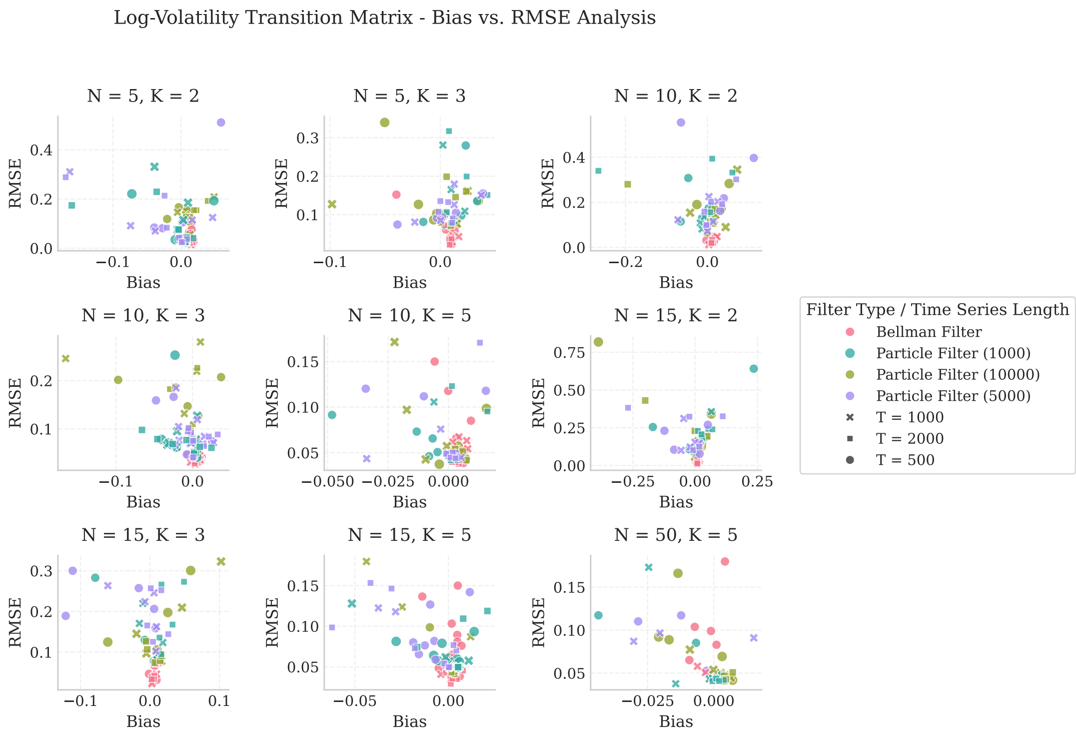
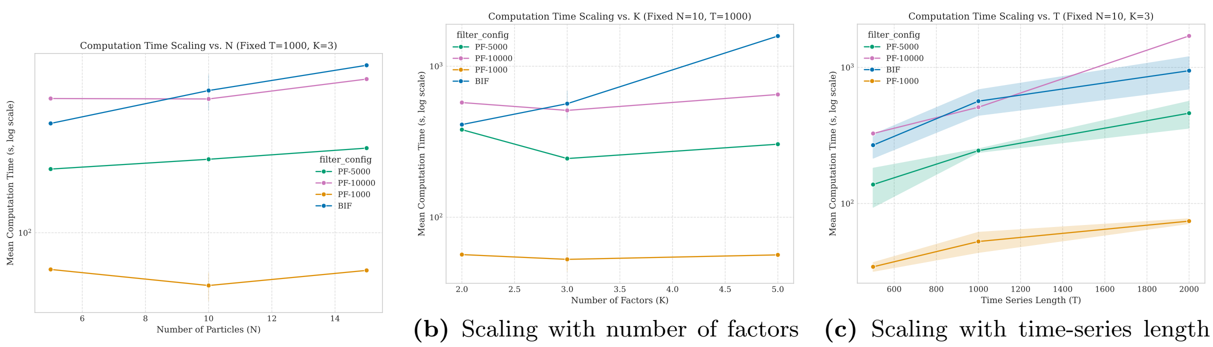


Figure 4.7: Bias versus RMSE for $\mathbf{\Phi}_h$ parameter estimates across different (N, K) pairs. Marker shape indicates time-series length T. BIF estimates (pink) cluster tightly near zero bias with low RMSE, while PF variants show higher dispersion.

4.2.5 Synthesis

Simulation Study 2 confirms the BIF's strong capabilities in joint parameter and state estimation for DFSV models. The BIF consistently provides superior recovery for most parameters (especially $\mathbf{\Lambda}$ and $\mathbf{\Phi}_h$), showing significantly lower RMSE and variance than PFs (Section 4.2.1, Table 4.2, Figure 4.7). While PFs can be competitive for estimating factor states in lower dimensions, the BIF provides substantially more accurate log-volatility states across all configurations (Section 4.2.2, Figure 4.6). Computationally, the BIF's runtime is comparable to the PF-10k in these dimensions, although its cubic scaling in K contrasts with the PF's dependence on particle count (Section 4.2.4).



(a) Scaling with number of assets N

(b) Scaling with number of factors K

(c) Scaling with time-series length T

Figure 4.8: Computational time scaling with model dimensions. BIF shows approximately linear growth with N and T, but cubic complexity with K. PF variants exhibit more gradual scaling with K but are heavily influenced by particle count.

This study validates the BIF as a robust and accurate method for estimating DFSV models. Its key advantages are precise parameter-recovery (loadings, volatility persistence) and superior log-volatility estimation. It offers a favorable bias-variance profile (Section 4.2.3) and greater reliability compared to PFs, which suffer from higher parameter variance and less accurate volatility tracking. The BIF's deterministic nature and reliable performance render it highly suitable for financial applications where volatility dynamics are crucial (e.g., recovering volatility states with 40% lower RMSE than the best PF (N=15, K=5)). The main trade-off involves computational scaling with K, which becomes relevant primarily when K exceeds 15–20 factors.

4.3 Overall Discussion: Synthesis of Simulation Findings

These simulation studies compare the Bellman Information Filter (BIF) against Particle Filter (PF) benchmarks for estimating DFSV models, directly addressing Research Question 2 (Section 1.5) regarding comparative performance across dimensions. Study 1 (Section 4.1) focused on computational efficiency and state estimation with known parameters, while Study 2 (Section 4.2) assessed joint parameter and state recovery.

Synthesizing the findings, the BIF's primary strength indicates superior accuracy in estimating latent log-volatilities ($\boldsymbol{h}_t$) and recovering parameters governing volatility dynamics ($\boldsymbol{\Phi}_h$) and factor structure ($\boldsymbol{\Lambda}$) (Figure 4.4, Figure 4.6, Table 4.2, Figure 4.7). This suggests greater reliability for inference compared to PFs, which exhibit higher variance in parameter recovery. While well-tuned PFs can achieve competitive factor state accuracy in lower dimensions (Figure 4.3, Figure 4.6a), they consistently struggle with volatility estimation.

Regarding runtime, the BIF scales linearly with the number of assets N and time T, but cubically with the number of factors K (Figure 4.1a, Figure 4.8), representing its main bottleneck. PF runtime scales more gently with K but linearly with N and the particle count P. Achieving acceptable PF accuracy often requires a large P, leading to comparable or even slower runtimes than the BIF in the configurations studied (Table 4.1, Table 4.3).

Numerically, the BIF's deterministic nature offers significant robustness against PF instability issues like filter divergence or particle degeneracy (Section 4.1.5). This represents a key practical advantage.

Answering Research Question 2 directly:

1. **Runtime scaling:** The BIF scales $O(NTK^3)$, while the PF scales roughly $O(NTP)$. The BIF remains competitive for dimensions typical in finance where $N \gg K$.

2. **Numerical stability:** The BIF provides superior, deterministic performance without the PF's risk of filter collapse.

3. **Factor-state accuracy:** The PF achieves slightly better accuracy in low dimensions; the BIF is competitive across the tested configurations.

4. **Volatility-state accuracy:** The BIF consistently outperforms PFs across all dimensions examined.

The choice between filters depends on priorities. For accurate volatility dynamics, reliable parameter inference, and predictable performance, the BIF is a strong candidate. If factor accuracy in low dimensions is the sole priority and potential PF instability is manageable, or if K is very large relative to N, a PF is considered. Note that the BIF's Gaussian approximation performs well in these simulations but is a limitation when dealing with extreme non-Gaussianities.

Based on this evidence, the BIF proves highly suitable for the empirical applications I present in subsequent chapters. It provides strengths in capturing volatility dynamics and factor structures, computational feasibility, and reliability for typical financial panel data dimensions, aligning well with modeling time-varying covariances and market co-movement. Limitations of this analysis include the reliance on specific Data Generating Processes (DGPs) and parameter values. The range of dimensions explored does not cover all possible scenarios. Future work can extend this analysis to alternative model specifications, larger dimensions, and different distributional assumptions. In the next chapter, I will validate the models on real-world data.

Chapter 5

Empirical Application: Factor Dynamics in Financial Markets using the Bellman Information Filter

This chapter applies the Dynamic Factor Stochastic Volatility (DFSV) model, which I estimate using the Bellman Information Filter (BIF), to 95 Fama-French portfolio returns (July 1963–December 2023). I evaluate if the DFSV-BIF framework empirically outperforms benchmarks in capturing return dynamics. The analysis reveals a tension: while DFSV-BIF yields plausible aggregate dynamics and factor structures, it struggles with conditional heteroskedasticity in individual returns compared to simpler benchmarks, presenting an empirical puzzle.

5.1 Model Specifications and Estimation Summary

This section recaps the models and estimation procedures I employ in this empirical analysis. I compare four models applied to the $N = 95$ Fama-French portfolio returns:

1. The Dynamic Factor Stochastic Volatility (DFSV) model, detailed in Chapter 2, featuring $K = 5$ latent factors with time-varying volatility. I estimate this model using two methods:
   - **DFSV-BIF:** Maximizing the pseudo-log-likelihood via the Bellman Information Filter (Section 2.2).
   - **DFSV-PF:** Maximizing an approximate marginal log-likelihood via a Particle Filter (Section 2.3) with $P = 10,000$ particles.
2. A benchmark Dynamic Factor Model (DFM) with constant volatility, which I estimate via standard Maximum Likelihood using the `statsmodels` Python package.
3. A benchmark DCC-GARCH(1,1) model, which I estimate via standard Maximum Likelihood using the `arch` and `mgarch` Python packages.

I estimate all models using the full sample period from July 1963 to December 2023.

5.1.1 Computational Performance

I first compare the computational resources estimation requires. Table 5.1 details the wall-clock time, convergence status, and optimizer iterations.

Table 5.1: Estimation Summary: Computational Performance and Stability (July 1963–Dec 2023). Convergence success indicates successful termination of the optimization procedure. Iterations refer to the number required by the optimizer, where applicable.

Model	Estimation Time (Wall Clock)	Convergence Success	Iterations
DFSV-BIF	175.18 min	Yes	289
DFSV-PF	28.75 min	Yes	466
DFM	4.53 min	Yes	137
DCC-GARCH	4.86 min	Yes	N/A

All optimizations converged successfully. Table 5.1 reveals significant estimation time differences: DFSV-BIF required the most time (approx. 175 min), followed by DFSV-PF (approx. 29 min). Both were substantially slower than DFM and DCC-GARCH (¡ 5 min each), reflecting the computational burden of estimating latent stochastic volatility. Interestingly, BIF required fewer optimizer iterations than PF despite its longer runtime, suggesting differences in iteration complexity or the optimization landscape encountered.

5.2 The DFSV-BIF Model: Internal Plausibility

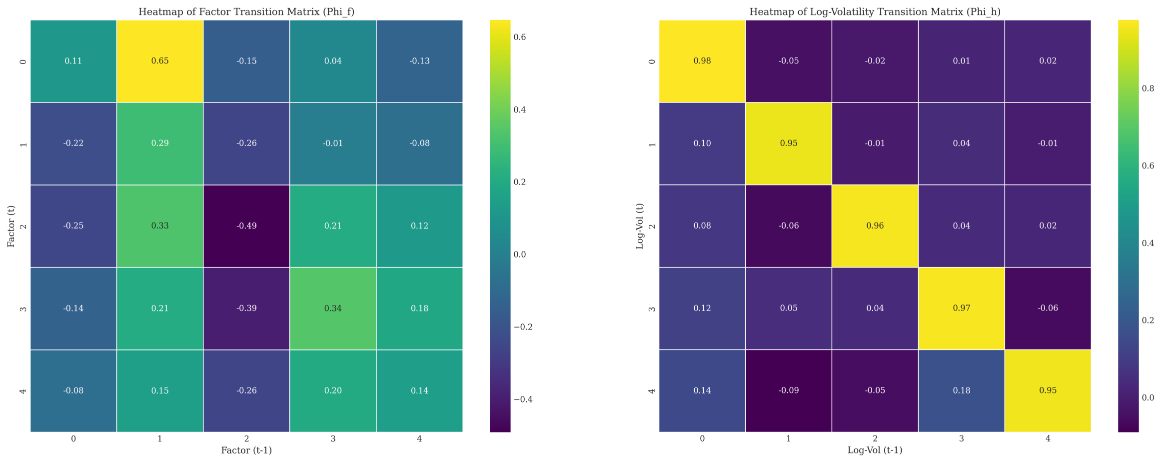
Before comparing models, I assess the internal plausibility of DFSV-BIF by examining if the estimated parameters and latent states align with economic intuition and established stylized facts.

Table 5.2 summarizes key DFSV-BIF parameter estimates. Both transition matrices satisfy stationarity conditions: the maximum absolute eigenvalue of $\hat{\mathbf{\Phi}}_f$ (0.3324) implies low factor persistence, while the maximum absolute eigenvalue of $\hat{\mathbf{\Phi}}_h$ (0.9857) indicates high volatility persistence, consistent with the well-documented phenomenon of volatility clustering.

Table 5.2: Summary of Key Parameter Estimates (DFSV-BIF). Eigenvalues indicate stationarity. $\hat{\boldsymbol{\mu}}$ represents the unconditional mean of log-volatilities. $\hat{\mathbf{\Sigma}}_\epsilon$ captures idiosyncratic variances, and $\hat{\boldsymbol{Q}}_h$ captures the volatility of volatilities.

Parameter Description	Value / Range
Max Eigenvalue($\hat{\mathbf{\Phi}}_f$)	0.3324
Max Eigenvalue($\hat{\mathbf{\Phi}}_h$)	0.9857
Mean($\hat{\boldsymbol{\mu}}$)	-6.0216
Range(diag($\hat{\mathbf{\Sigma}}_\epsilon$))	0.0002–0.0025
Mean(diag($\hat{\mathbf{\Sigma}}_\epsilon$))	0.0006
Mean(diag($\hat{\boldsymbol{Q}}_h$))	0.0292

Figure 5.1 visualizes these transition matrices. The heatmaps confirm the dynamics implied by the eigenvalues: weaker diagonal persistence in $\hat{\mathbf{\Phi}}_f$ contrasts with strong diagonal persistence in $\hat{\mathbf{\Phi}}_h$. Off-diagonal elements suggest some cross-factor interaction in $\hat{\mathbf{\Phi}}_f$ but minimal log-volatility spillover in $\hat{\mathbf{\Phi}}_h$.



(a) Factor transition matrix ($\hat{\mathbf{\Phi}}_f$) **(b)** Log-volatility transition matrix ($\hat{\mathbf{\Phi}}_h$)

Figure 5.1: Heatmaps of Estimated Transition Matrices from DFSV-BIF. The panels visualize the estimated transition matrices for latent factors ($\hat{\mathbf{\Phi}}_f$, left) and log-volatilities ($\hat{\mathbf{\Phi}}_h$, right). Diagonal elements indicate persistence, while off-diagonal elements capture cross-factor dependencies.

Other parameters reported in Table 5.2 are economically reasonable. The mean unconditional log-volatility ($\hat{\boldsymbol{\mu}} = -6.02$) implies a plausible unconditional monthly factor standard deviation (approximately 4.9%). The range and average of the diagonal elements of $\hat{\mathbf{\Sigma}}_\epsilon$ imply sensible non-systematic risk levels (corresponding to 1.4%-5.0% monthly standard deviation) relative to the total standard deviations observed in portfolio returns. Furthermore, the mean diagonal element of $\hat{\boldsymbol{Q}}_h$ allows for plausible variation in factor volatilities over time.

The estimated factor loadings $\hat{\mathbf{\Lambda}}$, visualized in Figure 5.2, further support the model's plausibility. The heatmap reveals a distinct block structure that aligns well with the known portfolio sorting characteristics (Size and Book-to-Market), indicating that the estimated factors successfully capture relevant systematic risk dimensions as intended by the model specification.

Finally, the smoothed latent states presented in Figure 5.3 appear reasonable. The estimated factors ($\hat{\boldsymbol{f}}_{t|T}$, top panel) exhibit varying levels of activity over the sample period, with Factor 1 clearly reflecting market stress periods such as the Global Financial Crisis (GFC). Concurrently, the estimated log-volatilities ($\hat{\boldsymbol{h}}_{t|T}$, bottom panel) exhibit high persistence and pronounced spikes during market crises, visually confirming the model's ability to capture volatility clustering.

In summary, the DFSV-BIF parameter estimates and the resulting latent states appear econometrically sound and align well with established stylized facts and economic intuition. This suggests the model possesses internal consistency and effectively captures key dynamic features of the financial market data.

5.3 Comparative Analysis of Market Dynamics

I now compare how the different estimated models capture aggregate market dynamics. This comparison focuses on the evolution of overall variance and correlation derived from the estimated conditional covariance matrix, $\widehat{\mathbf{\Sigma}}_t^{(m)}$ (defined in Equation 3.1). Figure 5.4 illustrates key metrics derived from these matrices.

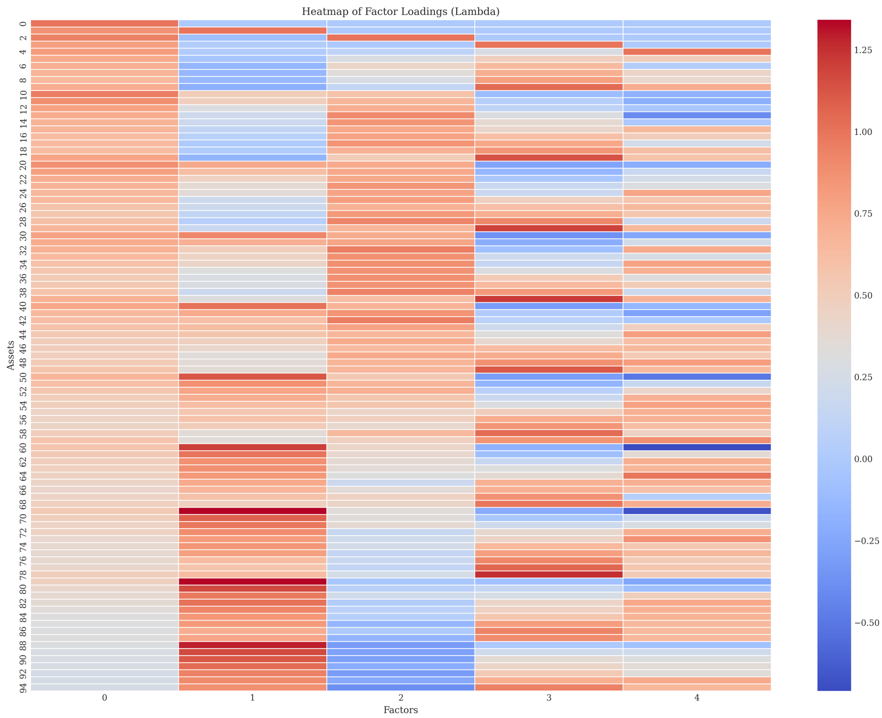


Figure 5.2: Heatmap of Estimated Factor Loadings ($\hat{\mathbf{\Lambda}}$) from DFSV-BIF on FF-95 Portfolios. The heatmap displays the estimated loadings of the $N = 95$ Fama-French portfolios (sorted by Size and Book-to-Market, y-axis) on the $K = 5$ latent factors (x-axis). Loadings are identified via the lower-triangular constraint discussed in Section 2.1.3.

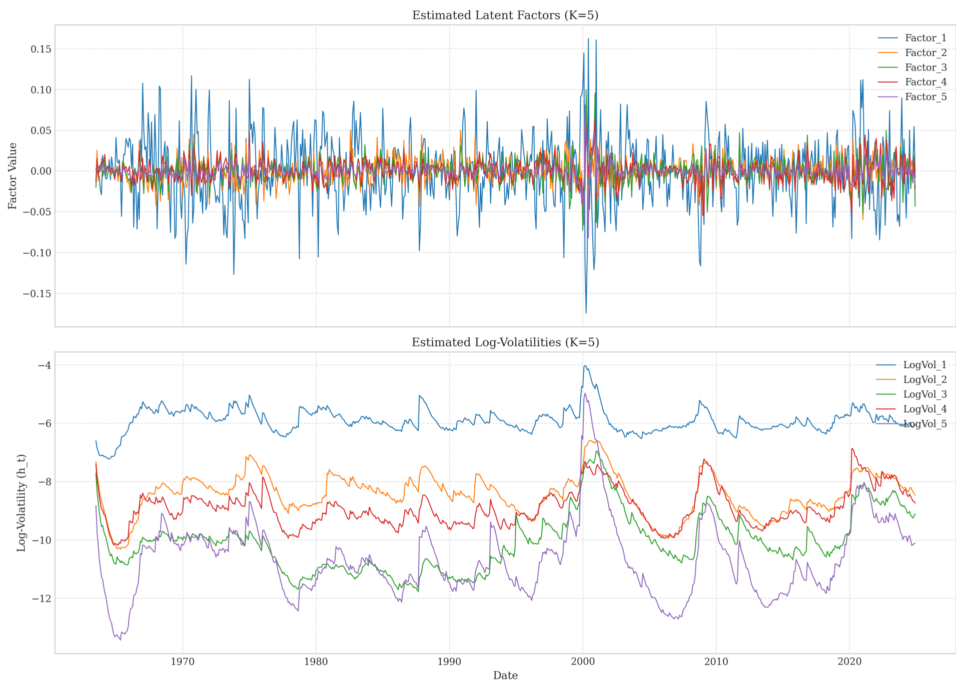
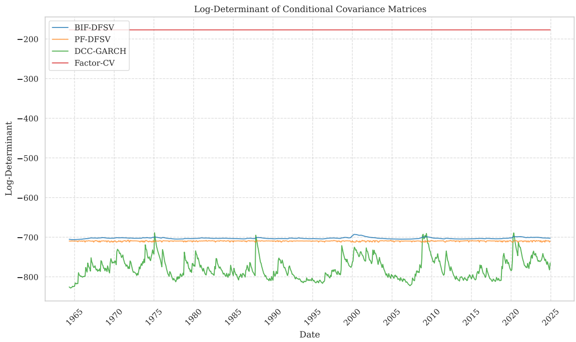
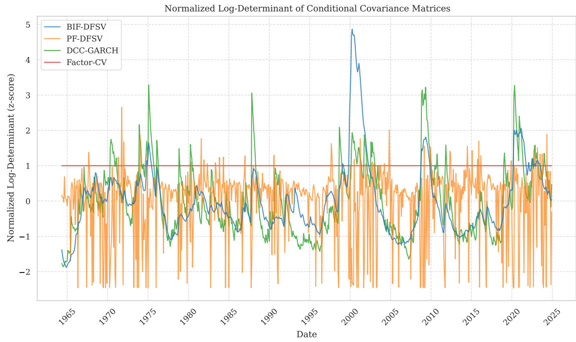


Figure 5.3: Estimated Latent Factors and Log-Volatilities from DFSV-BIF Model (July 1963–Dec 2023). The figure shows smoothed estimates of the $K = 5$ latent factors $\hat{\boldsymbol{f}}_{t|T}$ (top panel) and their corresponding log-volatilities $\hat{\boldsymbol{h}}_{t|T}$ (bottom panel). Individual factor and log-volatility plots are provided in Appendix D.

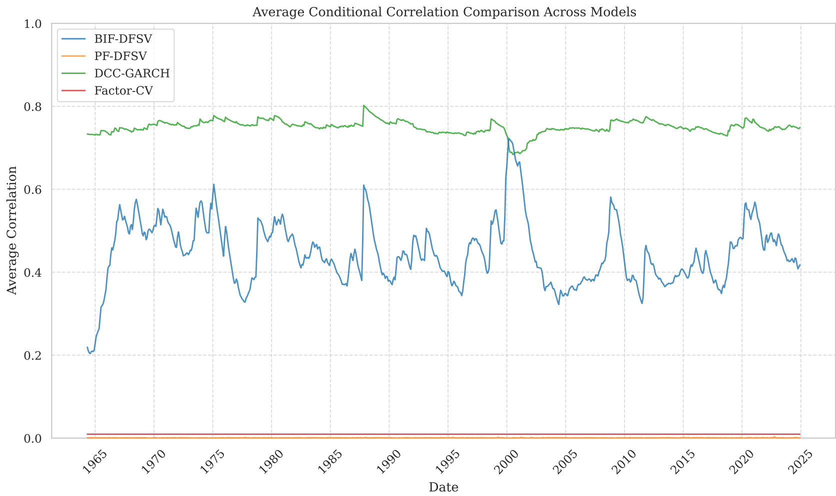
5.3.1 Covariance Dynamics (Overall Variance & Correlation)



(a) Log Generalized Variance ($\log|\hat{\mathbf{\Sigma}}_t|$)



(b) Normalized Log Generalized Variance



(c) Average Conditional Correlation ($\bar{\rho}_t$)

Figure 5.4: Comparative Time Series of Covariance Dynamics Metrics (July 1963–Dec 2023). Comparison of key covariance metrics derived from the estimated conditional covariance matrix $\widehat{\mathbf{\Sigma}}_t^{(m)}$ (Equation 3.1) for DFSV-BIF, DFSV-PF, DFM (Factor-CV), and DCC-GARCH models. Panel (a) shows Log Generalized Variance ($\log|\hat{\mathbf{\Sigma}}_t|$), reflecting overall systemic variance. Panel (b) shows Normalized Log Generalized Variance (standardized), highlighting the timing and relative intensity of volatility dynamics. Panel (c) shows Average Conditional Correlation ($\bar{\rho}_t$), measuring market co-movement.

Figure 5.4a presents the Log Generalized Variance ($\log|\hat{\mathbf{\Sigma}}_t|$), a measure of overall systemic variance. The DFM, by construction, exhibits a constant level due to its static covariance assumption. In contrast, DFSV-BIF, DFSV-PF, and DCC-GARCH all capture pronounced time variation, characterized by sharp spikes during market crises (e.g., GFC, COVID-19 pandemic), reflecting periods of heightened market volatility. Baseline variance levels differ across models, with DCC-GARCH implying generally higher systemic variance than the DFSV models.

Figure 5.4b displays the Normalized Log Generalized Variance, allowing comparison of dynamic patterns irrespective of baseline levels. DFSV-BIF and DCC-GARCH exhibit remarkably similar timing and relative magnitude of volatility fluctuations. This suggests broad agreement between these two structurally different models regarding the identification of volatility regimes. In contrast, the dynamics implied by DFSV-PF appear more chaotic and less clearly linked to known market events, indicating a potentially weaker capture of aggregate market volatility dynamics by the PF implementation used here.

Figure 5.4c illustrates the Average Conditional Correlation ($\bar{\rho}_t$), measuring the average co-movement across assets. Both DFSV-BIF and DCC-GARCH exhibit dynamically varying correlations, which typically increase during periods of market stress, consistent with established stylized facts. The DFM implies a constant average correlation. Strikingly, the DFSV-PF yields an extremely low and virtually flat average correlation throughout the sample period. This suggests a potential failure of the PF estimation, as implemented here, to capture time-varying market correlation structure, highlighting a possible weakness or sensitivity of the PF approach in this high-dimensional setting.

5.4 The Residual Diagnostics Puzzle

Having established the internal plausibility of the DFSV-BIF model's estimates in Section 5.2, I now evaluate its performance using standard time series diagnostics applied to its standardized residuals. The goal is to determine if these residuals adequately approximate the white noise process assumed by the model specification.

5.4.1 Diagnostic Test Results

Standard model evaluation often involves examining standardized residuals, defined as $\hat{\boldsymbol{z}}_t^{(m)} = (\widehat{\boldsymbol{\Sigma}}_t^{(m)})^{-1/2}(\boldsymbol{r}_t - \widehat{\boldsymbol{\Lambda}}^{(m)}\hat{\boldsymbol{f}}_t^{(m)})$, for any remaining serial correlation or conditional heteroskedasticity. Ideally, these residuals should behave like independent and identically distributed (i.i.d.) noise. Table 5.3 summarizes the results of univariate diagnostic tests applied to each of the $N = 95$ standardized residual series generated by the four competing models.

The diagnostic tests reveal stark performance differences among the models. The DCC-GARCH model successfully eliminated ARCH effects (as measured by ARCH-LM tests) in nearly 99% of the individual residual series, far exceeding the performance of the DFSV variants (Table 5.3). Similarly, for the Ljung-Box Q-statistic applied to squared residuals (testing for remaining autocorrelation in variance), DCC-GARCH achieved an 88.42% pass rate (at lag 10), compared to only 9.47% for DFSV-BIF. This strongly suggests that the DCC-GARCH specification captures the volatility clustering observed in individual asset returns much more effectively than the DFSV models.

Surprisingly, both DFSV variants performed poorly on these diagnostics, with pass rates generally falling below 20%. For the ARCH-LM test with 5 lags, DFSV-BIF (9.47% pass rate) performed even worse than the constant-volatility DFM (21.05%). This finding suggests that the complex factor stochastic volatility structure within the DFSV model, as specified here, contributes little towards capturing the conditional heteroskedasticity present in individual asset returns beyond what is captured by the factors themselves. As expected given the well-known heavy tails of financial return distributions, all models failed the Jarque-Bera normality tests for nearly all residual series.

5.4.2 Framing the Puzzle

The empirical results presented thus far constitute a puzzle. On one hand, the DFSV-BIF estimation produced plausible results: sensible parameter estimates, a clear factor loading structure

Table 5.3: Univariate Residual Diagnostic Pass Rates ($p > 0.05$). Percentage of the N=95 standardized residual series for which the null hypothesis (no serial correlation in squared residuals, no ARCH effects, normality) is not rejected at the 5% significance level. Higher percentages indicate better model specification regarding these aspects.

Test (Lag)	DFSV-BIF	DFSV-PF	DFM	DCC-GARCH
Ljung-Box Q(5) (Squared)	7.37%	11.58%	10.53%	88.42%
Ljung-Box Q(10) (Squared)	9.47%	15.79%	13.68%	88.42%
Ljung-Box Q(15) (Squared)	10.53%	12.63%	7.37%	88.42%
Ljung-Box Q(20) (Squared)	12.63%	17.89%	17.89%	88.42%
ARCH-LM(5)	9.47%	12.63%	21.05%	98.95%
ARCH-LM(10)	11.58%	23.16%	28.42%	98.95%
Jarque-Bera	2.11%	0.00%	0.00%	0.00%

consistent with economic priors (Figure 5.2), and estimated latent states that align well with major market events and crises (Figure 5.3). Furthermore, it captured aggregate market dynamics (variance and correlation) in a manner broadly similar to the DCC-GARCH model. On the other hand, standard diagnostic tests applied to the individual standardized residuals revealed high failure rates (often exceeding 90%) for tests related to conditional heteroskedasticity (Table 5.3). Why does a model that appears to capture market-wide dynamics reasonably well perform so poorly on standard tests designed to detect residual volatility dynamics at the individual asset level?

5.4.3 Diagnosing Residual Heteroskedasticity via GARCH Modelling

To investigate whether the poor diagnostic performance stems from unmodeled heteroskedasticity in the idiosyncratic errors (specifically, the assumption of a constant idiosyncratic covariance matrix $\boldsymbol{\Sigma}_\epsilon$), I conduct a further diagnostic exercise. I fit univariate GARCH(1,1) models to each of the 95 standardized residual series obtained from the DFSV-BIF estimation. Table 5.4 summarizes the key findings from this analysis.

Table 5.4: GARCH(1,1) Parameter Analysis for DFSV-BIF Residuals. Results from fitting GARCH(1,1) models (with Student's t distribution) to each of the 95 standardized residual series from the DFSV-BIF model. α_1 and β_1 refer to the ARCH and GARCH parameters, respectively. The Wald test examines $H_0 : \alpha_1 + \beta_1 = 0$ against $H_1 : \alpha_1 + \beta_1 \neq 0$.

Metric	Value
Total Series	95
Successful Fits	95
% α_1 Significant ($p < 0.05$)	77.89%
% β_1 Significant ($p < 0.05$)	92.63%
% Wald Test Rejects H0 ($\alpha_1 + \beta_1 = 0$)	92.63%
% Sum $\alpha_1 + \beta_1 > 0.9$	64.21%
% Sum $\alpha_1 + \beta_1 > 0.95$	31.58%
% Wald Rejects H0 AND Sum > 0.9	64.21%

The GARCH analysis presented in Table 5.4 reveals substantial remaining GARCH effects that the initial DFSV-BIF model failed to capture. The estimated GARCH parameters (α_1 and β_1) are statistically significant for a large majority of the residual series. Crucially, the Wald test rejected the null hypothesis of no GARCH effects ($H_0 : \alpha_1 + \beta_1 = 0$) for over 92% of the series.

Furthermore, 64% of the series exhibited high persistence in these residual GARCH dynamics, with $\alpha_1 + \beta_1 > 0.9$. This evidence strongly indicates the presence of significant and persistent volatility dynamics within the DFSV-BIF residuals that were not accounted for by the factor stochastic volatility structure alone.

To confirm that these GARCH effects explain the poor initial diagnostic results, I re-ran the Ljung-Box and ARCH-LM tests on the residuals obtained *after* filtering each original DFSV-BIF residual series through its estimated GARCH(1,1) model. Table 5.5 compares the diagnostic test pass rates before and after this GARCH correction.

Table 5.5: Diagnostic Test Pass Rate Comparison: Before vs. After GARCH Filtering. Comparison of diagnostic test pass rates (percentage of 95 series with $p > 0.05$) for DFSV-BIF residuals before ('Initial') and after applying GARCH(1,1) filtering ('Post-GARCH').

Test	Initial Pass Rate	Post-GARCH Pass Rate
Ljung-Box Sq (Lag 5)	7.37%	93.68%
Ljung-Box Sq (Lag 10)	9.47%	92.63%
Ljung-Box Sq (Lag 15)	10.53%	93.68%
Ljung-Box Sq (Lag 20)	12.63%	94.74%
ARCH-LM (Lag 5)	9.47%	92.63%
ARCH-LM (Lag 10)	11.58%	94.74%
Jarque-Bera	2.11%	9.47%

Table 5.5 clearly demonstrates the efficacy of the GARCH(1,1) correction. The pass rates for both the Ljung-Box test on squared residuals and the ARCH-LM test increased dramatically, from approximately 10% initially to over 92% post-correction. This confirms that standard univariate GARCH models effectively account for the conditional heteroskedasticity present in the original DFSV-BIF residuals. The Jarque-Bera pass rate improved only marginally (to 9.47%), reinforcing the finding that residual non-normality persists even after accounting for volatility dynamics.

These findings effectively resolve the diagnostic puzzle framed in Section 5.4.2. The GARCH parameter estimation confirms the presence of significant volatility dynamics missed by the DFSV-BIF model, while the post-GARCH diagnostic tests demonstrate that these missed dynamics are largely consistent with standard GARCH(1,1) effects. This supports the conclusion that the primary limitation of the DFSV specification employed here lies in its assumption of constant idiosyncratic variances ($\boldsymbol{\Sigma}_\epsilon$). The analysis validates the model's ability to capture common factor dynamics but strongly indicates the need to incorporate time-varying idiosyncratic volatility for a more complete description of asset return dynamics.

5.5 Estimation Method Comparison: DFSV-BIF vs. DFSV-PF

This section directly compares the performance of the Bellman Information Filter (BIF) and the Particle Filter (PF) as estimation methods for the same DFSV model specification. The comparison focuses on differences in the optimized statistical fit objective, the quality of latent state estimates, parameter estimates, residual diagnostic performance, and computational requirements.

5.5.1 Statistical Fit Objective

Table 5.6 presents the overall fit metrics achieved by each estimation method. DFSV-BIF attained a final pseudo-log-likelihood value of 222,084.10, substantially higher than the approximate marginal log-likelihood of -17,969.62 obtained by DFSV-PF. However, these values stem from optimizing fundamentally different objective functions and are therefore not directly comparable for model selection purposes (e.g., using AIC or BIC). The large difference observed might reflect inherent differences between the pseudo-likelihood and the true marginal likelihood, or it could indicate challenges faced by the PF optimization procedure in this high-dimensional application.

Table 5.6: Overall Model Fit Comparison (Full Sample, T=738). Log-likelihood for DFSV-BIF is the pseudo-log-likelihood, and for DFSV-PF it is the approximate marginal log-likelihood, both evaluated at the estimated parameters. AIC and BIC penalize for the number of parameters (p).

Model	(Pseudo) Log-Likelihood	Num Params (p)	AIC	BIC
DFSV-BIF	222,084.10	610	-442,948.21	-440,139.80
DFSV-PF	-17,969.62	610	37,159.23	39,967.64
DFM	-38,862.75	580	78,885.50	81,555.79
DCC-GARCH	162,516.36	288	-324,456.72	-323,130.79

5.5.2 Latent States

Comparing the estimated latent states reveals striking differences in quality between the two methods. As shown previously (Figure 5.3), BIF produced smooth and economically interpretable paths for both factors and their volatilities, clearly identifying periods of market stress. In stark contrast, the PF estimates appear erratic and jagged (Figure 5.5), offering little clear economic insight. The difference in estimated average conditional correlation starkly highlights this disparity: BIF shows the expected pattern of correlation increasing during crises, whereas PF produced near-zero and flat correlations throughout the sample (Figure 5.4c). This suggests that the PF, with the chosen number of particles ($P = 10,000$), struggled significantly with state estimation in this high-dimensional ($N = 95$) setting.

5.5.3 Parameter Estimates

Table 5.7 compares key parameter estimates obtained via BIF and PF. The PF estimated considerably higher persistence in the latent factors (max eigenvalue of $\hat{\mathbf{\Phi}}_f = 0.5974$ vs. 0.3324 for BIF) and near-unit root behavior in the log-volatility process (max eigenvalue of $\hat{\mathbf{\Phi}}_h = 0.9997$ vs. 0.9857 for BIF). Additionally, the PF estimated higher volatility of volatilities (mean diagonal of $\hat{\boldsymbol{Q}}_h = 0.0400$ vs. 0.0292 for BIF), consistent with the more erratic log-volatility paths observed in Figure 5.5. Despite these parameter differences, both methods yielded models that performed similarly poorly on residual diagnostics, as discussed next.

5.5.4 Residual Diagnostics

As established in Section 5.4, both DFSV specifications, whether estimated via BIF or PF, failed standard residual diagnostics related to conditional heteroskedasticity (Table 5.3). Both

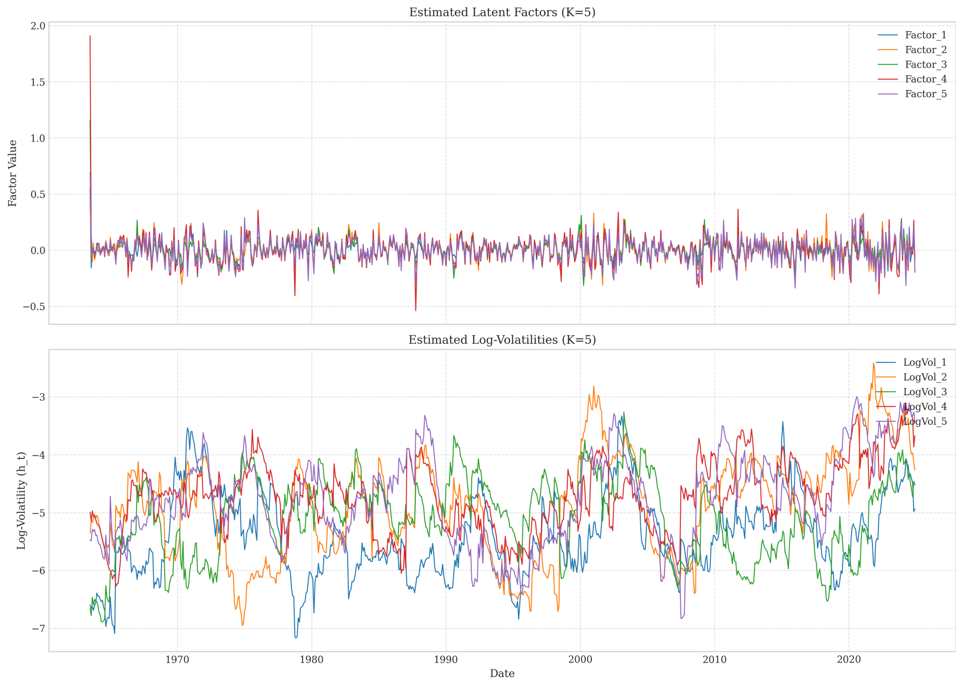


Figure 5.5: Estimated Latent Factors and Log-Volatilities from DFSV-PF Model (July 1963–Dec 2023). The figure shows smoothed estimates of the $K = 5$ latent factors $\hat{\boldsymbol{f}}_{t|T}$ (top panel) and their corresponding log-volatilities $\hat{\boldsymbol{h}}_{t|T}$ (bottom panel). Individual factor and log-volatility plots are provided in Appendix D.1.2 and Appendix D.2.2.

Table 5.7: Comparison of Key Parameter Estimates (DFSV-BIF vs DFSV-PF). Eigenvalues indicate stationarity. $\hat{\boldsymbol{\mu}}$ represents the unconditional mean of log-volatilities. $\hat{\boldsymbol{\Sigma}}_\epsilon$ captures idiosyncratic variances, and $\hat{\boldsymbol{Q}}_h$ captures the volatility of volatilities.

Parameter Description	BIF Value	PF Value
Max Eigenvalue($\hat{\boldsymbol{\Phi}}_f$)	0.3324	0.5974
Max Eigenvalue($\hat{\boldsymbol{\Phi}}_h$)	0.9857	0.9997
Mean($\hat{\boldsymbol{\mu}}$)	-6.0216	-5.9995
Range(diag($\hat{\boldsymbol{\Sigma}}_\epsilon$))	0.0002–0.0025	0.0003–0.0012
Mean(diag($\hat{\boldsymbol{\Sigma}}_\epsilon$))	0.0006	0.0006
Mean(diag($\hat{\boldsymbol{Q}}_h$))	0.0292	0.0400

methods yielded pass rates generally below 20% for Ljung-Box (squared) and ARCH-LM tests, performing significantly worse than the benchmark DCC-GARCH model. This indicates that neither estimation approach, when applied to this specific DFSV model structure with constant idiosyncratic variance, adequately addresses the conditional heteroskedasticity present in the individual portfolio returns.

5.5.5 Computation & Stability

The computational results I observed in this empirical application presented some nuances compared to the simulation study (Chapter 4). Notably, the BIF estimation required approximately 175 minutes, substantially longer than the PF's 29 minutes (Table 5.1). This finding contrasts with simulation results suggesting potential BIF advantages in higher dimensions, indicating that the complexity of evaluating the BIF's pseudo-likelihood and its gradient scales significantly with the number of assets N in practice. Consequently, each BIF iteration proved considerably more costly in this large-scale ($N = 95$) empirical setting, even though fewer iterations were required compared to the PF (289 vs. 466). Furthermore, I observed empirical instability in the PF, particularly in generating plausible latent state dynamics (Section 5.5.2); this challenge was not fully reflected in the median simulation results and highlights difficulties in applying the standard Bootstrap PF reliably in this context. Despite the BIF's higher computational cost in this application, its demonstrably superior quality of latent state estimates might justify the additional time when state interpretability is a primary objective.

5.6 Chapter Discussion

This chapter's empirical application of the DFSV model using the Bellman Information Filter revealed a central puzzle. On the one hand, the DFSV-BIF model produced plausible aggregate dynamics (Section 5.3), capturing overall market variance and correlation patterns similarly to a benchmark DCC-GARCH model. It also yielded interpretable parameters and latent states (Section 5.2) that align well with economic intuition and established stylized facts about financial markets, such as volatility clustering and time-varying factor importance.

On the other hand, the model failed standard residual diagnostics designed to detect remaining conditional heteroskedasticity at the individual asset level (Section 5.4). Its performance on these tests was significantly worse than simpler benchmarks like DCC-GARCH and, in some cases, no better than a constant-volatility DFM (Table 5.3). Further diagnostic analysis involving fitting GARCH models to the DFSV-BIF residuals (Section 5.4.3) strongly suggests that this failure stems primarily from the model's restrictive assumption of constant idiosyncratic variance ($\boldsymbol{\Sigma}_\epsilon$). This assumption effectively overlooks significant asset-specific GARCH effects that remain present in the residuals after accounting for the common factor dynamics.

This finding highlights a critical limitation of the current DFSV specification for capturing the nuances of individual asset dynamics, despite its apparent success at modeling aggregate market behavior. While the BIF estimation proved superior to the PF in generating interpretable latent states in this high-dimensional setting (Section 5.5), neither method could overcome the specification issue related to idiosyncratic volatility. The results underscore the potential need for extensions to the DFSV framework, such as incorporating stochastic volatility in the idiosyncratic components, to achieve a more comprehensive representation of financial asset return dynamics.

Chapter 6

Conclusion

This concluding chapter synthesizes the research undertaken in this thesis. It begins by summarizing the principal findings in direct response to the research questions posed in Chapter 1. Subsequently, it discusses the broader implications of these findings, particularly concerning the interplay between model specification, estimation filter choice, and the inherent trade-offs in high-dimensional financial modeling. The chapter then explicitly outlines the main contributions of this work, critically acknowledges its limitations, and proposes concrete directions for future research motivated by the results obtained. Finally, it offers brief concluding remarks on the overall endeavor.

6.1 Principal Findings

This section synthesizes the principal findings of the thesis by directly addressing the four research questions posed in Chapter 1.

First (RQ 1), the research demonstrates that the Bellman Information Filter (BIF) can be effectively customized for high-dimensional Dynamic Factor Stochastic Volatility (DFSV) models. Key adaptations detailed in Chapter 2 include a Block Coordinate Descent (BCD) optimization for the filter update, the integration of Vector Autoregressive (VAR) dynamics for both latent factors and their log-volatilities, and the use of the Expected Fisher Information Matrix (E-FIM) for numerically stable information matrix updates. This customized BIF framework achieves a computational complexity of $O(NK^2 + K^3)$, rendering deterministic state and hyperparameter estimation feasible for typical financial applications where the number of assets N greatly exceeds the number of factors K.

Second (RQ 2), simulation studies presented in Chapter 4 establish the BIF's comparative advantages and disadvantages against Particle Filter (PF) benchmarks. The BIF consistently provides superior accuracy in estimating latent log-volatility states and recovering key parameters (particularly factor loadings $\mathbf{\Lambda}$ and volatility persistence $\mathbf{\Phi}_h$), alongside greater numerical stability and a more favorable bias-variance trade-off. While PFs can achieve marginally better factor-state accuracy in lower dimensions, the BIF remains competitive across configurations. Computationally, the BIF scales cubically with K but linearly with N, whereas PF runtime depends linearly on N and the particle count P; the BIF proves competitive or faster than PFs requiring large particle counts for stability in the dimensions studied.

Third (RQ 3), the empirical application to 95 Fama-French portfolios (Chapter 5) reveals mixed performance for the DFSV-BIF model relative to benchmarks. The model successfully captures aggregate market dynamics (systemic variance, correlation) similarly to DCC-GARCH and produces economically plausible parameters, factor loadings, and interpretable latent states, outperforming the DFSV-PF which yielded unstable states empirically. However, the DFSV-BIF fails standard residual diagnostic tests for conditional heteroskedasticity, performing significantly worse than DCC-GARCH and often no better than a constant-volatility DFM in this regard. Empirically, the BIF estimation was also slower than the PF, contrary to simulation findings, likely due to the complexity of pseudo-likelihood evaluation with large N.

Fourth (RQ 4), the diagnostic analysis in Chapter 5 strongly indicates that the poor residual performance stems from model misspecification, specifically the assumption of constant, diagonal idiosyncratic error variance ($\mathbf{\Sigma}_\epsilon$). Fitting GARCH models to the DFSV-BIF residuals successfully removes the remaining conditional heteroskedasticity, confirming that significant asset-specific volatility dynamics are missed by the factor SV structure alone. This finding highlights a critical limitation of the employed DFSV specification and motivates extensions incorporating time-varying idiosyncratic volatility.

6.2 Discussion and Implications

The principal findings reveal important implications regarding both the DFSV model specification and the choice of estimation filter. A central empirical puzzle emerges from Chapter 5: the DFSV-BIF model captures aggregate market dynamics plausibly (Section 5.3) and yields interpretable states (Section 5.2), yet fails standard residual diagnostic tests for conditional heteroskedasticity (Section 5.4.1). The subsequent GARCH analysis of these residuals (Section 5.4.3) strongly suggests this discrepancy arises from the model's assumption of constant idiosyncratic variance ($\mathbf{\Sigma}_\epsilon$). This specification, while common (Aguilar & West, 2000), appears insufficient to capture asset-specific volatility dynamics remaining after accounting for common factor stochastic volatility, directly addressing the limitation highlighted in RQ 4.

This empirical puzzle informs the implications for filter selection. The simulation studies (Chapter 4) demonstrate the BIF's strengths: superior log-volatility state accuracy (Section 4.1.4), reliable parameter recovery (especially for $\mathbf{\Lambda}$ and $\mathbf{\Phi}_h$, Section 4.2.1), and deterministic stability (Section 4.1.5). Empirically, the BIF produced interpretable latent states, unlike the PF (Section 5.5). However, a distinction arises in computational scaling. While Simulation 1 showed the BIF filter step scales favorably (linearly in N, cubically in K, Section 4.1.2), the full BIF estimation procedure proved slower empirically than the PF in the high-N setting (Section 5.1.1). This suggests the pseudo-likelihood evaluation dominates the empirical runtime. The Particle Filter, conversely, exhibited significant instability and poor state estimation empirically (Section 5.5) and requires careful tuning of particle counts to avoid issues like weight degeneracy noted by Rebeschini and Handel (2015). The choice thus involves a trade-off: the BIF offers robustness and superior estimation quality, particularly for volatilities, but its current estimation approach can be slow for large N; the PF might offer faster estimation iterations but carries substantial risks regarding stability and the quality of results, echoing challenges with simulation methods in high dimensions (Kastner et al., 2017).

The findings underscore the persistent challenge in financial econometrics of balancing model realism against computational tractability, particularly for high-dimensional systems. This research utilizes the BIF, following Lange (2024), to push the tractability frontier for DFSV models, enabling estimation in dimensions ($N = 95, K = 5$) that challenge traditional methods. However, the empirical results demonstrate that computational feasibility alone is insufficient; model specification limitations, such as the constant $\boldsymbol{\Sigma}_\epsilon$ assumption, can ultimately constrain performance. Effectively modeling complex financial systems requires navigating this trade-off, selecting estimation methods like the BIF that are computationally viable while acknowledging and addressing the simplifying assumptions inherent in tractable model specifications.

6.3 Contributions

This thesis makes several contributions to the literature on high-dimensional financial time series modeling and state-space estimation:

1. **Methodological Adaptation of BIF for DFSV Models:** I develop and detail a practical framework for applying the Bellman Information Filter to estimate DFSV models with VAR dynamics for both factors and log-volatilities. Key adaptations include a Block Coordinate Descent (BCD) optimization strategy for the filter update (Section 2.2.2) and the use of the Expected Fisher Information Matrix (E-FIM) for numerically stable information matrix updates (Appendix A.4). This tailored framework, achieving $O(NK^2 + K^3)$ complexity, renders deterministic state and hyperparameter estimation feasible for typical financial applications (addressing RQ 1).

2. **Comparative Analysis of Estimation Filters:** I provide a rigorous comparative assessment of the BIF against Particle Filter benchmarks and standard models (DCC-GARCH, DFM) through extensive simulation studies (Chapter 4) and empirical application (Chapter 5). This analysis quantifies the trade-offs regarding computational scaling, numerical stability, state estimation accuracy (particularly the BIF's advantage in log-volatility estimation), and parameter recovery reliability (addressing RQ 2 and RQ 3).

3. **Empirical Findings and Diagnostic Insight:** The empirical application to Fama-French portfolios reveals a key puzzle: the DFSV-BIF captures aggregate market dynamics plausibly but fails standard residual diagnostics (Section 5.4). Subsequent GARCH analysis of residuals strongly identifies the model's assumption of constant, diagonal idiosyncratic variance ($\boldsymbol{\Sigma}_\epsilon$) as the primary source of misspecification (Section 5.4.3). This provides crucial diagnostic insight into the limitations of this common DFSV specification (addressing RQ 3 and RQ 4).

4. **Comprehensive Evaluation Approach:** I employ a multi-faceted evaluation framework (Section 3.3) encompassing statistical fit, covariance dynamics, factor loading interpretability, residual diagnostics, parameter plausibility, and computational cost. This provides a balanced approach for assessing complex financial models beyond simple statistical fit metrics.

6.4 Limitations

Despite the contributions, this research is subject to several limitations that define boundaries for the findings and suggest avenues for refinement:

1. **Constant Idiosyncratic Variance ($\boldsymbol{\Sigma}_\epsilon$):** The most significant limitation identified empirically (Section 5.4.3) appears to be the assumption of constant, diagonal idiosyncratic error variance. This simplification, while common (Aguilar & West, 2000), proved insufficient to capture asset-specific volatility dynamics remaining after accounting for common factor stochastic volatility. This directly led to the poor performance on residual diagnostic tests (Section 5.4.1) and appears to be the primary model specification issue identified in this analysis.

2. **Computational Scaling:** While the BIF filter step exhibits favorable scaling ($O(NK^2 + K^3)$, Section 4.1.2), the full estimation procedure based on direct pseudo-likelihood maximization proved computationally demanding in the empirical application with large N (Section 5.5.5). The cubic scaling in K within the filter step could still become prohibitive for models requiring a very large number of latent factors ($K \gg 15$), but the empirical bottleneck observed here was related to the overall estimation time for large N.

3. **Gaussian Approximation:** The BIF relies on Gaussian approximations for the filtering distributions (Section 2.2). While simulation studies (Chapter 4) did not indicate major issues for the specified DFSV model, this approximation could introduce bias or inaccuracy if the true underlying distributions exhibit strong non-Gaussian features (e.g., extreme skewness or heavy tails) not captured by the model's SV components. The empirical failure of Jarque-Bera tests (Table 5.3) hints at this possibility.

4. **Identification Constraint:** The use of a lower-triangular constraint on the factor loading matrix $\boldsymbol{\Lambda}$ to ensure model identification (Section 2.1.3) imposes an arbitrary ordering on the factors. Consequently, the estimated factors are identified statistically rather than corresponding directly to pre-defined economic factors, potentially complicating their direct interpretation.

5. **Empirical Scope:** The empirical analysis is based on a specific dataset (US Fama-French 95 portfolios) and focuses exclusively on in-sample estimation and evaluation (Chapter 5). The findings regarding model performance and limitations may not generalize directly to other asset classes, markets, or time periods. Furthermore, the lack of out-of-sample forecasting assessment limits conclusions about the model's practical predictive utility.

6.5 Future Research Directions

The findings and limitations of this thesis motivate several promising directions for future research:

1. **Incorporate Time-Varying Idiosyncratic Volatility:** This is the most critical extension, directly motivated by the empirical diagnostic failures (Section 5.4.3) attributed to

the constant $\mathbf{\Sigma}_\epsilon$ assumption. Future work should explore tractable ways to introduce dynamics in $\mathbf{\Sigma}_{\epsilon,t}$ within the BIF framework. Potential approaches include grouped structures (e.g., common idiosyncratic SV within industries or characteristic portfolios) or computationally efficient univariate SV models for the idiosyncratic terms, aiming to resolve the residual heteroskedasticity puzzle.

2. **Explore Alternative Dynamics:** The current model uses VAR(1) processes for factors and log-volatilities. Future research could investigate richer dynamic specifications to capture additional stylized facts. This includes incorporating leverage effects (asymmetric volatility responses) into the factor or volatility processes, exploring non-linear dynamics or regime-switching models to better capture crisis periods observed empirically (Figure 5.3), or considering long-memory processes if suggested by data.

3. **Employ Non-Gaussian Error Distributions:** Given the universal failure of Jarque-Bera normality tests on residuals (Table 5.3), exploring non-Gaussian distributions (e.g., Student's t) for the factor innovations ($\boldsymbol{\nu}_t$) or log-volatility innovations ($\boldsymbol{\eta}_t$) within the BIF framework could improve model fit and provide more realistic density forecasts.

4. **Conduct Out-of-Sample Forecasting Analysis:** Extending the empirical analysis to evaluate the out-of-sample forecasting performance of the DFSV-BIF model for covariance matrices and portfolio risk metrics is essential. Comparing its forecasting accuracy against benchmarks like DCC-GARCH would provide crucial insights into its practical utility for risk management and asset allocation.

5. **Investigate Alternative Estimation Strategies (EM Algorithm):** Motivated by the BIF's long empirical estimation runtime when using direct pseudo-likelihood maximization (Section 5.5.5), exploring alternative strategies like the Expectation-Maximization (EM) algorithm could be fruitful. The BIF, with its efficient filter scaling, could potentially be used effectively within the E-step to compute expected sufficient statistics, possibly offering computational advantages for parameter estimation compared to the direct optimization approach used in this thesis.

6. **Apply to Other Asset Classes and Problems:** Testing the generalizability of the DFSV-BIF framework by applying it to different asset classes (e.g., international equities, fixed income, commodities) or different financial problems (e.g., systemic risk measurement, derivative pricing) would further validate its usefulness and potentially reveal new modeling challenges.

6.6 Concluding Remarks

This thesis investigated the application of the Bellman Information Filter to estimate high-dimensional Dynamic Factor Stochastic Volatility models. The research demonstrates that the BIF provides a computationally feasible and numerically stable approach, offering significant advantages in estimating latent volatility states and recovering key model parameters compared to standard particle filters. However, the empirical analysis highlights a crucial limitation: the standard DFSV specification, even when estimated with the advanced BIF, struggles to capture asset-specific volatility dynamics, with strong evidence pointing to the assumption of constant idiosyncratic variance as a major source of this misspecification. Ultimately, this work underscores the ongoing tension between model realism and computational tractability in financial econometrics. While the BIF pushes the boundary of what is computationally feasible for complex factor models, achieving a truly comprehensive description of high-dimensional financial risk likely requires further innovations in both model specification and estimation techniques to bridge the remaining gap between tractable approximations and market reality.

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Appendix A

Mathematical Derivations

This appendix provides detailed mathematical derivations that support the methodology presented in Chapter 2. It includes matrix identities, log-likelihood derivations, optimization algorithms, and Fisher Information Matrix calculations that are essential to the implementation of the Bellman Information Filter.

A.1 Matrix Identities for Efficient Computation

This section provides a comprehensive overview of the matrix identities used throughout the thesis for efficient computation in high-dimensional settings. These identities are fundamental to the implementation of the Bellman Information Filter and are referenced in various parts of the methodology.

A.1.1 Key Matrix Identities

Two key matrix identities are used throughout the implementation of the Bellman Information Filter to enable efficient computation in high-dimensional settings.

The Woodbury Matrix Identity: For matrices $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{U} \in \mathbb{R}^{n\times k}$, $\boldsymbol{C} \in \mathbb{R}^{k\times k}$, and $\boldsymbol{V} \in \mathbb{R}^{k\times n}$, the Woodbury identity states:

$$(\boldsymbol{A} + \boldsymbol{UCV})^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1}\boldsymbol{U}(\boldsymbol{C}^{-1} + \boldsymbol{VA}^{-1}\boldsymbol{U})^{-1}\boldsymbol{VA}^{-1}. \tag{A.1}$$

The Matrix Determinant Lemma: For matrices $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{U}, \boldsymbol{V} \in \mathbb{R}^{n\times k}$, the determinant can be computed as:

$$\det(\boldsymbol{A} + \boldsymbol{UV}^\top) = \det(\boldsymbol{A}) \cdot \det(\boldsymbol{I}_k + \boldsymbol{V}^\top \boldsymbol{A}^{-1}\boldsymbol{U}). \tag{A.2}$$

Both identities reduce computational complexity from $O(n^3)$ to $O(nk^2 + k^3)$ when $k \ll n$, which is a significant improvement for our DFSV models where n represents the number of assets N and k represents the number of factors K, with $K \ll N$.

A.1.2 Application in DFSV Models

In the context of DFSV models, these identities are applied to:

1. Observation Covariance Matrix Inverse: The observation covariance matrix in the DFSV model has the form:

$$\mathbf{\Sigma}_t = \mathbf{\Lambda}\,\mathrm{diag}(e^{\boldsymbol{h}_t})\mathbf{\Lambda}^\top + \mathbf{\Sigma}_\epsilon. \tag{A.3}$$

This can be written in the form $\boldsymbol{A} + \boldsymbol{UCV}$ where:

$$\boldsymbol{A} = \mathbf{\Sigma}_\epsilon \quad \text{(diagonal matrix of idiosyncratic variances)} \tag{A.4}$$
$$\boldsymbol{U} = \mathbf{\Lambda} \quad \text{(factor loading matrix)} \tag{A.5}$$
$$\boldsymbol{C} = \mathrm{diag}(e^{\boldsymbol{h}_t}) \quad \text{(diagonal matrix of factor variances)} \tag{A.6}$$
$$\boldsymbol{V} = \mathbf{\Lambda}^\top \tag{A.7}$$

Applying the Woodbury identity, we get:

$$\mathbf{\Sigma}_t^{-1} = (\mathbf{\Sigma}_\epsilon + \mathbf{\Lambda}\,\mathrm{diag}(e^{\boldsymbol{h}_t})\mathbf{\Lambda}^\top)^{-1} \tag{A.8}$$
$$= \mathbf{\Sigma}_\epsilon^{-1} - \mathbf{\Sigma}_\epsilon^{-1}\mathbf{\Lambda}(\mathrm{diag}(e^{-\boldsymbol{h}_t}) + \mathbf{\Lambda}^\top\mathbf{\Sigma}_\epsilon^{-1}\mathbf{\Lambda})^{-1}\mathbf{\Lambda}^\top\mathbf{\Sigma}_\epsilon^{-1}. \tag{A.9}$$

2. Log-Determinant Calculation: Similarly, for the log-determinant term in the likelihood calculation:

$$\log|\mathbf{\Sigma}_t| = \log|\mathbf{\Sigma}_\epsilon + \mathbf{\Lambda}\,\mathrm{diag}(e^{\boldsymbol{h}_t})\mathbf{\Lambda}^\top| \tag{A.10}$$
$$= \log|\mathbf{\Sigma}_\epsilon| + \log|\boldsymbol{I}_K + \mathrm{diag}(e^{\boldsymbol{h}_t})\mathbf{\Lambda}^\top\mathbf{\Sigma}_\epsilon^{-1}\mathbf{\Lambda}| \tag{A.11}$$
$$= \sum_{i=1}^{N}\log(\sigma_i^2) + \log|\,\mathrm{diag}(e^{\boldsymbol{h}_t})| + \log|\,\mathrm{diag}(e^{-\boldsymbol{h}_t}) + \mathbf{\Lambda}^\top\mathbf{\Sigma}_\epsilon^{-1}\mathbf{\Lambda}|. \tag{A.12}$$

3. Information Matrix Prediction: In the BIF prediction step, the Woodbury identity is used to efficiently compute the predicted information matrix:

$$\mathbf{\Omega}_{t|t-1} = \boldsymbol{Q}_t^{-1} - \boldsymbol{Q}_t^{-1}\boldsymbol{T}\boldsymbol{M}^{-1}\boldsymbol{T}^\top\boldsymbol{Q}_t^{-1}. \tag{A.13}$$

Where $\boldsymbol{M} = \mathbf{\Omega}_{t-1|t-1} + \boldsymbol{T}^\top\boldsymbol{Q}_t^{-1}\boldsymbol{T}$.

4. Fisher Information Matrix Calculation: The Woodbury identity is also used in computing the Fisher Information Matrix for the update step, particularly in the calculation of the blocks $\boldsymbol{J}_{ff}$, $\boldsymbol{J}_{fh}$, and $\boldsymbol{J}_{hh}$. See Section A.4 for a detailed derivation of the Fisher Information Matrix.

A.2 Log-Likelihood Derivation and Efficient Computation

This appendix provides a detailed derivation of the log-likelihood for the DFSV model and the efficient computation methods used to evaluate it. The log-likelihood is a key component in both the Bellman Information Filter and Particle Filter implementations, and its efficient computation is crucial for handling high-dimensional data.

A.2.1 Derivation of the DFSV Model Log-Likelihood Formulations

The log-likelihood of the DFSV model consists of several components, with the observation log-likelihood being particularly important for filtering. In this section, I derive two key formulations of the observation log-likelihood that are used in different filtering approaches: the marginal log-likelihood used in the Bellman Information Filter and the conditional log-likelihood used in the Particle Filter.

Observation Equation and Probability Model

The observation equation of the DFSV model is given by:

$$\boldsymbol{r}_t = \boldsymbol{\Lambda} \boldsymbol{f}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim N\big(\mathbf{0},\ \boldsymbol{\Sigma}_\epsilon\big). \tag{A.14}$$

This equation defines a hierarchical probability model where the distribution of returns depends on both the factors and log-volatilities. We can express this in two different ways, leading to two different log-likelihood formulations.

Conditional Log-Likelihood (Used in Particle Filter)

The first formulation is the conditional log-likelihood of observations given only the factors. This is derived directly from the observation equation by considering the distribution of the idiosyncratic error term $\boldsymbol{\epsilon}_t$:

$$\boldsymbol{r}_t | \boldsymbol{f}_t \sim N(\boldsymbol{\Lambda} \boldsymbol{f}_t, \boldsymbol{\Sigma}_\epsilon) \tag{A.15}$$

$$\log p(\boldsymbol{r}_t | \boldsymbol{f}_t) = \log \left[\frac{1}{(2\pi)^{N/2} |\boldsymbol{\Sigma}_\epsilon|^{1/2}} \exp\left(-\frac{1}{2} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t)' \boldsymbol{\Sigma}_\epsilon^{-1} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t) \right) \right] \tag{A.16}$$

$$= -\frac{N}{2} \log(2\pi) - \frac{1}{2} \log |\boldsymbol{\Sigma}_\epsilon| - \frac{1}{2} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t)' \boldsymbol{\Sigma}_\epsilon^{-1} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t). \tag{A.17}$$

This can be written more compactly as:

$$\log p(\boldsymbol{r}_t | \boldsymbol{f}_t) = -\frac{1}{2} \left[N \log(2\pi) + \log |\boldsymbol{\Sigma}_\epsilon| + (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t)' \boldsymbol{\Sigma}_\epsilon^{-1} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t) \right]. \tag{A.18}$$

This is the conditional log-likelihood defined in Equation 2.6 in the main text. It is used in the Particle Filter to compute importance weights for each particle, as it directly evaluates how well the observed returns match the predicted returns given a specific realization of the factors.
The key characteristic of this formulation is that it uses only the idiosyncratic covariance matrix $\boldsymbol{\Sigma}_\epsilon$, which is diagonal and constant over time. This makes it computationally efficient to evaluate for each particle in the filter.

Marginal Log-Likelihood (Used in Bellman Information Filter)

The second formulation is the marginal log-likelihood of observations given the complete state vector $\boldsymbol{\alpha}_t = [\boldsymbol{f}_t', \boldsymbol{h}_t']'$. This considers the joint distribution of returns and factors, marginalizing over the uncertainty in the factors (conditional on the log-volatilities).

To derive this, we first note that the factors $\boldsymbol{f}_t$ have a distribution conditional on the log-volatilities $\boldsymbol{h}_t$:

$$\boldsymbol{f}_t | \boldsymbol{h}_t \sim N(\hat{\boldsymbol{f}}_{t|t-1}, \mathrm{diag}(e^{\boldsymbol{h}_t})). \tag{A.19}$$

Where $\hat{\boldsymbol{f}}_{t|t-1}$ is the predicted factor mean from the previous time step. Combining this with the observation equation, we get the marginal distribution of returns conditional on log-volatilities:

$$\begin{aligned}
\boldsymbol{r}_t | \boldsymbol{h}_t &\sim N(\boldsymbol{\Lambda} \hat{\boldsymbol{f}}_{t|t-1}, \boldsymbol{\Lambda}\, \mathrm{diag}(e^{\boldsymbol{h}_t}) \boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_\epsilon) && \text{(A.20)} \\
&\sim N(\boldsymbol{\Lambda} \hat{\boldsymbol{f}}_{t|t-1}, \boldsymbol{\Sigma}_t). && \text{(A.21)}
\end{aligned}$$

Where $\boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}\, \mathrm{diag}(e^{\boldsymbol{h}_t}) \boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_\epsilon$ is the marginal observation covariance matrix that accounts for both factor uncertainty and idiosyncratic noise.

The log-likelihood of this multivariate normal distribution is:

$$\begin{aligned}
\log p(\boldsymbol{r}_t | \boldsymbol{\alpha}_t) &= \log \left[\frac{1}{(2\pi)^{N/2} |\boldsymbol{\Sigma}_t|^{1/2}} \exp\left(-\frac{1}{2} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t)' \boldsymbol{\Sigma}_t^{-1} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t) \right) \right] && \text{(A.22)} \\
&= -\frac{N}{2} \log(2\pi) - \frac{1}{2} \log |\boldsymbol{\Sigma}_t| - \frac{1}{2} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t)' \boldsymbol{\Sigma}_t^{-1} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t). && \text{(A.23)}
\end{aligned}$$

This can be written more compactly as:

$$\ell(\boldsymbol{r}_t | \boldsymbol{\alpha}_t) = -\frac{1}{2} \left[N \log(2\pi) + \log |\boldsymbol{\Sigma}_t| + (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t)' \boldsymbol{\Sigma}_t^{-1} (\boldsymbol{r}_t - \boldsymbol{\Lambda} \boldsymbol{f}_t) \right]. \tag{A.24}$$

This is the marginal log-likelihood defined in Equation 2.7 in the main text. It is used in the Bellman Information Filter's objective function and its pseudo-likelihood calculation.

Comparison of the Two Formulations

The key difference between these two formulations lies in the covariance matrix used:

- **Conditional Log-Likelihood:** Uses only $\boldsymbol{\Sigma}_\epsilon$ (idiosyncratic covariance)
- **Marginal Log-Likelihood:** Uses $\boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}\, \mathrm{diag}(e^{\boldsymbol{h}_t}) \boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_\epsilon$ (marginal covariance)

The marginal formulation accounts for both the uncertainty in the factors (through the term $\boldsymbol{\Lambda}\, \mathrm{diag}(e^{\boldsymbol{h}_t}) \boldsymbol{\Lambda}'$) and the idiosyncratic noise (through $\boldsymbol{\Sigma}_\epsilon$). This makes it more appropriate for the Bellman filter, which aims to find the posterior mode of the complete state vector $\boldsymbol{\alpha}_t$ by maximizing the log-posterior density.

The conditional formulation, on the other hand, is more appropriate for the Particle Filter, which represents the full posterior distribution through a set of weighted particles. Each particle already represents a specific realization of the state, so the conditional likelihood is used to evaluate how well each particle explains the observed returns.

Both formulations play crucial roles in their respective filtering approaches and contribute to the overall parameter estimation process.

Factor Transition Log-Likelihood

The factor transition equation is:

$$\boldsymbol{f}_{t+1} = \boldsymbol{\Phi}_f\, \boldsymbol{f}_t + \boldsymbol{\nu}_{t+1}, \quad \boldsymbol{\nu}_{t+1} \sim N\Big(\boldsymbol{0},\ \mathrm{diag}\Big(e^{h_{1,t+1}}, \ldots, e^{h_{K,t+1}}\Big)\Big). \tag{A.25}$$

The conditional distribution of $\boldsymbol{f}_{t+1}$ given $\boldsymbol{f}_t$ and $\boldsymbol{h}_{t+1}$ is:

$$\boldsymbol{f}_{t+1}|\boldsymbol{f}_t, \boldsymbol{h}_{t+1} \sim N(\boldsymbol{\Phi}_f \boldsymbol{f}_t, \mathrm{diag}(e^{\boldsymbol{h}_{t+1}})). \tag{A.26}$$

The log-likelihood of this distribution is:

$$\log p(\boldsymbol{f}_{t+1}|\boldsymbol{f}_t, \boldsymbol{h}_{t+1}) = -\frac{K}{2}\log(2\pi) - \frac{1}{2}\log|\,\mathrm{diag}(e^{\boldsymbol{h}_{t+1}})| \tag{A.27}$$

$$- \frac{1}{2}(\boldsymbol{f}_{t+1} - \boldsymbol{\Phi}_f \boldsymbol{f}_t)'\,\mathrm{diag}(e^{-\boldsymbol{h}_{t+1}})(\boldsymbol{f}_{t+1} - \boldsymbol{\Phi}_f \boldsymbol{f}_t). \tag{A.28}$$

Since $\log|\,\mathrm{diag}(e^{\boldsymbol{h}_{t+1}})| = \sum_{k=1}^{K} h_{k,t+1}$ and the quadratic form with a diagonal matrix can be simplified, we get:

$$\log p(\boldsymbol{f}_{t+1}|\boldsymbol{f}_t, \boldsymbol{h}_{t+1}) = -\frac{K}{2}\log(2\pi) - \frac{1}{2}\sum_{k=1}^{K} h_{k,t+1} \tag{A.29}$$

$$- \frac{1}{2}\sum_{k=1}^{K} e^{-h_{k,t+1}}(f_{k,t+1} - [\boldsymbol{\Phi}_f \boldsymbol{f}_t]_k)^2. \tag{A.30}$$

This formulation avoids the numerical issues that can arise from computing $\log(\exp(\cdot))$ directly.

Log-Volatility Transition Log-Likelihood

The log-volatility transition equation is:

$$\boldsymbol{h}_{t+1} = \boldsymbol{\mu} + \boldsymbol{\Phi}_h\,(\boldsymbol{h}_t - \boldsymbol{\mu}) + \boldsymbol{\eta}_{t+1}, \quad \boldsymbol{\eta}_{t+1} \sim N(\boldsymbol{0},\, \boldsymbol{Q}_h)\,. \tag{A.31}$$

The conditional distribution of $\boldsymbol{h}_{t+1}$ given $\boldsymbol{h}_t$ is:

$$\boldsymbol{h}_{t+1}|\boldsymbol{h}_t \sim N(\boldsymbol{\mu} + \boldsymbol{\Phi}_h(\boldsymbol{h}_t - \boldsymbol{\mu}), \boldsymbol{Q}_h). \tag{A.32}$$

The log-likelihood of this distribution is:

$$\log p(\boldsymbol{h}_{t+1}|\boldsymbol{h}_t) = -\frac{K}{2}\log(2\pi) - \frac{1}{2}\log|\boldsymbol{Q}_h| \tag{A.33}$$

$$- \frac{1}{2}(\boldsymbol{h}_{t+1} - \boldsymbol{\mu} - \boldsymbol{\Phi}_h(\boldsymbol{h}_t - \boldsymbol{\mu}))'\boldsymbol{Q}_h^{-1}(\boldsymbol{h}_{t+1} - \boldsymbol{\mu} - \boldsymbol{\Phi}_h(\boldsymbol{h}_t - \boldsymbol{\mu})). \tag{A.34}$$

Joint Log-Likelihood

The joint log-likelihood of the entire observation sequence $\{\boldsymbol{r}_t\}_{t=1}^T$ and state sequence $\{\boldsymbol{\alpha}_t\}_{t=1}^T$ is the sum of the individual components:

$$\log p(\{\boldsymbol{r}_t\}_{t=1}^T, \{\boldsymbol{\alpha}_t\}_{t=1}^T) = \log p(\boldsymbol{\alpha}_1) + \sum_{t=1}^{T} \log p(\boldsymbol{r}_t|\boldsymbol{\alpha}_t) \tag{A.35}$$

$$+ \sum_{t=1}^{T-1} \log p(\boldsymbol{f}_{t+1}|\boldsymbol{f}_t, \boldsymbol{h}_{t+1}) + \sum_{t=1}^{T-1} \log p(\boldsymbol{h}_{t+1}|\boldsymbol{h}_t). \tag{A.36}$$

Where $\log p(\boldsymbol{\alpha}_1)$ is the log-likelihood of the initial state, typically assumed to be a diffuse prior or specified based on stationarity properties of the model.
In the context of filtering, we are often interested in the marginal log-likelihood of the observations, which requires integrating out the states:

$$\log p(\{\boldsymbol{r}_t\}_{t=1}^T) = \log \int p(\{\boldsymbol{r}_t\}_{t=1}^T, \{\boldsymbol{\alpha}_t\}_{t=1}^T) d\{\boldsymbol{\alpha}_t\}_{t=1}^T. \tag{A.37}$$

This integration is analytically intractable for the DFSV model due to the non-linear relationship between the log-volatilities and the observation covariance matrix. Both the Bellman Information Filter and Particle Filter provide approximations to this marginal likelihood.

A.2.2 Efficient Computation via Matrix Identities

Direct computation of the observation log-likelihood (Equation A.24) is challenging due to the high-dimensional matrix operations involved, particularly when $N \gg K$. The computational complexity would be $O(N^3)$ for the determinant and matrix inversion operations.
To address this, we use the matrix identities described in Section A.1. Specifically, we apply the Matrix Determinant Lemma (Equation A.2) and the Woodbury Matrix Identity (Equation A.1) to efficiently compute the log-likelihood.

Efficient Computation of $\log|\boldsymbol{\Sigma}_t|$

Using the Matrix Determinant Lemma, we can compute the log-determinant term as:

$$\log|\boldsymbol{\Sigma}_t| = \log|\boldsymbol{\Sigma}_\epsilon + \boldsymbol{\Lambda}\,\mathrm{diag}(e^{\boldsymbol{h}_t})\boldsymbol{\Lambda}'| \tag{A.38}$$

$$= \log|\boldsymbol{\Sigma}_\epsilon| + \log|\mathrm{diag}(e^{\boldsymbol{h}_t})| + \log|\mathrm{diag}(e^{-\boldsymbol{h}_t}) + \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_\epsilon^{-1}\boldsymbol{\Lambda}| \tag{A.39}$$

$$= \sum_{i=1}^{N} \log(\sigma_i^2) + \sum_{k=1}^{K} h_{k,t} + \log|\mathrm{diag}(e^{-\boldsymbol{h}_t}) + \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_\epsilon^{-1}\boldsymbol{\Lambda}|. \tag{A.40}$$

This reduces the computational complexity from $O(N^3)$ to $O(NK^2 + K^3)$ since we only need to compute the determinant of a $K \times K$ matrix instead of an $N \times N$ matrix.

Efficient Computation of the Quadratic Form

Using the Woodbury Matrix Identity, we can compute the inverse of the observation covariance matrix as:

$$\begin{aligned}\boldsymbol{\Sigma}_t^{-1} &= (\boldsymbol{\Sigma}_\epsilon + \boldsymbol{\Lambda}\,\mathrm{diag}(e^{\boldsymbol{h}_t})\boldsymbol{\Lambda}')^{-1} &\text{(A.41)}\\ &= \boldsymbol{\Sigma}_\epsilon^{-1} - \boldsymbol{\Sigma}_\epsilon^{-1}\boldsymbol{\Lambda}(\mathrm{diag}(e^{-\boldsymbol{h}_t}) + \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_\epsilon^{-1}\boldsymbol{\Lambda})^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Sigma}_\epsilon^{-1}. &\text{(A.42)}\end{aligned}$$

This allows us to compute the quadratic form in the log-likelihood without explicitly forming the $N \times N$ inverse matrix:

$$\begin{aligned}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_t^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t) &= (\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_\epsilon^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t) &\text{(A.43)}\\ &\quad - (\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_\epsilon^{-1}\boldsymbol{\Lambda}(\mathrm{diag}(e^{-\boldsymbol{h}_t}) + \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_\epsilon^{-1}\boldsymbol{\Lambda})^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Sigma}_\epsilon^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t). &\text{(A.44)}\end{aligned}$$

Again, this reduces the computational complexity from $O(N^3)$ to $O(NK^2 + K^3)$.

A.2.3 Efficient Implementation

By applying these matrix identities, the computational complexity of key operations in the BIF is reduced from $O(N^3)$ to $O(NK^2 + K^3)$, which is a significant improvement when $N \gg K$. The implementation includes numerical stability enhancements such as Cholesky decomposition for matrix inversions and small jitter terms to prevent numerical issues.

The JAX implementation of these computations leverages automatic differentiation and just-in-time compilation for further efficiency gains. The log-likelihood calculation is a core component of both the filtering algorithms and the parameter estimation procedures, making these optimizations crucial for the practical application of DFSV models to high-dimensional financial data.

A.3 Block Coordinate Descent for Bellman Filter State Update

This section provides a detailed mathematical derivation of the Block Coordinate Descent (BCD) algorithm used in the Bellman Information Filter to find the posterior mode. While the main text in Section 2.2.2 outlines the general approach and algorithm, here we focus on the mathematical details and computational considerations that make BCD particularly efficient for the DFSV model.

The Bellman filter update step aims to find the posterior mode $\hat{\boldsymbol{\alpha}}_{t|t}$ by maximizing the log posterior density. This is equivalent to maximizing the following objective function (as defined in Equation 2.17 in the main text):

$$J(\boldsymbol{\alpha}_t) = \ell(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) - \frac{1}{2}\left(\boldsymbol{\alpha}_t - \hat{\boldsymbol{\alpha}}_{t|t-1}\right)'\boldsymbol{\Omega}_{t|t-1}\left(\boldsymbol{\alpha}_t - \hat{\boldsymbol{\alpha}}_{t|t-1}\right). \tag{A.45}$$

Here, $\boldsymbol{\alpha}_t = [\boldsymbol{f}_t', \boldsymbol{h}_t']'$ is the state vector containing both factors $\boldsymbol{f}_t$ and log-volatilities $\boldsymbol{h}_t$, $\boldsymbol{\Omega}_{t|t-1}$ is the predicted information matrix, and $\ell(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t)$ is the observation log-likelihood. Due to the non-linear relationship between $\boldsymbol{h}_t$ and the observation covariance matrix $\boldsymbol{\Sigma}_t$, direct optimization of this objective function with respect to the full state vector $\boldsymbol{\alpha}_t$ is challenging. The BCD approach addresses this by alternating between optimizing the factors $\boldsymbol{f}_t$ and log-volatilities $\boldsymbol{h}_t$, which leads to more tractable subproblems.

A.3.1 Factor Update Step: Closed-Form Solution

The key advantage of the BCD approach for DFSV models is that the factor update step has a closed-form solution. When maximizing the Bellman objective function $J(\boldsymbol{\alpha}_t)$ from Equation 2.17 with respect to $\boldsymbol{f}_t$ while keeping $\boldsymbol{h}_t = \boldsymbol{h}^{(k)}$ fixed, we can derive a linear system by setting the gradient to zero:

$$\frac{\partial J(\boldsymbol{\alpha}_t)}{\partial \boldsymbol{f}_t} = \frac{\partial \ell(\boldsymbol{r}_t \mid \boldsymbol{f}_t, \boldsymbol{h}^{(k)})}{\partial \boldsymbol{f}_t} - \boldsymbol{\Omega}_{ff}(\boldsymbol{f}_t - \boldsymbol{f}_{t|t-1}) - \boldsymbol{\Omega}_{fh}(\boldsymbol{h}^{(k)} - \boldsymbol{h}_{t|t-1}) = \mathbf{0}. \tag{A.46}$$

With fixed $\boldsymbol{h}^{(k)}$, the observation covariance matrix $\boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}\ \mathrm{diag}(e^{\boldsymbol{h}^{(k)}})\,\boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_\epsilon$ is constant with respect to $\boldsymbol{f}_t$. The log-likelihood gradient simplifies to $\frac{\partial \ell(\boldsymbol{r}_t|\boldsymbol{f}_t,\boldsymbol{h}^{(k)})}{\partial \boldsymbol{f}_t} = \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{r}_t - \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Lambda}\boldsymbol{f}_t$. Substituting and rearranging terms leads to the linear system:

$$(\boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Lambda} + \boldsymbol{\Omega}_{ff})\boldsymbol{f}_t = \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{r}_t + \boldsymbol{\Omega}_{ff}\boldsymbol{f}_{t|t-1} - \boldsymbol{\Omega}_{fh}(\boldsymbol{h}^{(k)} - \boldsymbol{h}_{t|t-1}). \tag{A.47}$$

The solution gives the updated factors:

$$\boldsymbol{f}^{(k+1)} = (\boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Lambda} + \boldsymbol{\Omega}_{ff})^{-1}(\boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{r}_t + \boldsymbol{\Omega}_{ff}\boldsymbol{f}_{t|t-1} - \boldsymbol{\Omega}_{fh}(\boldsymbol{h}^{(k)} - \boldsymbol{h}_{t|t-1})). \tag{A.48}$$

This is the same update equation presented in the main text, but here we've provided the detailed derivation showing how it arises from the gradient of the objective function.

A.3.2 Log-Volatility Update Step: Numerical Optimization

Unlike the factor update, the log-volatility update step requires numerical optimization due to the non-linear relationship between $\boldsymbol{h}_t$ and the observation covariance matrix. This section details the mathematical challenges and solutions for this step.

Non-linearity Challenge: The key challenge in the log-volatility update is that the observation covariance matrix $\boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}\ \mathrm{diag}(e^{\boldsymbol{h}_t})\,\boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_\epsilon$ depends exponentially on $\boldsymbol{h}_t$. This creates non-linearities in both the log-determinant term and the quadratic form in the log-likelihood (Equation 2.7), where the factors are fixed at $\boldsymbol{f}^{(k+1)}$.
When optimizing the objective function $J(\boldsymbol{\alpha}_t)$ from Equation 2.17 with respect to $\boldsymbol{h}_t$ while keeping $\boldsymbol{f}_t = \boldsymbol{f}^{(k+1)}$ fixed, the optimization problem becomes:

$$
\begin{aligned}
\boldsymbol{h}^{(k+1)} = \arg\max_{\boldsymbol{h}_t} \Bigg\{ & \ell(\boldsymbol{r}_t \mid \boldsymbol{f}^{(k+1)}, \boldsymbol{h}_t) && \text{(A.49)} \\
& - \frac{1}{2}(\boldsymbol{f}^{(k+1)} - \hat{\boldsymbol{f}}_{t|t-1})' \boldsymbol{\Omega}_{ff} (\boldsymbol{f}^{(k+1)} - \hat{\boldsymbol{f}}_{t|t-1}) && \text{(A.50)} \\
& - (\boldsymbol{f}^{(k+1)} - \hat{\boldsymbol{f}}_{t|t-1})' \boldsymbol{\Omega}_{fh} (\boldsymbol{h}_t - \hat{\boldsymbol{h}}_{t|t-1}) && \text{(A.51)} \\
& - \frac{1}{2}(\boldsymbol{h}_t - \hat{\boldsymbol{h}}_{t|t-1})' \boldsymbol{\Omega}_{hh} (\boldsymbol{h}_t - \hat{\boldsymbol{h}}_{t|t-1}) \Bigg\}. && \text{(A.52)}
\end{aligned}
$$

Since $\boldsymbol{f}^{(k+1)}$ is fixed in this step, the term involving $\boldsymbol{\Omega}_{ff}$ is constant and can be omitted from the optimization. The problem then simplifies to:

$$
\begin{aligned}
\boldsymbol{h}^{(k+1)} = \arg\max_{\boldsymbol{h}_t} \Bigg\{ & \ell(\boldsymbol{r}_t \mid \boldsymbol{f}^{(k+1)}, \boldsymbol{h}_t) && \text{(A.53)} \\
& - (\boldsymbol{f}^{(k+1)} - \hat{\boldsymbol{f}}_{t|t-1})' \boldsymbol{\Omega}_{fh} (\boldsymbol{h}_t - \hat{\boldsymbol{h}}_{t|t-1}) && \text{(A.54)} \\
& - \frac{1}{2}(\boldsymbol{h}_t - \hat{\boldsymbol{h}}_{t|t-1})' \boldsymbol{\Omega}_{hh} (\boldsymbol{h}_t - \hat{\boldsymbol{h}}_{t|t-1}) \Bigg\}. && \text{(A.55)}
\end{aligned}
$$

Gradient-Based Optimization and Implementation: To solve this non-linear optimization problem, we use a trust-region variant of the BFGS algorithm to maximize the objective function. The implementation leverages JAX's automatic differentiation capabilities rather than manually computing derivatives. This approach defines the objective function for the log-volatility update, and JAX automatically computes the gradients needed for optimization, improving accuracy and maintainability while simplifying the codebase.

A.3.3 Computational Efficiency Considerations

The BCD algorithm alternates between the factor update and log-volatility update steps until convergence (typically within 10 iterations). This approach offers significant computational advantages over direct optimization of the full state vector $\boldsymbol{\alpha}_t$ for several reasons:

First, by optimizing over K dimensions at a time rather than $2K$ dimensions simultaneously, each sub-problem becomes more tractable. The factor update step yields a closed-form solution, avoiding expensive iterative optimization for half of the state vector. This reduces computational complexity from $O(N^3)$ to $O(NK^2 + K^3)$ for high-dimensional problems with many assets ($N \gg K$).

Second, several numerical stability techniques are employed, including Cholesky decomposition for solving linear systems, small jitter terms to ensure positive definiteness of matrices, and the Woodbury identity to efficiently compute $\boldsymbol{\Sigma}_t^{-1}$ without directly inverting high-dimensional matrices.

For the log-volatility optimization specifically, a trust-region approach constrains step sizes to maintain validity of the quadratic approximation in the BFGS algorithm, with fallback strategies ensuring robustness if optimization fails to converge.

These efficiency considerations make the BCD approach particularly well-suited for DFSV models in high-dimensional settings, allowing the Bellman Information Filter to efficiently handle problems that would be computationally prohibitive with direct optimization methods.

A.4 Fisher Information Matrix in DFSV Models

This section provides a detailed explanation of the Fisher Information Matrix (FIM) as used in the Bellman Information Filter for DFSV models. The FIM plays a crucial role in the BIF update step, where it represents the curvature of the log-likelihood function at the posterior mode.

A.4.1 Definition and Role in Bellman Filtering

In the context of the Bellman Information Filter, the Fisher Information Matrix is used to update the information matrix in the filtering recursion:

$$\boldsymbol{\Omega}_{t|t} = \boldsymbol{\Omega}_{t|t-1} + \boldsymbol{\mathcal{I}}_t. \tag{A.56}$$

Where $\boldsymbol{\Omega}_{t|t}$ is the posterior information matrix, $\boldsymbol{\Omega}_{t|t-1}$ is the predicted information matrix, and $\boldsymbol{\mathcal{I}}_t$ is the Fisher Information Matrix evaluated at the posterior mode $\hat{\boldsymbol{\alpha}}_{t|t}$.

A.4.2 Observed vs. Expected Fisher Information

There are two main formulations of the Fisher Information Matrix:

1. Observed Fisher Information Matrix (O-FIM): The Observed Fisher Information Matrix is defined as the negative Hessian of the log-likelihood function evaluated at the posterior mode:

$$\boldsymbol{\mathcal{I}}_t^{\text{obs}} = -\nabla^2 \log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t)|_{\boldsymbol{\alpha}_t = \hat{\boldsymbol{\alpha}}_{t|t}}. \tag{A.57}$$

This matrix captures the local curvature of the log-likelihood function at the posterior mode.

2. Expected Fisher Information Matrix (E-FIM): The Expected Fisher Information Matrix is defined as the expectation of the negative Hessian of the log-likelihood function:

$$\boldsymbol{\mathcal{I}}_t^{\text{exp}} = \mathbb{E}[-\nabla^2 \log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t)]|_{\boldsymbol{\alpha}_t = \hat{\boldsymbol{\alpha}}_{t|t}}. \tag{A.58}$$

The expectation is taken with respect to the distribution of the observations $\boldsymbol{r}_t$ given the state $\boldsymbol{\alpha}_t$.

Choice in BIF Implementation: The BIF implementation uses the E-FIM rather than the O-FIM. This choice is motivated by several considerations:

- The E-FIM is guaranteed to be positive semi-definite (PSD), which ensures numerical stability in the information matrix update.

- The O-FIM may not be PSD in practice due to numerical issues or model misspecification, requiring eigenvalue clipping or other regularization techniques.
- The E-FIM often has a simpler analytical form that can be computed more efficiently.

A.4.3 Block Structure of the Fisher Information Matrix

For the DFSV model, the Fisher Information Matrix has a block structure corresponding to the partitioning of the state vector $\boldsymbol{\alpha}_t = [\boldsymbol{f}_t', \boldsymbol{h}_t']'$:

$$\boldsymbol{\mathcal{I}}_t = \begin{bmatrix} \boldsymbol{\mathcal{I}}_{ff} & \boldsymbol{\mathcal{I}}_{fh} \\ \boldsymbol{\mathcal{I}}_{fh}' & \boldsymbol{\mathcal{I}}_{hh} \end{bmatrix}. \tag{A.59}$$

Where:

- $\boldsymbol{\mathcal{I}}_{ff}$ represents the curvature with respect to the factors $\boldsymbol{f}_t$
- $\boldsymbol{\mathcal{I}}_{hh}$ represents the curvature with respect to the log-volatilities $\boldsymbol{h}_t$
- $\boldsymbol{\mathcal{I}}_{fh}$ represents the cross-derivatives between factors and log-volatilities

A.4.4 Derivation of the E-FIM

For the DFSV model with observation equation $\boldsymbol{r}_t = \boldsymbol{\Lambda}\boldsymbol{f}_t + \boldsymbol{\epsilon}_t$, where $\boldsymbol{\epsilon}_t \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_\epsilon)$, we need to compute the E-FIM for the state vector $\boldsymbol{\alpha}_t = [\boldsymbol{f}_t', \boldsymbol{h}_t']'$. This derivation follows the standard approach for Gaussian models where one set of parameters affects only the mean and another set affects only the covariance.

The Log-Likelihood Function

The log-likelihood for the observation model $\boldsymbol{r}_t \sim N(\boldsymbol{\Lambda}\boldsymbol{f}_t, \boldsymbol{\Sigma}_t)$ is given by:

$$\log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) = -\frac{1}{2}\left[\log|\boldsymbol{\Sigma}_t| + (\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_t^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)\right]. \tag{A.60}$$

Where $\boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}\,\mathrm{diag}(e^{\boldsymbol{h}_t})\boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_\epsilon$ is the marginal observation covariance matrix.

Derivatives with respect to $\boldsymbol{f}_t$

The factors $\boldsymbol{f}_t$ affect only the mean of the observation distribution, not the covariance. The first derivative of the log-likelihood with respect to $\boldsymbol{f}_t$ is:

$$\frac{\partial}{\partial \boldsymbol{f}_t}\log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) = \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t). \tag{A.61}$$

Taking the second derivative:

$$\frac{\partial^2}{\partial \boldsymbol{f}_t \partial \boldsymbol{f}_t'}\log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) = -\boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Lambda}. \tag{A.62}$$

The E-FIM is the negative expectation of this Hessian:

$$\mathcal{I}_{ff} = \mathbb{E}\left[-\frac{\partial^2}{\partial \boldsymbol{f}_t \partial \boldsymbol{f}_t'} \log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t)\right] = \boldsymbol{\Lambda}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Lambda}. \tag{A.63}$$

Derivatives with respect to $\boldsymbol{h}_t$

The log-volatilities $\boldsymbol{h}_t$ affect only the covariance of the observation distribution, making their derivatives more complex than those for the factors. For the (i,j)-th element of the $\mathcal{I}_{hh}$ block, we have:

$$[\mathcal{I}_{hh}]_{i,j} = \mathbb{E}\left[-\frac{\partial^2}{\partial h_{i,t}\partial h_{j,t}} \log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t)\right]. \tag{A.64}$$

To compute this, we need to differentiate both the log-determinant term and the quadratic form in the log-likelihood. Let's start with the first derivatives with respect to $h_{i,t}$:

$$\frac{\partial}{\partial h_{i,t}} \log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) = -\frac{1}{2}\frac{\partial}{\partial h_{i,t}} \log|\boldsymbol{\Sigma}_t| - \frac{1}{2}\frac{\partial}{\partial h_{i,t}}\left[(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_t^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)\right] \tag{A.65}$$

$$= -\frac{1}{2}\mathrm{tr}\left(\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\right) + \frac{1}{2}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\boldsymbol{\Sigma}_t^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t). \tag{A.66}$$

Where we've used the matrix calculus identities:

$$\frac{\partial}{\partial h_{i,t}} \log|\boldsymbol{\Sigma}_t| = \mathrm{tr}\left(\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\right) \tag{A.67}$$

$$\frac{\partial}{\partial h_{i,t}}\left[\boldsymbol{x}'\boldsymbol{\Sigma}_t^{-1}\boldsymbol{x}\right] = -\boldsymbol{x}'\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\boldsymbol{\Sigma}_t^{-1}\boldsymbol{x}. \tag{A.68}$$

The partial derivatives of the covariance matrix with respect to log-volatilities are:

$$\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}} = \boldsymbol{\Lambda}\,\mathrm{diag}(0,\ldots,e^{h_{i,t}},\ldots,0)\boldsymbol{\Lambda}'. \tag{A.69}$$

Where the diagonal matrix has $e^{h_{i,t}}$ in the i-th position and zeros elsewhere. This reflects how a change in log-volatility $h_{i,t}$ affects only the variance of the i-th factor.
Now, taking the second derivative with respect to $h_{j,t}$:

$$\frac{\partial^2}{\partial h_{i,t}\partial h_{j,t}} \log p(\boldsymbol{r}_t \mid \boldsymbol{\alpha}_t) = -\frac{1}{2}\frac{\partial}{\partial h_{j,t}}\mathrm{tr}\left(\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\right) \tag{A.70}$$

$$+\frac{1}{2}\frac{\partial}{\partial h_{j,t}}\left[(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\boldsymbol{\Sigma}_t^{-1}(\boldsymbol{r}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)\right]. \tag{A.71}$$

The first term involves differentiating the trace expression, which yields:

$$\frac{\partial}{\partial h_{j,t}}\mathrm{tr}\left(\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\right) = \mathrm{tr}\left(\frac{\partial \boldsymbol{\Sigma}_t^{-1}}{\partial h_{j,t}}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\right) + \mathrm{tr}\left(\boldsymbol{\Sigma}_t^{-1}\frac{\partial^2 \boldsymbol{\Sigma}_t}{\partial h_{j,t}\partial h_{i,t}}\right) \tag{A.72}$$

$$= -\mathrm{tr}\left(\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{j,t}}\boldsymbol{\Sigma}_t^{-1}\frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}}\right) + \mathrm{tr}\left(\boldsymbol{\Sigma}_t^{-1}\frac{\partial^2 \boldsymbol{\Sigma}_t}{\partial h_{j,t}\partial h_{i,t}}\right). \tag{A.73}$$

Where we've used the identity $\frac{\partial \boldsymbol{\Sigma}_t^{-1}}{\partial h_{j,t}} = -\boldsymbol{\Sigma}_t^{-1} \frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{j,t}} \boldsymbol{\Sigma}_t^{-1}$.
The second derivative of the covariance matrix is:

$$\frac{\partial^2 \boldsymbol{\Sigma}_t}{\partial h_{j,t} \partial h_{i,t}} = \begin{cases} \boldsymbol{\Lambda} \, \mathrm{diag}(0, \ldots, e^{h_{i,t}}, \ldots, 0) \boldsymbol{\Lambda}' & \text{if } i = j \\ \mathbf{0} & \text{if } i \neq j \end{cases}. \tag{A.74}$$

The second term in the Hessian involves differentiating a quadratic form, which is more complex. After taking the expectation, the terms involving $(\boldsymbol{r}_t - \boldsymbol{\Lambda f}_t)$ vanish because $\mathbb{E}[\boldsymbol{r}_t - \boldsymbol{\Lambda f}_t] = \mathbf{0}$. The final result for the E-FIM is:

$$[\boldsymbol{\mathcal{I}}_{hh}]_{i,j} = \frac{1}{2} \mathrm{tr} \left(\boldsymbol{\Sigma}_t^{-1} \frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{i,t}} \boldsymbol{\Sigma}_t^{-1} \frac{\partial \boldsymbol{\Sigma}_t}{\partial h_{j,t}} \right). \tag{A.75}$$

Substituting the expressions for the partial derivatives and working through the matrix algebra:

$$[\boldsymbol{\mathcal{I}}_{hh}]_{i,j} = \frac{1}{2} \mathrm{tr} \left(\boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda} \, \mathrm{diag}(0, \ldots, e^{h_{i,t}}, \ldots, 0) \boldsymbol{\Lambda}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda} \, \mathrm{diag}(0, \ldots, e^{h_{j,t}}, \ldots, 0) \boldsymbol{\Lambda}' \right) \tag{A.76}$$

$$= \frac{1}{2} e^{h_{i,t}} e^{h_{j,t}} \mathrm{tr} \left(\boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda} \boldsymbol{e}_i \boldsymbol{e}_i' \boldsymbol{\Lambda}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda} \boldsymbol{e}_j \boldsymbol{e}_j' \boldsymbol{\Lambda}' \right). \tag{A.77}$$

Where $\boldsymbol{e}_i$ is the i-th standard basis vector (a vector with 1 in the i-th position and 0 elsewhere). Using properties of the trace and matrix multiplication, we can further simplify this expression:

$$[\boldsymbol{\mathcal{I}}_{hh}]_{i,j} = \frac{1}{2} e^{h_{i,t}} e^{h_{j,t}} \mathrm{tr} \left((\boldsymbol{e}_i' \boldsymbol{\Lambda}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda} \boldsymbol{e}_j)(\boldsymbol{e}_j' \boldsymbol{\Lambda}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda} \boldsymbol{e}_i) \right) \tag{A.78}$$

$$= \frac{1}{2} e^{h_{i,t} + h_{j,t}} [\boldsymbol{\Lambda}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda}]_{i,j}^2. \tag{A.79}$$

Since $\boldsymbol{\mathcal{I}}_{ff} = \boldsymbol{\Lambda}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Lambda}$, we can express this more compactly as:

$$[\boldsymbol{\mathcal{I}}_{hh}]_{i,j} = \frac{1}{2} e^{h_{i,t} + h_{j,t}} [\boldsymbol{\mathcal{I}}_{ff}]_{i,j}^2. \tag{A.80}$$

For $i, j = 1, \ldots, K$. This elegant result reveals a direct relationship between the FIM for log-volatilities and the FIM for factors, showing how the exponential nature of the volatility parameterization affects the curvature of the log-likelihood.

Cross-Derivatives

The cross-derivatives between $\boldsymbol{f}_t$ and $\boldsymbol{h}_t$ involve mixing derivatives of the mean and covariance terms. Since $\boldsymbol{f}_t$ affects only the mean and $\boldsymbol{h}_t$ affects only the covariance, these cross-terms involve expressions containing $(\boldsymbol{r}_t - \boldsymbol{\Lambda f}_t)$. When taking the expectation, since $\mathbb{E}[\boldsymbol{r}_t - \boldsymbol{\Lambda f}_t] = \mathbf{0}$, these terms vanish:

$$\boldsymbol{\mathcal{I}}_{fh} = \boldsymbol{\mathcal{I}}_{hf} = \mathbf{0}. \tag{A.81}$$

Block-Diagonal Structure

Putting these results together, the E-FIM has a block-diagonal structure:

$$\mathcal{I}_t = \begin{bmatrix} \mathcal{I}_{ff} & \mathbf{0} \\ \mathbf{0} & \mathcal{I}_{hh} \end{bmatrix} = \begin{bmatrix} \mathbf{\Lambda}'\mathbf{\Sigma}_t^{-1}\mathbf{\Lambda} & \mathbf{0} \\ \mathbf{0} & \frac{1}{2}\left[e^{h_{i,t}+h_{j,t}}[\mathcal{I}_{ff}]_{i,j}^2\right]_{i,j} \end{bmatrix}. \tag{A.82}$$

This block-diagonal structure is a direct consequence of the fact that in the DFSV model, the factors affect only the mean of the observation distribution, while the log-volatilities affect only the covariance. The elegant relationship between $\mathcal{I}_{hh}$ and $\mathcal{I}_{ff}$ allows for efficient computation, as $\mathcal{I}_{hh}$ can be derived directly from $\mathcal{I}_{ff}$ once it has been calculated for the factor update.

A.4.5 Computational Efficiency Note

The E-FIM can be computed efficiently by leveraging the Woodbury identity to avoid direct inversion of the $N \times N$ observation covariance matrix $\mathbf{\Sigma}_t$. This reduces computational complexity from $O(N^3)$ to $O(NK^2+K^3)$, which is particularly important in high-dimensional settings where $N \gg K$. The elegant relationship between $\mathcal{I}_{hh}$ and $\mathcal{I}_{ff}$ allows for further efficiency, as $\mathcal{I}_{hh}$ can be derived directly from $\mathcal{I}_{ff}$ using the formula $[\mathcal{I}_{hh}]_{i,j} = \frac{1}{2}e^{h_{i,t}+h_{j,t}}[\mathcal{I}_{ff}]_{i,j}^2$.

A.4.6 Significance of the FIM Derivatives

The analytical derivation of the FIM derivatives with respect to log-volatilities $\boldsymbol{h}_t$ is significant for both theoretical and practical reasons. The $\mathcal{I}_{hh}$ block captures the curvature of the log-likelihood with respect to log-volatilities, providing accurate uncertainty estimates for volatility states. This is particularly important in financial applications where volatility estimation and its uncertainty are key for risk management.

The block-diagonal structure of the FIM supports the BCD approach by allowing separate updates for factors and log-volatilities, which is a key feature of the Bellman filter for DFSV models. Additionally, these derivatives are essential for parameter estimation via maximum likelihood, as they contribute to the score function and observed information matrix used in optimization algorithms.

A.5 Optimizer Comparison for DFSV Models

This section provides a detailed comparison of the optimization algorithms used for parameter estimation in the DFSV model. The choice of optimizer significantly impacts both the convergence reliability and the final parameter accuracy, particularly given the complex likelihood surface of DFSV models.

A.5.1 Comprehensive Optimizer Comparison Results

To provide a detailed view of optimizer performance, we present comprehensive comparison results for various optimization algorithms applied to the DFSV-BIF and DFSV-PF models with unfixed mean parameters (μ). This comparison evaluates each optimizer based on convergence success, final loss value achieved, number of optimization steps required, and total computation time.

Table A.1 presents the results for the DFSV-BIF model.

Table A.1: Optimizer Comparison Results (DFSV-BIF, Unfixed μ / Fix_mu = No)

Optimizer	Success	Final Loss	Steps	Time (s)
DampedTrustRegionBFGS	Yes	-1,380.15	194	250.6
IndirectTrustRegionBFGS	Yes	-1,380.15	174	227.5
DogLegBFGS	Yes	-1,380.15	367	482.6
ArmijoBFGS	Yes	-1,380.15	808	643.3
AdamW	Yes	-1,380.15	991	1,474.8
Adam	Yes	-1,380.15	992	1,375.7
RMSProp	Yes	-1,379.46	990	1,358.4
Lion	Yes	-1,365.36	997	1,397.2
Adafactor	Yes	-1,352.82	972	1,362.8
GradientDescent	Yes	-1,349.00	983	1,297.0
BFGS	Yes	-803.28	385	100.0
Adagrad	Yes	1,059.65	995	1,313.0
Adadelta	Yes	11,121.37	978	1,302.4
SGD	No	-	0	1,360.5

Table A.2: Optimizer Comparison Results (DFSV-PF, Unfixed μ / Fix_mu = No) - Revised

Optimizer	Success	Final Loss	Steps	Time (s)	Notes
BFGS	Yes	22,120.80	299	146.0	-
DampedTrustRegionBFGS	Yes	22,841.28	30	37.0	-
DogLegBFGS	Yes	22,841.28	52	65.3	-
ArmijoBFGS	Yes	22,841.28	81	63.1	-
AdamW	No	Inf/NaN	0	1,991.2	The maximum number of steps was reached in the nonlinear solver.
IndirectTrustRegionBFGS	No	Inf/NaN	0	1,474.3	The maximum number of steps was reached in the nonlinear solver.

As shown in Table A.1, most optimizers successfully converge, but with significant differences in efficiency and final loss values. The trust region variants of BFGS (DampedTrustRegionBFGS and IndirectTrustRegionBFGS) achieved the optimal final loss value with substantially fewer steps and lower computation time compared to gradient-based methods like Adam and AdamW. This supports the selection of DampedTrustRegionBFGS as the primary optimizer for the BIF approach in the main simulation studies. Standard SGD failed to converge, underscoring the necessity of adaptive step sizing for the complex likelihood surface of DFSV models.

Table A.2 presents a similar comparison for the DFSV-PF model.

Table A.2 shows a different performance pattern for the DFSV-PF model. While several BFGS variants converged successfully, standard BFGS achieved the lowest final loss value, albeit requiring more steps than the trust region variants. Notably, AdamW and IndirectTrustRegionBFGS failed to converge, indicating the challenging nature of optimizing the particle filter likelihood. The ArmijoBFGS optimizer, selected for the PF approach, demonstrates a good balance of convergence reliability, final loss value, and computational efficiency among the tested methods that successfully converged.

A.5.2 Selected Optimizer Algorithms

Based on the comprehensive comparison results presented above, DampedTrustRegionBFGS was selected for the BIF approach and ArmijoBFGS was selected for the PF approach. The following paragraphs provide a detailed description of these two algorithms.

DampedTrustRegionBFGS: This algorithm, used for the BIF approach, is a specialized variant of the BFGS quasi-Newton method that combines two key enhancements for robust optimization. First, it employs a damped Newton descent strategy that modifies the Hessian approximation to ensure positive definiteness, which is crucial for maintaining descent properties when optimizing non-convex functions. Second, it incorporates a classical trust region approach that adaptively adjusts step sizes based on the agreement between predicted and actual objective function improvements.
The trust region mechanism uses configurable high and low cutoff thresholds (0.995 and 0.1 by default) with corresponding expansion and contraction constants (2.5 and 0.2) to dynamically resize the trust region during optimization. When the actual improvement closely matches or exceeds the predicted improvement (ratio > 0.995), the trust region expands by a factor of 2.5, allowing for more aggressive steps. Conversely, when the actual improvement falls significantly short of predictions (ratio < 0.1), the trust region contracts by a factor of 0.2, forcing more conservative steps. This adaptive approach is particularly effective for the non-convex likelihood surfaces encountered in DFSV models with stochastic volatility components.

ArmijoBFGS: This algorithm, used for the PF approach, combines BFGS with a backtracking Armijo line search strategy. The line search begins with an initial step size (0.1 by default) and progressively reduces it until the Armijo condition is satisfied, which ensures sufficient decrease in the objective function relative to the gradient magnitude. This approach is particularly well-suited for the particle filter likelihood, which can exhibit irregularities due to the Monte Carlo approximation of the likelihood function.
The Armijo condition specifically requires that the actual decrease in the objective function is at least a fraction (typically 0.0001) of the decrease predicted by a linear approximation. This conservative approach helps prevent steps that might lead to numerical instabilities or divergence, which is especially important when optimizing the particle filter likelihood where the gradient information may be noisy due to the stochastic nature of the particle approximation.

Appendix B

Supplementary Data Analysis

This appendix provides additional details and visualizations related to the exploratory data analysis presented in Chapter 3.

B.1 Detailed Portfolio Return Statistics

Table B.1: Concise Summary Statistics for Demeaned Portfolio Returns ($\boldsymbol{r}'_t$)

Statistic	SMALL LoBM	ME5 BM5	BIG LoBM	Avg_Return
Mean	0.000	0.000	0.000	0.000
Std Dev	0.085	0.058	0.049	0.053
Skewness	0.229	-0.416	-0.118	-0.488
Kurtosis	2.592	1.633	1.100	2.335

Table B.2: Summary Statistics for Demeaned Portfolio Returns ($\boldsymbol{r}'_t$)

Statistic	SMALL LoBM	ME1 BM2	SMALL HiBM	ME5 BM5	BIG LoBM	ME9 BM2	ME10 BM2	Avg_Return
Mean	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Std Dev	0.085	0.080	0.065	0.058	0.049	0.052	0.046	0.053
Skewness	0.229	0.201	-0.006	-0.416	-0.118	-0.263	-0.297	-0.488
Kurtosis	2.592	2.126	3.510	1.633	1.100	1.702	1.532	2.335
Minimum	-0.346	-0.324	-0.305	-0.268	-0.195	-0.256	-0.249	-0.277
Maximum	0.441	0.363	0.337	0.249	0.215	0.216	0.195	0.226
Autocorr(1)	0.211	0.182	0.222	0.057	0.025	0.027	0.009	0.101

B.2 Portfolio Characteristics Analysis

Figure B.1: Time Series Plots of Selected Demeaned Portfolio Returns

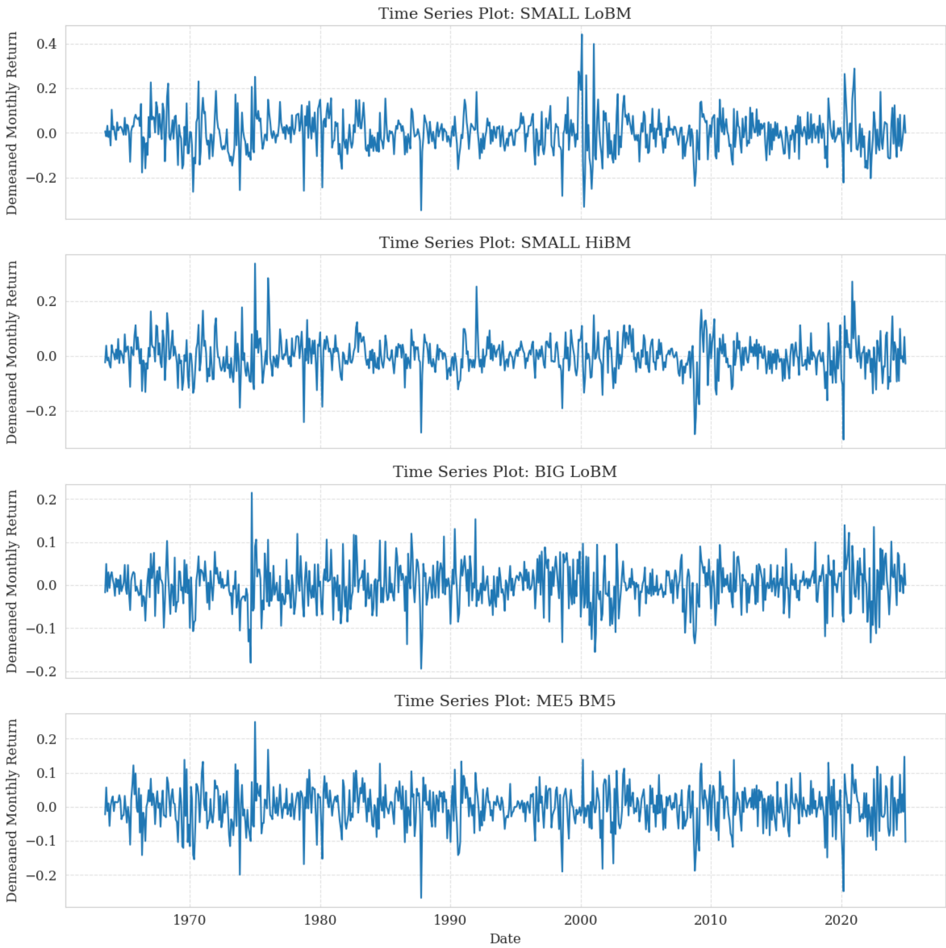


Table B.3: Summary Statistics for Portfolio Characteristics (Number of Firms, Average Market Cap)

Portfolio	Mean Number of Firms	Mean Market Cap ($ Millions)
Average	38.6	4,513.4
SMALL LoBM	208.0	53.5
ME1 BM2	115.0	55.9
SMALL HiBM	356.4	38.8
ME5 BM5	22.1	947.0
BIG LoBM	37.8	50,947.3
ME9 BM2	22.5	7,399.3
ME10 BM2	26.3	41,880.2

Figure B.2: Time Series of Average Portfolio Characteristics

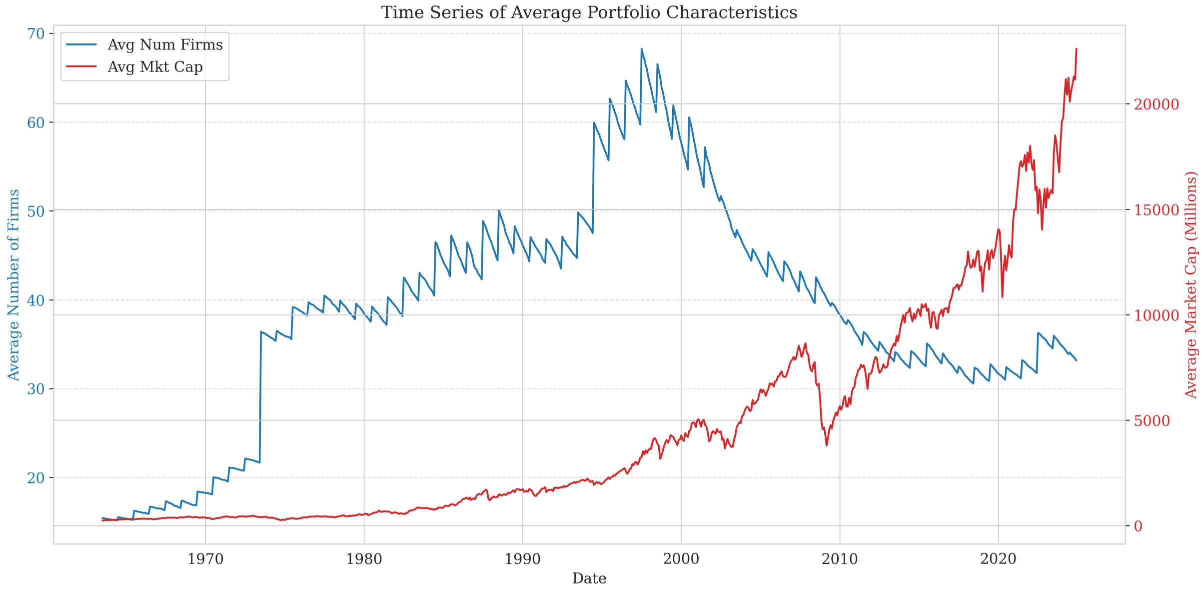
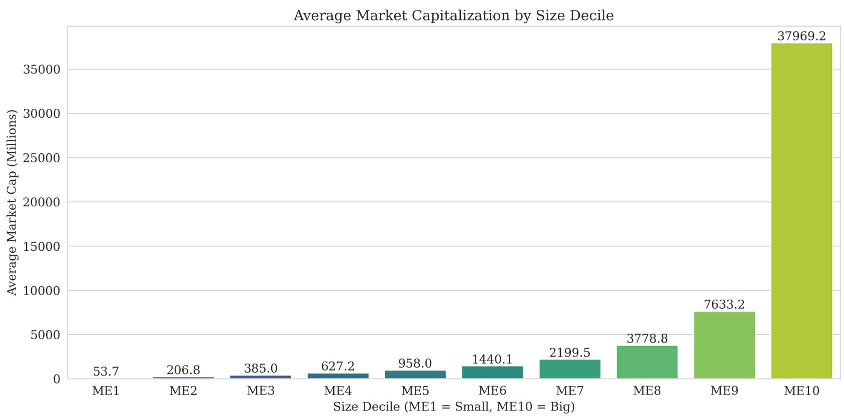


Figure B.3: Average Market Capitalization by Size Decile



Appendix C

Supplementary material for simulation studies

C.1 Simulation Study 1: Parameter Generation Details

Table C.1 details the parameter generation process for Simulation Study 1, focusing on computational scaling.

Table C.1: Parameter Generation for Simulation Study 1 (Computational Scaling)

Parameter	**Dimensions**	**Distribution/Method**	**Constraints**
Factor loadings ($\mathbf{\Lambda}$)	$N \times K$	$\mathcal{N}(0, 1)$	None
Factor transition ($\mathbf{\Phi}_f$)	$K \times K$	$\mathcal{N}(0, 0.5)$	$\lvert\text{eig}(\mathbf{\Phi}_f)\rvert < 1$
Log-volatility transition ($\mathbf{\Phi}_h$)	$K \times K$	$\mathcal{N}(0, 0.5)$	$\lvert\text{eig}(\mathbf{\Phi}_h)\rvert < 1$
Log-volatility mean ($\boldsymbol{\mu}$)	K	$\mathcal{N}(-1, 0.5)$	None
Idiosyncratic variance (σ_ϵ^2)	N (diagonal)	$\exp(z)$ where $z \sim \mathcal{N}(-1, 0.5)$	Positive by construction
Log-volatility innovation ($\boldsymbol{Q}_h$)	$K \times K$	$\boldsymbol{Q}_{raw} \sim \mathcal{N}(0, 0.2)$, $\boldsymbol{Q}_h = \boldsymbol{Q}_{raw}\boldsymbol{Q}_{raw}^\top$	PD by construction

This section provides supplementary material for Simulation Study 1, discussed in Section 4. It includes detailed stability analysis tables and additional figures illustrating performance metrics across various configurations.

C.1.1 Stability Analysis Tables

Tables C.2 and C.3 summarize the stability analysis for the BIF and PF, respectively, based on the mean-to-median RMSE ratios. Ratios significantly greater than 1 indicate potential instability or skewed error distributions.

The stability analysis reveals striking differences between the BIF and PF approaches. While the BIF exhibits moderate factor ratio instability (maximum 10.99), the PF shows extreme instability with factor ratios reaching orders of magnitude higher (up to 4.26e+12). This indicates that the PF occasionally produces catastrophic estimation failures that dramatically skew the mean RMSE upward, despite having comparable or even better median performance in some configurations. Interestingly, both methods show relatively stable log-volatility estimation (ra-

Table C.2: BIF Stability Analysis: Mean vs Median RMSE (Top 10 by Factor Ratio).

N	K	Mean RMSE (f)	Median RMSE (f)	Ratio (f)	Mean RMSE (h)	Median RMSE (h)	Ratio (h)
20	5	5.19e+00	4.72e-01	10.99	9.34e-01	8.56e-01	1.09
150	15	9.75e+00	1.14e+00	8.57	2.27e+00	1.36e+00	1.67
100	10	3.53e+00	8.53e-01	4.13	1.27e+00	1.16e+00	1.09
50	15	4.68e+00	1.22e+00	3.84	6.67e+00	1.30e+00	5.14
50	10	3.10e+00	8.42e-01	3.68	1.27e+00	1.19e+00	1.07
20	15	4.74e+00	1.41e+00	3.37	1.62e+00	1.28e+00	1.26
10	10	3.30e+00	1.04e+00	3.18	1.30e+00	1.18e+00	1.11
150	10	1.90e+00	7.22e-01	2.63	1.26e+00	1.15e+00	1.09
20	10	1.47e+00	7.33e-01	2.01	1.11e+00	1.04e+00	1.06
5	5	1.03e+00	5.93e-01	1.74	8.60e-01	8.09e-01	1.06

Table C.3: PF Stability Analysis: Mean vs Median RMSE (Top 10 by Factor Ratio).

N	K	Particles	Mean RMSE (f)	Median RMSE (f)	Ratio (f)	Mean RMSE (h)	Median RMSE (h)	Ratio (h)
50	10	5000	4.02e+12	9.44e-01	4.26e+12	2.06e+00	1.69e+00	1.21
100	5	20000	5.61e+07	2.05e-01	2.73e+08	1.16e+00	9.33e-01	1.24
50	10	20000	9.27e+06	6.43e-01	1.44e+07	1.79e+00	1.46e+00	1.22
20	15	1000	2.18e+06	2.30e+00	9.51e+05	2.63e+00	2.39e+00	1.10
10	5	5000	9.45e+03	3.33e-01	2.84e+04	1.05e+00	8.51e-01	1.23
100	15	1000	1.93e+04	1.55e+00	1.25e+04	2.29e+00	2.10e+00	1.09
50	15	1000	1.62e+04	1.84e+00	8.83e+03	2.52e+00	2.19e+00	1.15
10	10	20000	4.22e+03	8.92e-01	4.73e+03	1.85e+00	1.56e+00	1.18
100	10	1000	2.37e+03	9.32e-01	2.54e+03	2.02e+00	1.70e+00	1.19
150	10	1000	1.75e+03	1.00e+00	1.74e+03	2.00e+00	1.73e+00	1.16

tios mostly between 1.0-1.7), suggesting that volatility estimates are more robust than factor estimates across filtering approaches. The BIF's deterministic nature provides more consistent performance across replications, while the PF's stochastic approach leads to occasional but severe estimation failures, particularly in higher-dimensional settings ($N \geq 50$, $K \geq 10$) or with insufficient particles. These findings highlight the importance of considering not just median performance but also stability when selecting a filtering approach for practical applications.

C.1.2 Supplementary Figures

The following figures provide additional visualizations of the performance metrics discussed in the main text, organized to tell a comprehensive story about filter performance across different dimensions and metrics.

Heatmap Visualizations: The Complete Dimensional Landscape

While the main text presents selected heatmaps, the complete set below offers a more comprehensive view of how both filters perform across the entire dimensional grid. These visualizations reveal patterns that might be missed when examining individual slices of the parameter space.

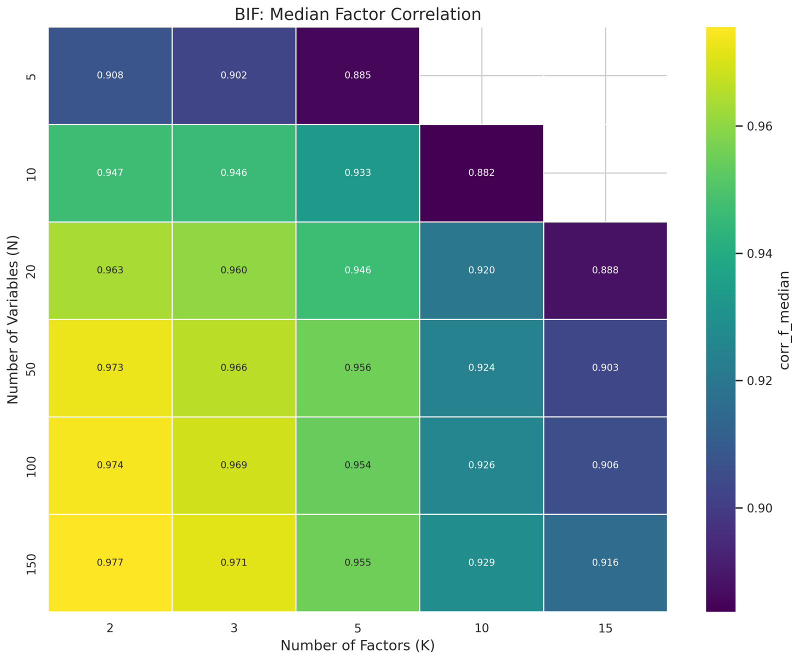


Figure C.1: Heatmap of BIF Median Factor Correlation vs. N and K. The relatively uniform coloring across most of the grid indicates that the BIF maintains consistent factor estimation quality even as dimensions increase, with only slight degradation in the highest-dimensional corner (N=150, K=15). This stability is a key advantage of the deterministic BIF approach.

C.2 Simulation Study 2: Parameter Generation Details

Table C.4 details the parameter generation process for Simulation Study 2, focusing on parameter estimation performance.

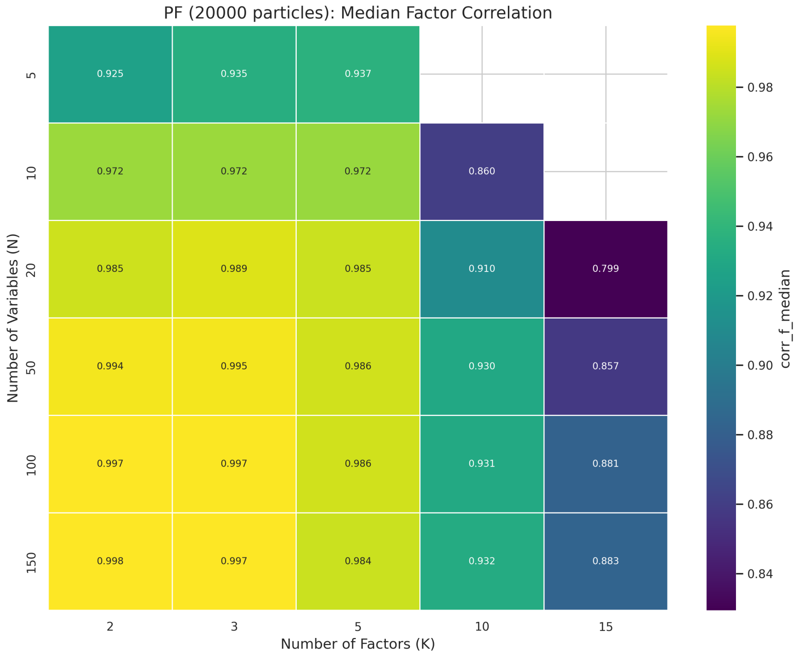


Figure C.2: Heatmap of PF-20k Median Factor Correlation vs. N and K. Compared to the BIF heatmap in Figure C.1, the PF shows stronger performance (lighter colors) in the lower-dimensional regions but experiences more pronounced degradation as K increases. This illustrates the PF's strength in lower dimensions and its challenges in higher-dimensional state-spaces, even with 20,000 particles.

Table C.4: Parameter Generation for Simulation Study 2 (Parameter Estimation)

Parameter	**Dimensions**	**Distribution/Method**	**Constraints**
Factor loadings ($\mathbf{\Lambda}$)	$N \times K$	$\mathcal{N}(0.5, 0.5)$	Lower triangular with diagonal elements fixed to 1.0 for identification.
Factor transition ($\mathbf{\Phi}_f$)	$K \times K$	Off-diagonals $\sim \mathcal{U}(0.01, 0.1)$, diagonals $\sim \mathcal{U}(0.15, 0.35)$	Stationarity ensured (stable by construction).
Log-volatility transition ($\mathbf{\Phi}_h$)	$K \times K$	Off-diagonals $\sim \mathcal{U}(0.01, 0.1)$, diagonals $\sim \mathcal{U}(0.9, 0.99)$	Stationarity ensured (spectral norm < 0.97).
Log-volatility mean ($\boldsymbol{\mu}$)	K	Fixed values, e.g., $[-1.0, -0.5]$ for $K = 2$ or $[-1.0, \ldots, -1.0]$ for $K > 2$.	None.
Idiosyncratic variance (σ_ϵ^2)	N (diagonal)	$\mathcal{U}(0.05, 0.1)$	Positive by construction.
Log-volatility innovation ($\boldsymbol{Q}_h$)	K (diagonal)	$\mathcal{U}(0.8, 1.0)$	Positive by construction.

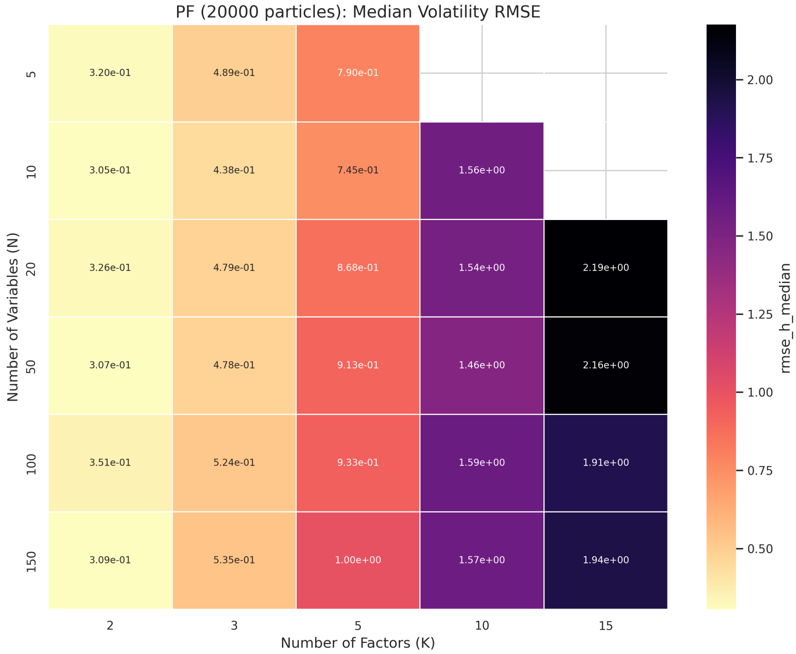


Figure C.3: Heatmap of PF-20k Median Log-Volatility RMSE vs. N and K. The darker regions in the upper-right corner reveal the PF's difficulty in accurately estimating log-volatilities as both N and K increase. When compared with the BIF's log-volatility RMSE heatmap in Figure 4.2b of the main text, this visualization highlights one of the BIF's key advantages: more robust log-volatility estimation in higher dimensions.

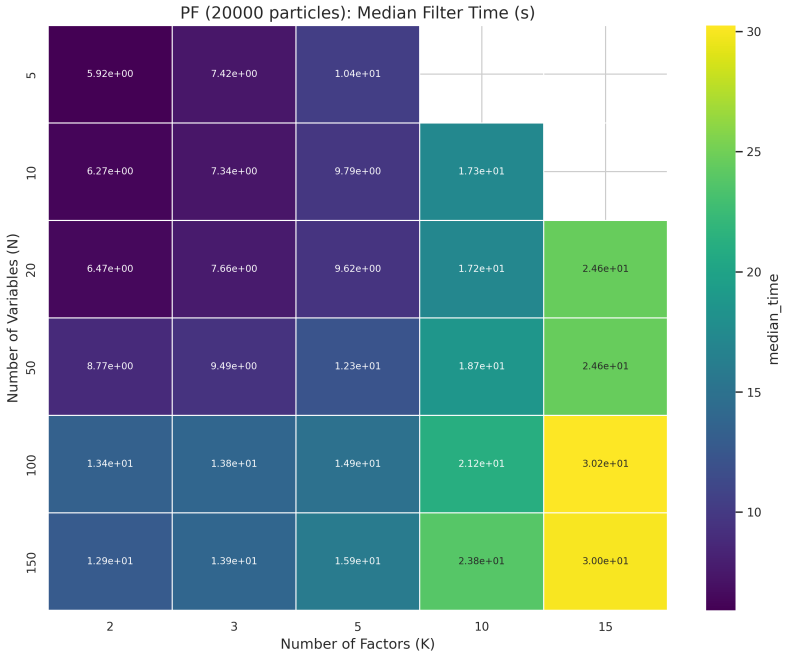


Figure C.4: Heatmap of PF-20k Median Computation Time (s) vs. N and K. The gradual darkening from left to right and bottom to top illustrates the PF's computational scaling with both N and K. Unlike the BIF's heatmap in Figure 4.2a of the main text, which shows more dramatic darkening along the K axis, the PF exhibits more balanced scaling with both dimensions, confirming its theoretical $O(NK)$ complexity. This different scaling behavior is a key consideration when choosing between filters for specific dimensional configurations.

Unique Supplementary Visualizations

While the main text presents the key performance metrics with detailed visualizations, the appendix focuses on complementary analyses that provide additional insights not covered in the main text. The following figures offer unique perspectives on filter performance that help complete the overall evaluation picture.

Stability and Trade-off Analysis: The final set of figures moves beyond individual metrics to examine broader patterns: the numerical stability of each filter and the fundamental trade-offs between computational cost and estimation accuracy.

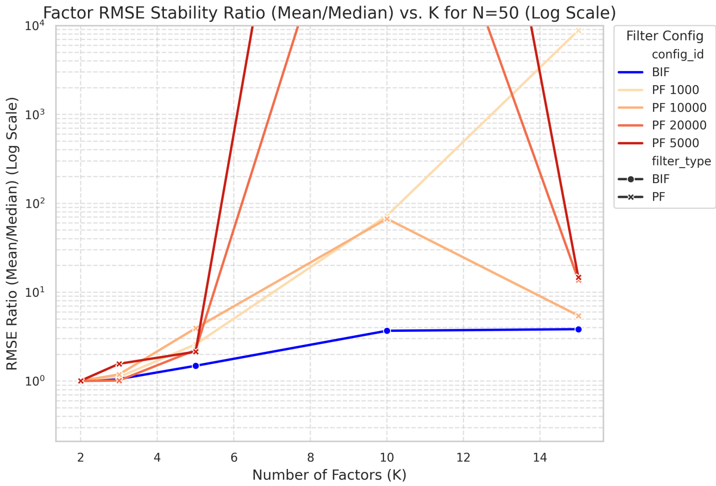


Figure C.5: Mean/Median Factor RMSE Ratio vs. K (N=50, Log Scale). This figure provides a direct measure of filter stability by showing the ratio of mean to median RMSE. Values close to 1 indicate consistent performance across replications, while higher values suggest occasional poor performance (outliers) that skews the mean upward. The BIF maintains a ratio near 1 across all K values, reflecting its deterministic nature and consistent performance. In contrast, the PF variants show increasing ratios with K, particularly for lower particle counts, indicating growing instability as the state dimension increases. This instability is a key practical limitation of the PF approach that is not captured by median metrics alone.

Summary of Supplementary Analysis: Taken together, these supplementary figures tell a nuanced story about the relative strengths of the BIF and PF approaches. The BIF offers superior log-volatility estimation, more consistent performance across replications, and favorable accuracy-to-computation trade-offs for most configurations. The PF provides slightly better factor estimation in lower dimensions but struggles with stability and computational scaling as dimensions increase. These findings complement and extend the analysis in the main text, providing a more complete picture of how these filtering approaches perform across the full range of tested configurations.

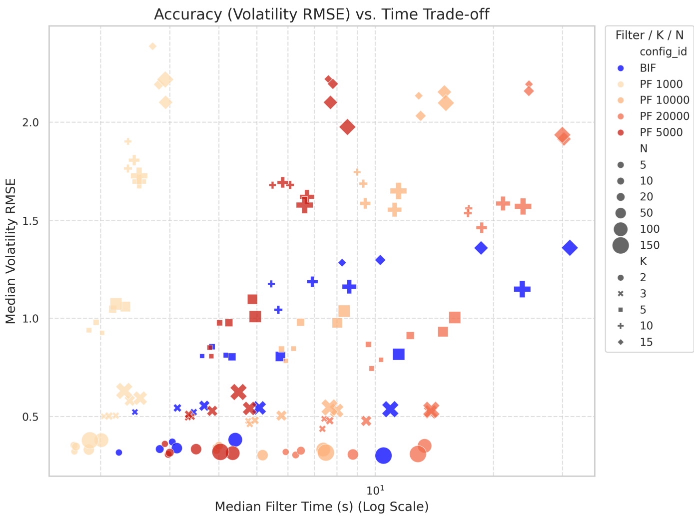


Figure C.6: Trade-off between Median Log-Volatility RMSE and Median Computation Time. This figure synthesizes the key performance dimensions into a single visualization, plotting accuracy (log-volatility RMSE) against computational cost for different filter configurations and model dimensions. Points toward the bottom-left corner represent the ideal combination of low error and fast runtime.

C.3 Supplementary Material for Simulation Study 2: Estimation Algorithm Details

This section provides detailed information on the optimization algorithms and configurations used for parameter estimation in Simulation Study 2, complementing the concise description in Section 3.4.4.
Parameter estimation is performed via maximum likelihood using both BIF and PF.

C.3.1 Selected Estimation Algorithms

Based on preliminary studies (see Appendix A.5), DampedTrustRegionBFGS was selected for the BIF approach and ArmijoBFGS was selected for the PF approach due to their convergence reliability and accuracy on the respective likelihood surfaces.

DampedTrustRegionBFGS (for BIF): This algorithm is a specialized variant of the BFGS quasi-Newton method that combines a damped Newton descent strategy with a classical trust region approach. The damped Newton descent modifies the Hessian approximation to ensure positive definiteness, crucial for optimizing non-convex functions. The trust region approach adaptively adjusts step sizes based on the agreement between predicted and actual objective function improvements. The trust region mechanism uses configurable high and low cutoff thresholds (0.995 and 0.1 by default) with corresponding expansion and contraction constants (2.5 and 0.2) to dynamically resize the trust region during optimization. This adaptive approach is particularly effective for the non-convex pseudo-likelihood surfaces encountered in DFSV models.

ArmijoBFGS (for PF): This algorithm combines BFGS with a backtracking Armijo line search strategy. The line search begins with an initial step size (0.1) and progressively reduces it until the Armijo condition is satisfied, ensuring sufficient decrease in the objective function relative to the gradient magnitude. This strategy helps navigate the potentially rugged likelihood landscapes that arise from the particle-based approximation of the likelihood function.

C.3.2 Estimation Configurations

Both optimizers were implemented using the Optimistix library Rader et al., 2024. Parameter transformations (e.g., tanh for diagonals, softplus for variances) and eigenvalue penalties ($\lambda = 10^4$) were employed to enforce stability constraints. Optimizations were run for a maximum of 2000 iterations.

For the PF approach, two particle counts are used: PF-1000 ($P = 1000$) and PF-5000 ($P = 5000$). The primary BIF variant estimates all parameters, including $\boldsymbol{\mu}$.

Optimizations are initialized with plausible parameter values generated similarly to the true parameters but with different random seeds, simulating a realistic scenario without prior knowledge of the true values.

C.4 Simulation Study 2: Supplementary Material (Bias Analysis)

This section provides a comprehensive analysis of parameter estimation bias patterns observed in Simulation Study 2, expanding on the discussion in Section 4.2.3 of the main text. While the main text focuses specifically on the bias patterns for the log-volatility transition matrix $\mathbf{\Phi}_h$, this appendix presents bias versus RMSE scatter plots for all model parameters across different model configurations.

C.4.1 Bias Analysis Methodology

For each parameter and model configuration, we analyze two key metrics:

- **Bias**: The average difference between estimated and true parameter values (Est - True), capturing systematic estimation errors.
- **RMSE**: The root mean squared error, capturing the overall accuracy of parameter estimates by combining both bias and variance.

The scatter plots presented in this section visualize the relationship between these metrics across different filter configurations (BIF, PF with varying particle counts) and time series lengths. Each point represents a specific simulation run, with marker shape indicating time series length and color indicating filter configuration. This visualization approach allows us to identify patterns in the bias-variance trade-off for different parameters and filtering approaches.

C.4.2 Factor Loading Parameters (Λ)

Figure C.7 presents the bias versus RMSE scatter plots for factor loading parameters across different model configurations. These parameters determine how the latent factors influence the observed returns.

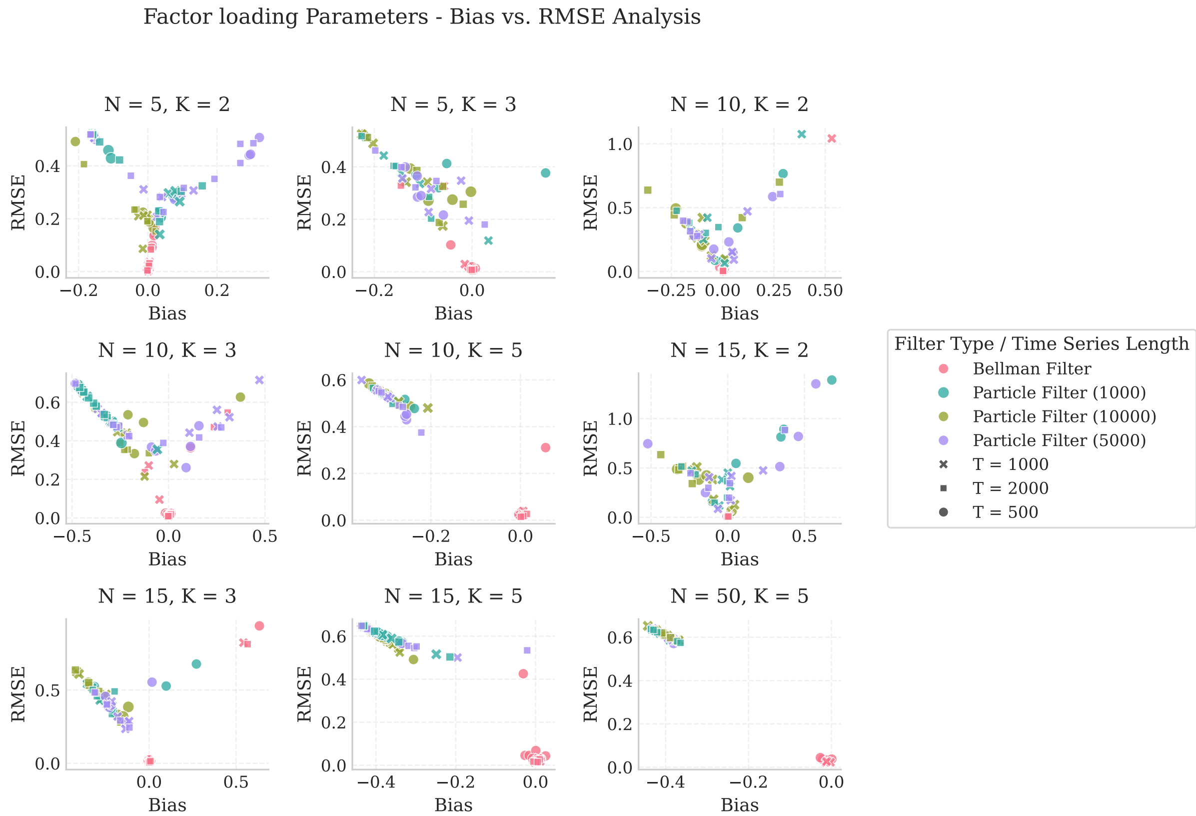


Figure C.7: Bias versus RMSE for factor loading parameters (Λ) across different model configurations. Each panel represents a specific (N, K) pair, with marker shape indicating time series length and color indicating filter configuration.

The scatter plots reveal that BIF estimates (blue crosses/triangles) typically exhibit lower RMSE compared to PF estimates, particularly for larger model dimensions. For all configurations, they demonstrate higher variance and bias, resulting in larger overall estimation errors. This pattern suggests that the BIF's deterministic approach provides more consistent factor loading estimates, even if they occasionally show small systematic biases.

C.4.3 Factor Transition Matrix (Φ_f)

Figure C.8 shows the bias versus RMSE scatter plots for the factor transition matrix parameters, which govern the temporal dynamics of the latent factors.

For the factor transition matrix, both BIF and PF approaches show relatively similar performance patterns, with the PF-10K having superior performance in terms of RMSE for the smaller model configurations. The bias patterns are more mixed compared to other parameters, suggesting that the temporal dynamics of factors present similar estimation challenges for both filtering approaches.

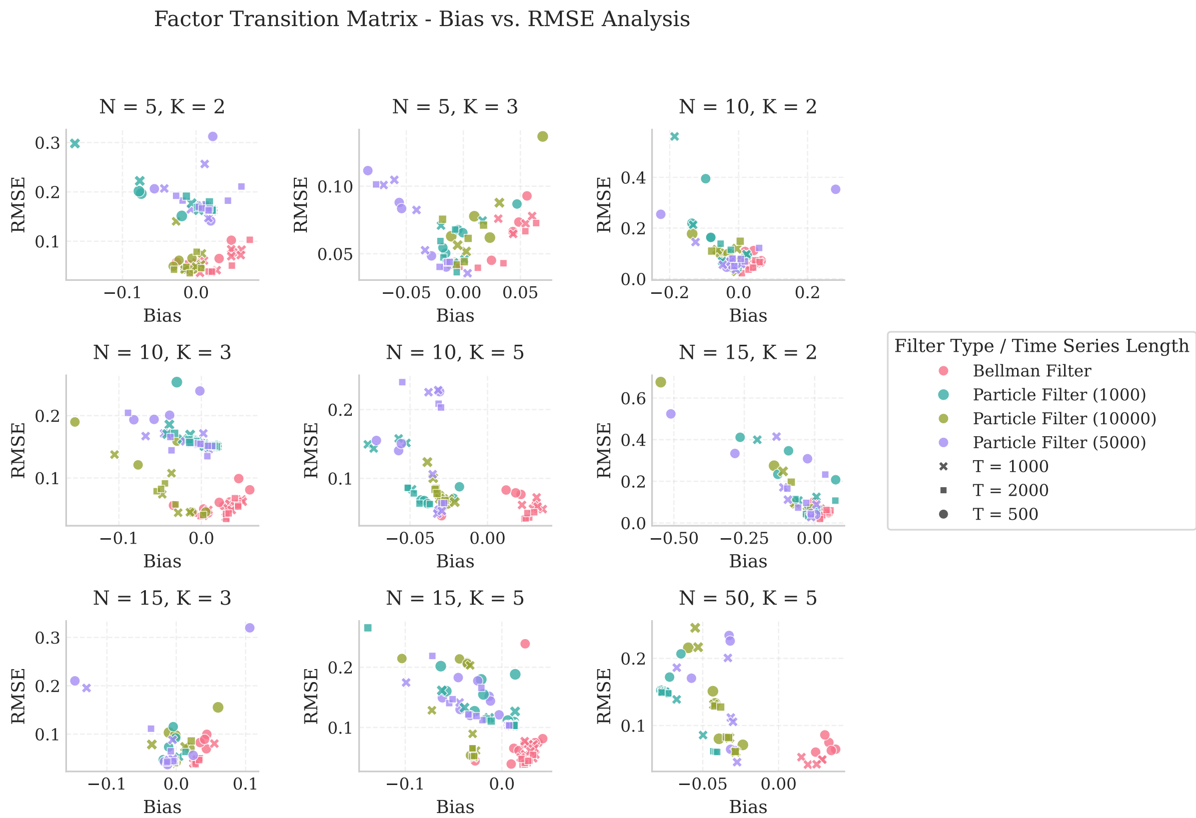


Figure C.8: Bias versus RMSE for factor transition matrix parameters ($\mathbf{\Phi}_f$) across different model configurations. Each panel represents a specific (N, K) pair, with marker shape indicating time series length and color indicating filter configuration.

C.4.4 Log-Volatility Transition Matrix ($\mathbf{\Phi}_h$)

Figure C.9 presents the bias versus RMSE scatter plots for the log-volatility transition matrix parameters, which was highlighted in the main text as showing a distinctive bias pattern for the BIF.

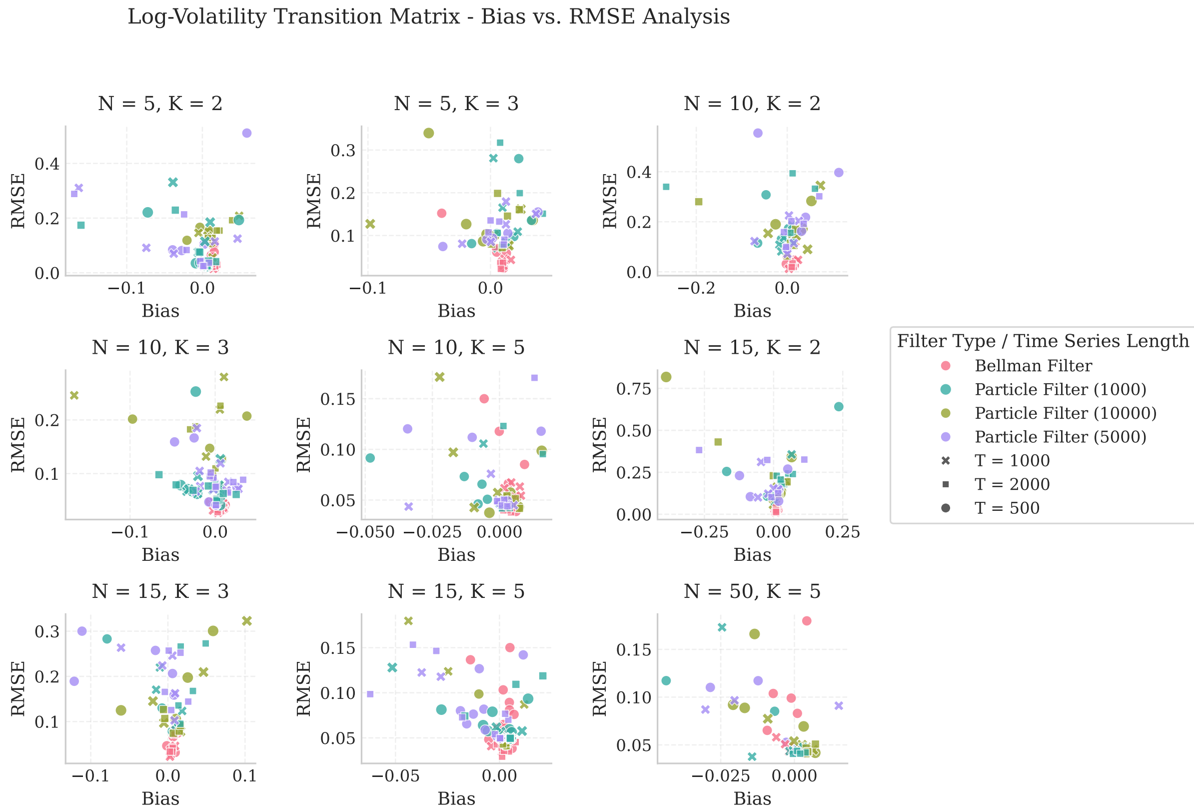


Figure C.9: Bias versus RMSE for log-volatility transition matrix parameters ($\mathbf{\Phi}_h$) across different model configurations. Each panel represents a specific (N, K) pair, with marker shape indicating time series length and color indicating filter configuration.

As discussed in the main text, the BIF estimates for $\mathbf{\Phi}_h$ show a consistent small negative bias across configurations, indicating a tendency to slightly underestimate the persistence of log-volatilities. However, the BIF achieves substantially lower RMSE compared to PF variants, demonstrating a favorable bias-variance trade-off. The PF estimates show wider dispersion around zero bias, with higher overall estimation errors, particularly for configurations with fewer particles.

C.4.5 Volatility Covariance Matrix ($\boldsymbol{Q}_h$)

Figure C.10 shows the bias versus RMSE scatter plots for the volatility covariance matrix parameters, which determine the magnitude and correlation structure of log-volatility innovations. The volatility covariance matrix parameters show some of the most pronounced differences between filtering approaches. The PF estimates consistently achieve lower RMSE and a smaller bias compared to the BIF. The BIF estimates show a consistent negative bias pattern, suggesting a tendency to underestimate the volatility of log-volatility processes. This underestimation may be related to the BIF's mode-seeking nature, which can lead to smoother volatility estimates

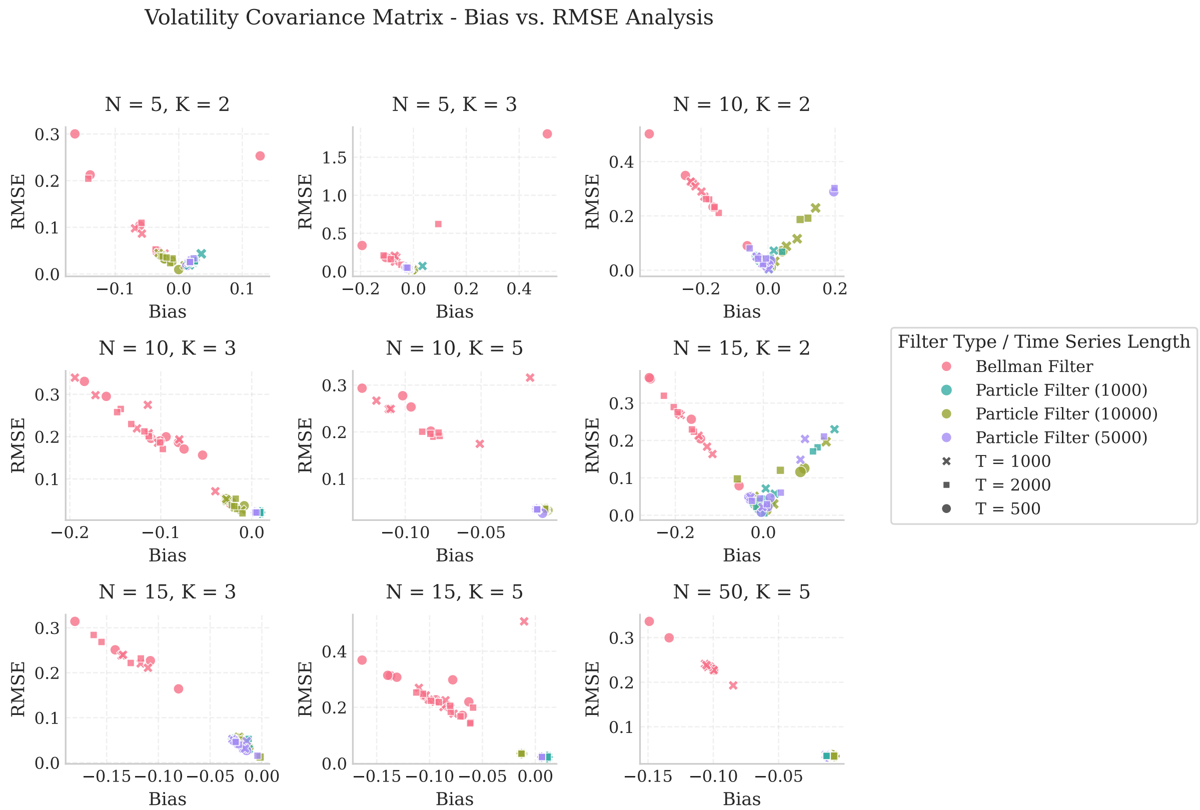


Figure C.10: Bias versus RMSE for volatility covariance matrix parameters ($\boldsymbol{Q}_h$) across different model configurations. Each panel represents a specific (N, K) pair, with marker shape indicating time series length and color indicating filter configuration.

compared to the full posterior representation of the PF.

C.4.6 Long-run Mean of Log-Volatilities ($\boldsymbol{\mu}$)

Figure C.11 presents the bias versus RMSE scatter plots for the long-run mean parameters of the log-volatilities, which determine the central tendency of the volatility processes.

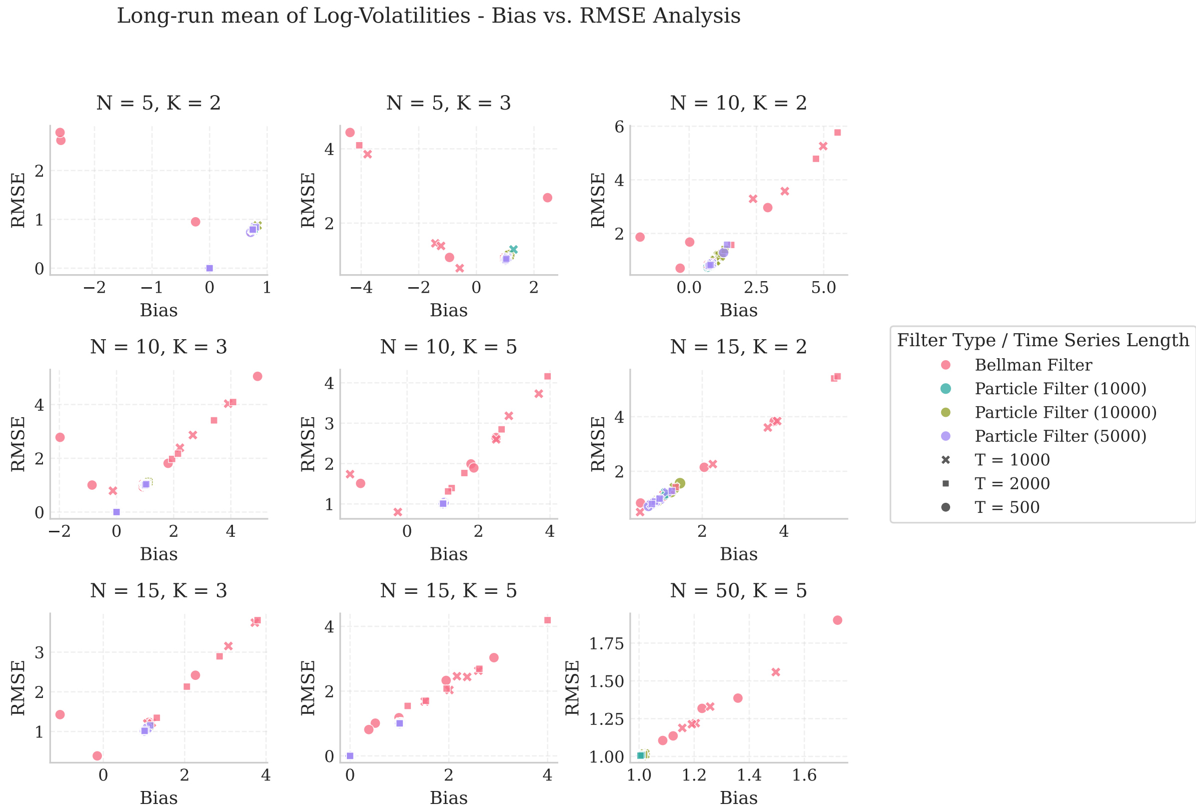


Figure C.11: Bias versus RMSE for long-run mean parameters of log-volatilities ($\boldsymbol{\mu}$) across different model configurations. Each panel represents a specific (N, K) pair, with marker shape indicating time series length and color indicating filter configuration.

The long-run mean parameters show some of the most challenging estimation patterns among all parameters, with relatively high RMSE for both filtering approaches. This difficulty aligns with the parameter difficulty ranking presented in Table 4.2 of the main text, where $\boldsymbol{\mu}$ components are among the most difficult to estimate accurately. The various PF configurations appear to be more consistent in their estimation of $\boldsymbol{\mu}$ compared to the BIF.

C.4.7 Idiosyncratic Noise Variance (Σ_ϵ)

Figure C.12 shows the bias versus RMSE scatter plots for the idiosyncratic noise variance parameters, which determine the magnitude of asset-specific return variations not explained by the common factors.

For the smaller model configurations both filtering approaches show relatively similar performance patterns, with the BIF generally achieving slightly lower RMSE. As the model dimensions increase, the PF estimates show significantly increased RMSE and bias, particularly for configurations with fewer particles.

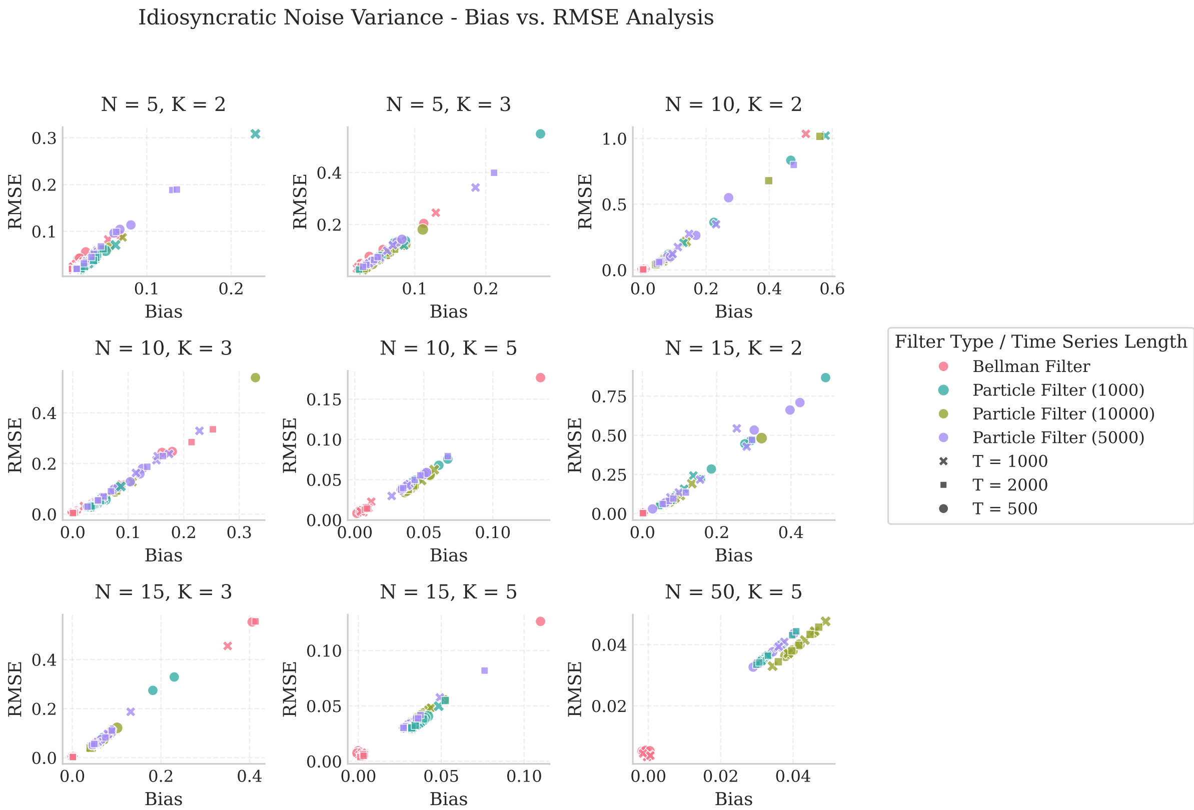


Figure C.12: Bias versus RMSE for idiosyncratic noise variance parameters ($\mathbf{\Sigma}_\epsilon$) across different model configurations. Each panel represents a specific (N, K) pair, with marker shape indicating time series length and color indicating filter configuration.

C.4.8 Synthesis of Bias Analysis Findings

The comprehensive bias analysis presented in this appendix reveals several important patterns:

1. **Bias-Variance Trade-off**: The BIF consistently demonstrates lower overall estimation error (RMSE) across most parameters and configurations, but often shows small systematic biases. In contrast, PF estimates typically cluster around zero bias but exhibit higher variance, resulting in larger overall estimation errors.

2. **Parameter-Specific Patterns**: Different parameters show distinct bias patterns. The log-volatility parameters ($\boldsymbol{Q}_h$, $\boldsymbol{\mu}$) show the most pronounced differences between filtering approaches, while factor-related parameters ($\boldsymbol{\Lambda}$, $\boldsymbol{\Phi}_f$) and observation parameters ($\boldsymbol{\Sigma}_\epsilon$) show more similar patterns.

3. **Dimensional Effects**: As model dimensions increase (higher N and K), the performance gap between BIF and PF approaches generally widens, with the BIF maintaining more consistent estimation quality.

4. **Time Series Length Effects**: Longer time series (T=2000 vs. T=1000) generally lead to improved estimation accuracy for both approaches, but the relative patterns between filtering methods remain consistent.

These findings extend the discussion in the main text by providing a more comprehensive view of parameter estimation patterns across the full DFSV model specification. The observed bias-variance trade-off between BIF and PF approaches represents an important consideration for practitioners when choosing between these filtering methods for empirical applications.

C.4.9 Parameter Estimation Error Scaling

This section presents additional figures illustrating how the median Root Mean Squared Error (RMSE) for various model parameters scales with model dimensions (N, K, and T) in Simulation Study 2. These figures complement Figure 4.5 in the main text and provide a more complete picture of parameter estimation accuracy across the full set of parameters.

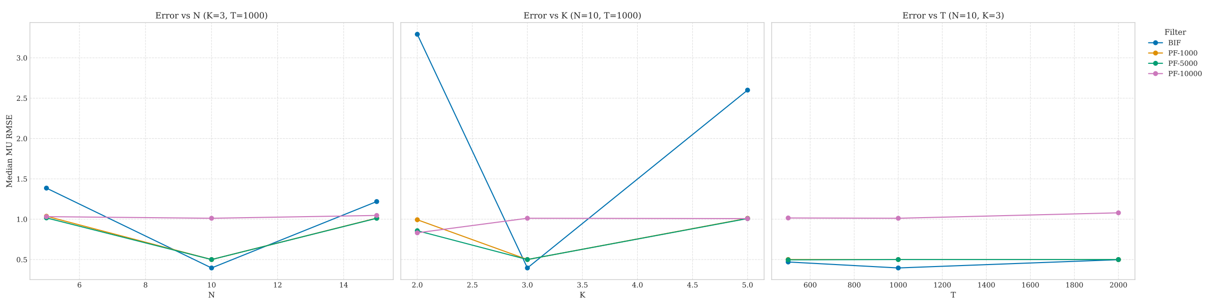


Figure C.13: Median RMSE for log-volatility long-run mean parameters ($\boldsymbol{\mu}$) scaling with model dimensions (N, K, T). BIF and PF variants show relatively high RMSE, indicating difficulty in estimating this parameter, with performance generally improving with increasing T.

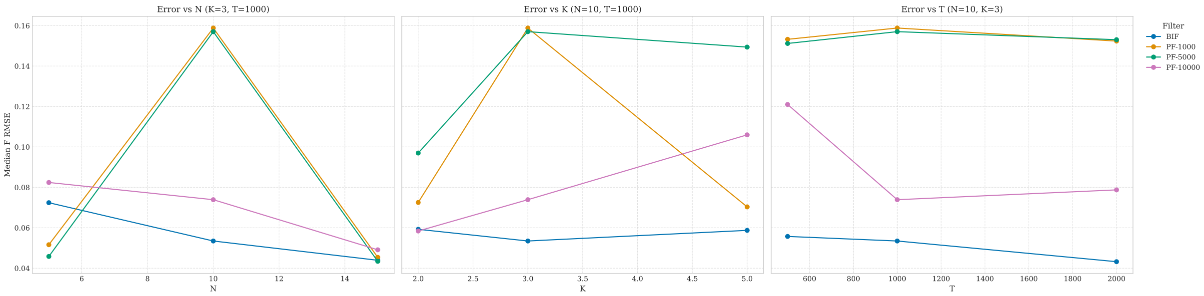


Figure C.14: Median RMSE for factor transition matrix parameters ($\mathbf{\Phi}_f$) scaling with model dimensions (N, K, T). BIF generally achieves lower RMSE than PF variants, with accuracy improving with increasing T.

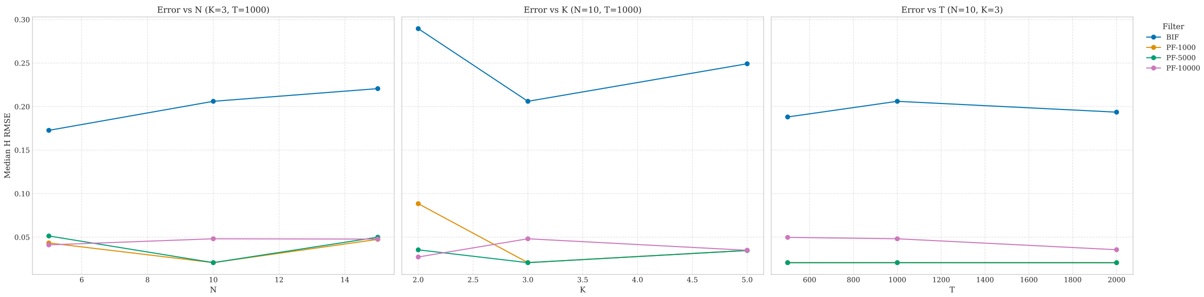


Figure C.15: Median RMSE for log-volatility innovation covariance parameters ($\boldsymbol{Q}_h$) scaling with model dimensions (N, K, T). PF-10k shows the lowest RMSE for this parameter, while BIF exhibits higher RMSE, particularly with increasing K.

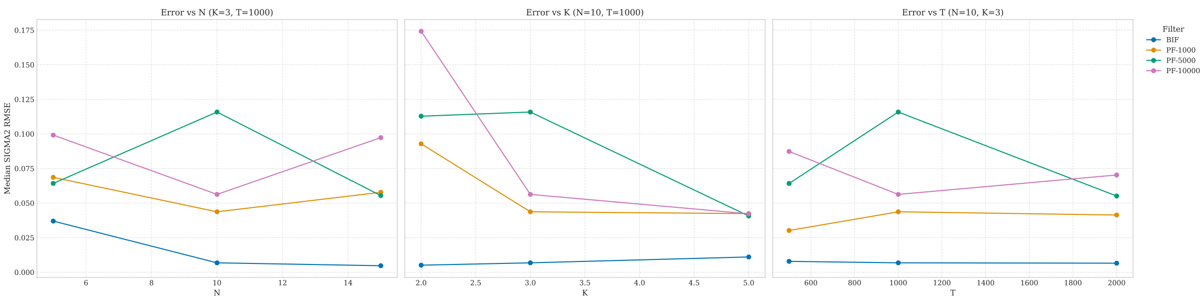


Figure C.16: Median RMSE for idiosyncratic noise variance parameters ($\mathbf{\Sigma}_\epsilon$) scaling with model dimensions (N, K, T). Both BIF and PF variants show similar RMSE patterns, with accuracy improving with increasing T.

Appendix D

Supplementary Material for Empirical Application

This appendix provides supplementary figures and detailed results for the empirical application discussed in Chapter 5. It includes individual factor and log-volatility plots that complement the aggregated figures presented in the main text.

D.1 Individual Factor Estimates

This section presents the individual factor estimates from both the DFSV-BIF and DFSV-PF models. These detailed plots complement the aggregated overviews presented in the main text.

D.1.1 BIF Factor Estimates

The following figures show the individual factor estimates from the DFSV-BIF model. These detailed plots complement the overview presented in Figure 5.3 of the main text.

D.1.2 PF Factor Estimates

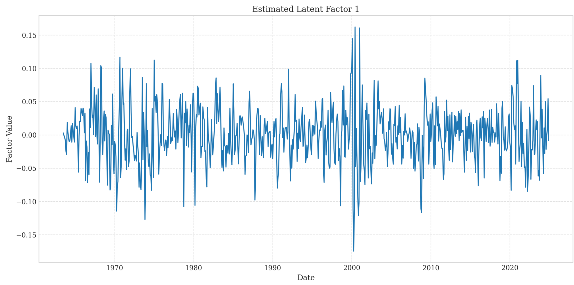
The following figures show the individual factor estimates from the DFSV-PF model. These detailed plots provide a comparison with the BIF estimates and complement the overview presented in the main text.

D.2 Individual Log-Volatility Estimates

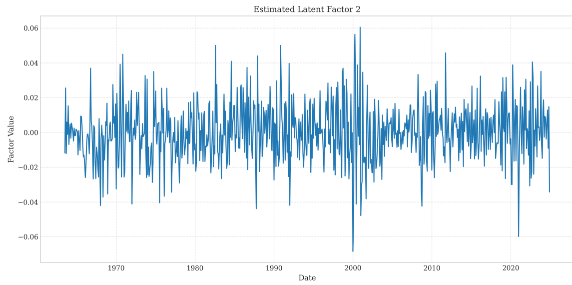
This section presents the individual log-volatility estimates from both the DFSV-BIF and DFSV-PF models. These detailed plots complement the aggregated overviews presented in the main text.

D.2.1 BIF Log-Volatility Estimates

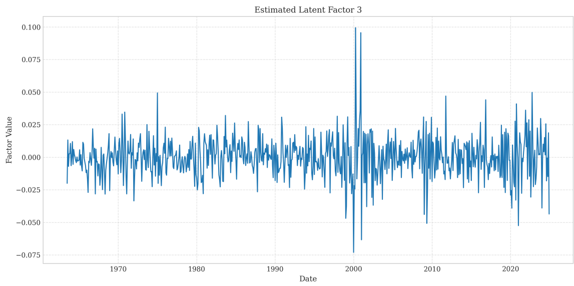
The following figures show the individual log-volatility estimates from the DFSV-BIF model. These detailed plots complement the overview presented in Figure 5.3 of the main text.



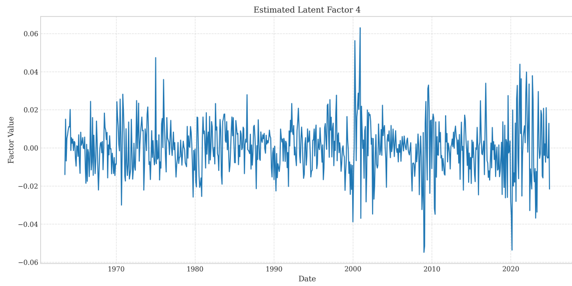
(a) Factor 1



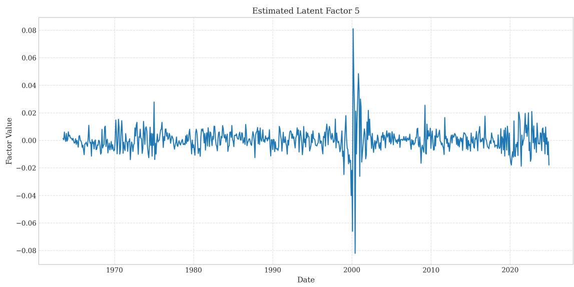
(b) Factor 2



(c) Factor 3



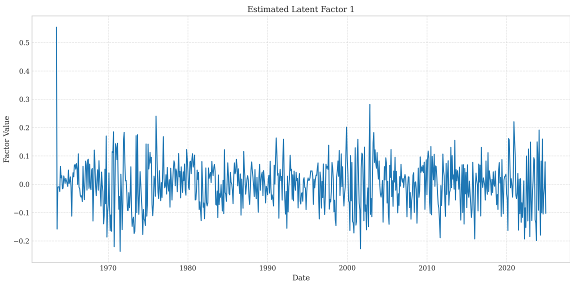
(d) Factor 4



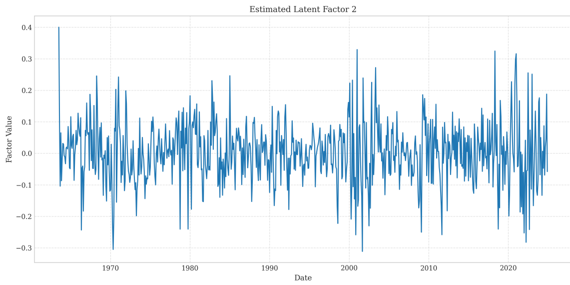
(e) Factor 5

Figure D.1: Individual Estimated Latent Factors ($\hat{\boldsymbol{f}}_{t|T}$) (DFSV-BIF)

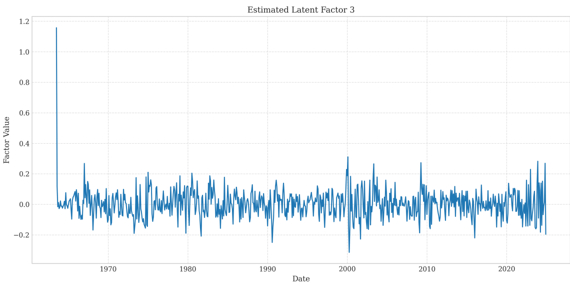
Note: Smoothed estimates of the K=5 latent factors obtained from the DFSV-BIF model over the full sample period (July 1963–Dec 2023).



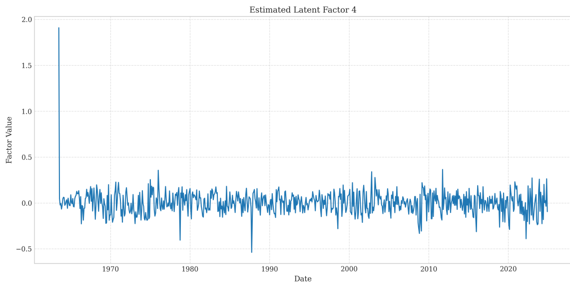
(a) Factor 1



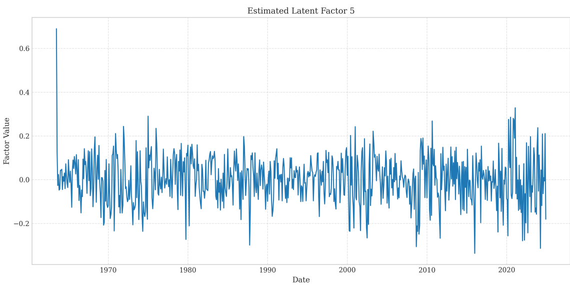
(b) Factor 2



(c) Factor 3



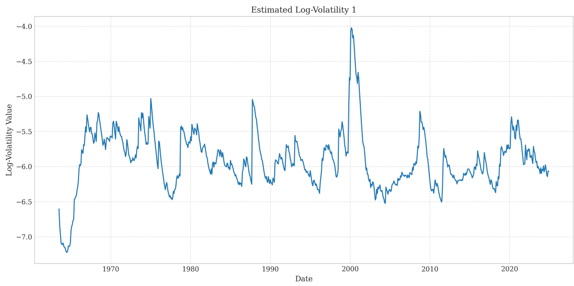
(d) Factor 4



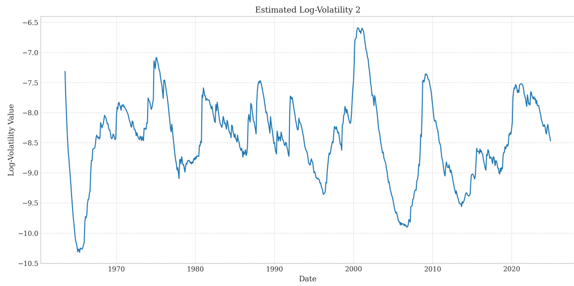
(e) Factor 5

Figure D.2: Individual Estimated Latent Factors ($\hat{\boldsymbol{f}}_{t|T}$) (DFSV-PF)

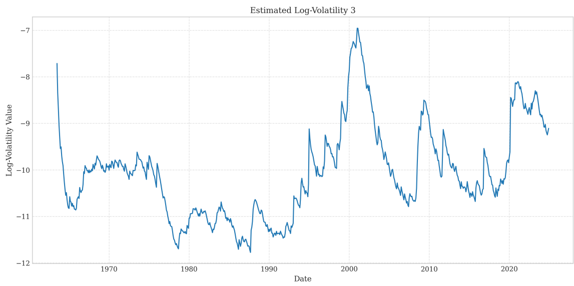
Note: Smoothed estimates of the K=5 latent factors obtained from the DFSV-PF model over the full sample period (July 1963–Dec 2023).



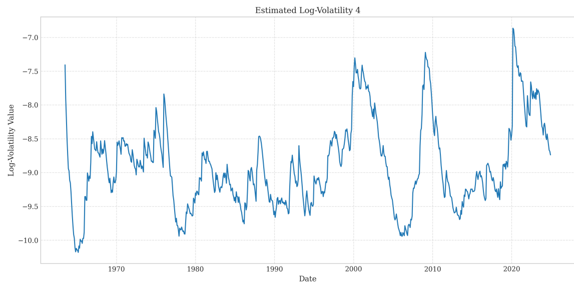
(a) Log-Volatility Factor 1



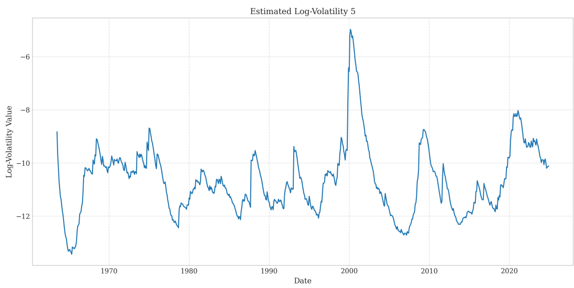
(b) Log-Volatility Factor 2



(c) Log-Volatility Factor 3



(d) Log-Volatility Factor 4



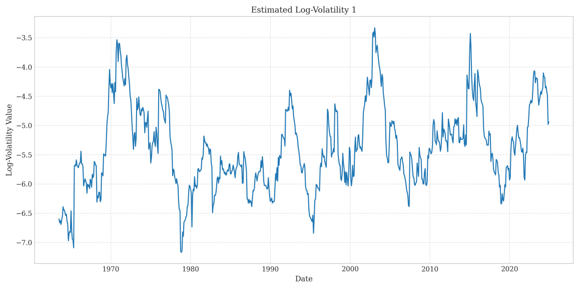
(e) Log-Volatility Factor 5

Figure D.3: Individual Estimated Factor Log-Volatilities ($\hat{\boldsymbol{h}}_{t|T}$) (DFSV-BIF)

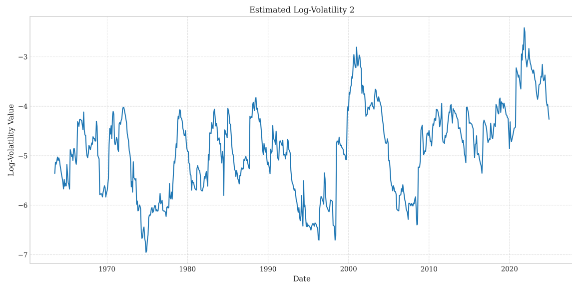
Note: Smoothed estimates of the K=5 factor log-volatilities obtained from the DFSV-BIF model over the full sample period.

D.2.2 PF Log-Volatility Estimates

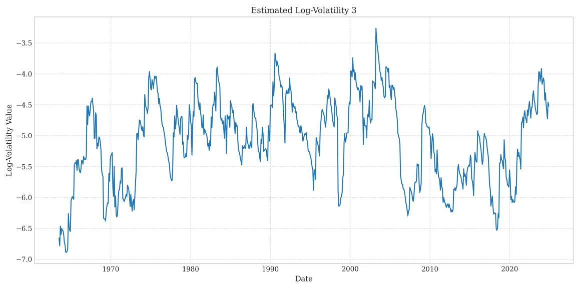
The following figures show the individual log-volatility estimates from the DFSV-PF model. These detailed plots provide a comparison with the BIF estimates and complement the overview presented in the main text.



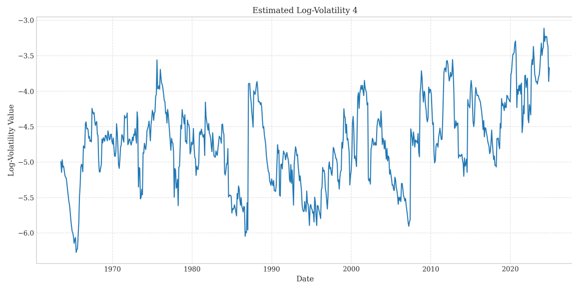
(a) Log-Volatility Factor 1



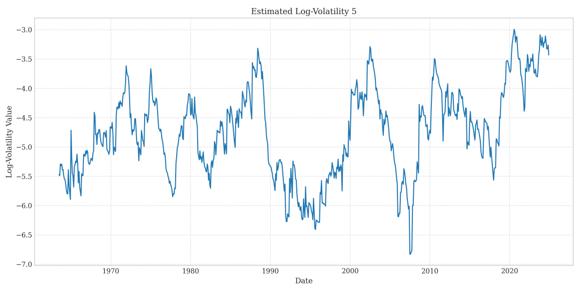
(b) Log-Volatility Factor 2



(c) Log-Volatility Factor 3



(d) Log-Volatility Factor 4



(e) Log-Volatility Factor 5

Figure D.4: Individual Estimated Factor Log-Volatilities ($\hat{\boldsymbol{h}}_{t|T}$) (DFSV-PF)

Note: Smoothed estimates of the K=5 factor log-volatilities obtained from the DFSV-PF model over the full sample period.

D.3 Factor Structure Dynamics

This section presents the analysis of factor structure dynamics that complements the covariance dynamics analysis in the main text. It examines how the proportion of total variance attributed to common factors varies over time in the DFSV models compared to the constant-volatility DFM.

Figure D.5 displays the Factor Contribution Ratio (FCR) over time for the factor-based models. The FCR is defined as:

$$\mathrm{FCR}_t = \frac{\operatorname{tr}(\hat{\boldsymbol{\Lambda}} \operatorname{Var}(\hat{\boldsymbol{f}}_t) \hat{\boldsymbol{\Lambda}}')}{\operatorname{tr}(\hat{\boldsymbol{\Sigma}}_t)}. \tag{D.1}$$

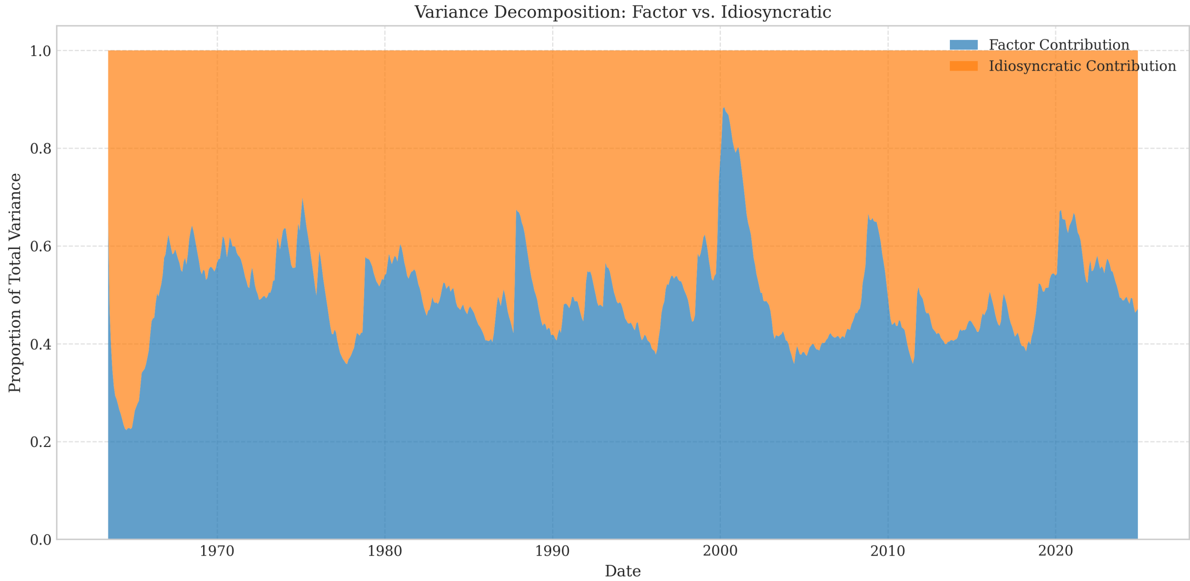


Figure D.5: Factor Variance Contribution Ratio (FCR) Time Series. Time series of the Factor Variance Contribution Ratio (FCR_t), defined as the trace of the common component variance divided by the trace of the total conditional variance, for DFM, DFSV-BIF, and DFSV-PF models. It measures the proportion of total variance explained by common factors.

This ratio measures the proportion of total variance explained by common factors at each point in time. As expected, the DFM (Factor-CV) shows a constant FCR due to its static covariance assumption. In contrast, the DFSV-BIF model exhibits notable time variation in the proportion of variance attributed to common factors, with the ratio generally decreasing during periods of market stress. This suggests that idiosyncratic risks become relatively more important during crisis periods, a finding consistent with some previous studies on factor structures during market turbulence. The DFSV-PF model shows more erratic patterns in its FCR, consistent with its generally less stable state estimates observed in other metrics.